PART C ENVIRONMENTAL MANAGEMENT

Balancing the protection of the natural environment and development

Revision: Amendment 8 (November 2025)



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PART CI
BIODIVERSITY AND NATURAL RESOURCES



Introduction

Biodiversity (biological diversity) is the variety of life: the different species of plants, animals and microorganisms, the genes they contain and the various populations, communities and ecosystems of which they form a part. Biodiversity, and the natural functions that occur in healthy ecosystems, provide essential processes including nutrient, carbon and energy cycling, atmospheric oxygen production, pollination, seed dispersal and climate regulation. These processes are referred to as ecosystem services and provide us with fresh water, clean air, soil fertility and biological pest control.

Conservation of biodiversity and ecological integrity is a fundamental principle of ecologically sustainable development. The Blue Mountains contains rich biodiversity values and natural resources due to its unique setting within the Greater Blue Mountains World Heritage Area, and the network of bushland connecting the towns and villages. The natural areas are comprised of unique landscape and geological features, environmentally sensitive areas including escarpments, caves, rock outcrops, watercourses, wetlands, swamps, groundwater seepage zones, heath and scrub, woodland and forest vegetation. Flora and fauna species, vegetation communities and ecosystem composition vary according to aspect, slope, underlying geology, soil type and depth, groundwater and surface water conditions.

There are many threats to biodiversity including vegetation and habitat loss and fragmentation, nutrient enrichment, water pollution, altered microclimate and drainage patterns, soil disturbance, introduced and invasive species, altered fire regimes, climate change and the unsustainable use of natural resources. This part details requirements for the protection of biodiversity, environmentally sensitive or significant natural features and natural resources occurring on or adjacent to sites proposed for development and provides controls for the protection of: threatened species and populations, threatened ecological communities or their habitat; significant native vegetation communities and rare or threatened Australian plants; fauna corridors; ecological buffer areas; watercourses, wetlands and riparian land; escarpment areas and sandstone geological features; and the National Park and the World Heritage Areas.

In assessing a development proposal, Council will consider all likely environmental impacts, including those which may be caused by future vegetation clearing to the full entitlement permissible under the NSW Rural Fire Service 10/50 Vegetation Clearing Code of Practice (where this entitlement is applicable).

The term *site* is used within this Part to describe a property and/or area affected by the proposed development.



Part C1 - Figure 1: Aerial of Katoomba - 'City within a World Heritage Area'

Read in conjunction with:



Submission requirements:

One or more of the following types of reports may need to accompany a development application:

- *Site Analysis Plan*
- *Flora and Fauna Assessment*
- *Vegetation Survey and Species Inventory*
- *Vegetation Management Plan*

C1.1. Biodiversity & Natural Resources

Objectives

O1. To protect and enhance biodiversity, environmentally sensitive or significant features and natural resources where they occur on or adjacent to, sites proposed for development.

O2. To apply the principles of ecologically sustainable development including: conserving biological diversity and ecological integrity, exercising the precautionary principle and ensuring inter-generational and intra-generational equity.

O3. To balance the conservation of the natural environment with the protection of life and property from bush fire.

Controls

C1. Protect and enhance biodiversity, environmentally sensitive or significant natural features and natural resources by:

(a) implementing sustainable development design, layout and practices, and

(b) maintaining native flora and fauna and their habitat, native vegetation diversity, structure, function and linkages, and the ecological processes and services provided by an ecosystem, and

(c) maintaining natural geological, landscape and escarpment features and habitat areas, and

(d) identifying all likely environmental impacts that may result from the proposed development, and

(e) avoiding, minimising, mitigating and offsetting environmental impacts in an acceptable way using effective measures, ensuring all practical opportunities to avoid environmental impact are utilised.

Note: The consideration of all likely environmental impacts that may result from the proposed development includes 10/50 vegetation clearing entitlement areas (NSW RFS Vegetation Clearing Code of Practice).

C2. Provide a site responsive development and protect the environmental values of the site by ensuring the location and design of the development:

(a) effectively integrates with the natural topography and reflects and compliments natural site features through the location of structures, outlook, design and materials, and

(b) is located as far as practicable within existing cleared areas and avoids unnecessary clearing of native trees and other native vegetation, and

(c) minimises site disturbance and cut and fill, and

(d) is located outside of environmentally sensitive land as far as practicable, and

(e) provides adequately sized buffer areas to protect environmentally sensitive or significant natural features, and

(f) maximises the retention of native vegetation linkages across the landscape, and

(g) incorporates best practice water management techniques to protect the surface and groundwater regimes and water quality of the site.

C3. Development is to balance the conservation of the natural environment with the protection of life and property from bushfire by:

(a) locating and designing developments and asset protection zones that minimise impacts on native flora and fauna and where possible, are located outside of environmentally sensitive areas, and

(b) considering measures to avoid and minimise environmental impact when implementing all appropriate bushfire protection measures, and

(c) utilising environmentally sensitive methods for fuel reduction in bushland areas.

Note: With reference to C3(b), these measures may include radiant heat shields; sprinklers; increased building construction Bushfire Attack Levels (BAL) or alternative development location

C4. Undertake all reasonable environmental protection works including conservation, regeneration, restoration, management and maintenance to protect and enhance the condition, structure and function of retained bushland and natural areas within the site, including specific environmental protection works necessary for the protection of:

(a) threatened species, threatened populations, and threatened ecological communities and their habitat, and

(b) significant vegetation communities or rare flora species, and

(c) fauna corridors, and

(d) ecological buffer areas, and

(e) watercourses, wetlands and riparian land, and

(f) escarpment areas and geological features, and

(g) National Park and World Heritage Areas.

C5. To protect biodiversity, best management practices are to be implemented in accordance with:

(a) Part C2 - Bushland and weed management - environmental protection works and environmental and priority weed control, and

(b) Part C3 - Landscaping - environmentally sensitive landscaping, and

(c) Part C4 - Bushfire - design and vegetation management in asset protection zones, and

(d) Part C6 - Water management - water sensitive urban design, and

(e) Part E4 - Site management - erosion and sedimentation controls.

C6. Prior to granting consent to development that has the potential to impact upon biodiversity and environmentally sensitive features or areas, a *Flora and Fauna Assessent* prepared in accordance with Council's guideline may be required. In some circumstances, depending on the scale and impact of a proposed development, the condition of the site and history of environmental assessment, a *Vegetation Survey and Species Inventory* may be required to be prepared instead of a *Flora and Fauna Assessment*.

Note 1: A *Flora and Fauna Assessment* is often required in (though not limited to) the following circumstances where:

- areas of intact bushland will be removed or modified by the development, or
- LEP 2015 environmental protection zones or mapped protected areas are likely to be impacted, or
- threatened species, threatened ecological communities, significant vegetation communities or their habitat are likely to be directly or indirectly impacted, or
- surface or groundwater changes may result, particularly if on-site effluent disposal or groundwater interception is proposed within the catchment of a groundwater dependent ecosystem, or

- proposed works may directly or indirectly impact upon a habitat features such as watercourses or waterbodies, trees with hollows, caves, crevices or other rock habitat features.

Note 2: Where Council determines that a significant effect on threatened species, threatened populations or threatened ecological communities and their habitats are likely a Species Impact Statement is to be prepared in accordance with State government guidelines.

C7. A *Vegetation Management Plan* (VMP) prepared in accordance with Council's guideline may be required to guide the development of the site in circumstances where:

(a) environmental protection works including but not limited to conservation, regeneration, restoration, management and maintenance are necessary, or

(b) the construction or future operation of the development will require removal, or ongoing management of, native vegetation and/or habitat

C1.2. Threatened Species, Threatened Populations and Threatened Ecological Communities

Explanation

The Blue Mountains is home to many threatened flora and fauna species, populations and communities listed for protection under the following legislation:

- *NSW Biodiversity Conservation (BC) Act, 2016* and records identified on the BioNet Atlas,
- *NSW Fisheries Management Act, 1994*
- *Commonwealth Environment Protection and Biodiversity Conservation (EPBC) Act, 1999.*

Threatened flora found within the Blue Mountains LGA includes *Leionema lachnoides*, *Pultenaea glabra* (Smooth Bush Pea), *Acacia bynoeana* (Bynoe's Wattle), *Persoonia acerosa* (Needle Geebung), *Pherosphaera fitzgeraldii* (Dwarf Mountain Pine) and *Leucopogon fletcheri* subsp. *Fletcheri*.



Part C1 - Figure 2: Pherosphaera fitzgeraldii

Threatened fauna species found within the Blue Mountains LGA include *Petalura gigantea* (Giant Dragonfly), *Eulamprus leuraensis* (Blue Mountains Water Skink), *Hopocephalus bungaroides* (Broad-headed Snake), *Calyptorhynchus lathami* (Glossy Black-Cockatoo), *Cercartetus nanus* (Eastern Pygmy-Possum) and *Dasyurus maculatus* (Spotted-tailed Quoll).



Part C1 - Figure 3: Giant Dragonfly

A threatened population of *Pultenaea villifera* var. *villifera* located in the Springwood, Winmalee and Warrimoo areas is listed in the NSW BC Act 2016.

Critically endangered, endangered and vulnerable ecological communities (CEEC, EEC or VEC) listed in Schedule 2 of the NSW BC Act 2016 and located within the Blue Mountains LGA include:

- Blue Mountains Shale Cap Forest in the Sydney Basin Bioregion (EEC),
- Blue Mountains Swamps in the Sydney Basin Bioregion (VEC),
- Montane Peatlands and Swamps of the New England Tableland, NSW North Coast, Sydney Basin, South East Corner, South Eastern Highlands and Australian Alps Bioregions (EEC),
- Newnes Plateau Shrub Swamp in the Sydney Basin Bioregion (EEC),
- Shale/Sandstone Transition Forest (CEEC),
- Sun Valley Cabbage Gum Forest in the Sydney Basin Bioregion (CEEC),
- Sydney Turpentine-Ironbark Forest (CEEC),
- Blue Mountains Basalt Forest (EEC),
- River-flat Eucalypt Forest on Coastal Flood Plains (EEC).

Threatened ecological communities (TECs) listed under the Commonwealth Environment Protection and Biodiversity Conservation Act, 1999 include:

- Shale/Sandstone Transition Forest (CEEC),
- Temperate Highland Peat Swamps on Sandstone (EEC)
- Turpentine-Ironbark Forest, in the Sydney Basin Bioregion (CEEC)
- Upland Basalt Eucalypt Forest of the Sydney Basin Bioregion (EEC)

The above listed threatened communities are also recognised by the Local Environment Plan as being significant vegetation communities.

Many key threatening processes, including the removal of vegetation and habitat loss or change, predation by pest animals, weed invasion and infection by diseases, have been identified in the Act 2016 and the EPBC Act, 1999 as leading to the decline or threat of extinction of threatened species, populations or ecological communities.

Clause 1.7 of the *Environmental Planning and Assessment Act 1979* gives effect to Parts 7 of the Biodiversity Conservation Act 2016 and Part 7A of the Fisheries Management Act 1994. These parts require an assessment to determine whether a development proposal is likely to result in a significant adverse effect on threatened species, populations or ecological communities or their habitats. Council must consider these parts in the determination of all development applications. A Flora and Fauna Assessment may be required to be prepared in accordance with Council's guideline.

Note: Commonwealth assessment and approval under the EPBC Act, 1999 is

independent of the Council development application process. It is the responsibility of the applicant to meet obligations under the EPBC Act. However, it is recommended that potential impacts under the EPBC Act are considered concurrently by a Flora and Fauna Assessment, where this is being undertaken for the purpose of a development application.

Objectives

O1. To protect, enhance and restore threatened species, threatened populations and threatened ecological communities (TECs) and their habitat.

Controls

C1. Identify the location, extent and nature of all threatened species, threatened populations and TECs that occur within the site, relative to the development proposal, on a Site Analysis Plan. Refer to B1 Site Analysis and Part I – Submission Requirements.

C2. The location, design and implementation of the development and any asset protection zone is to demonstrate achievement of the following:

(a) in satisfying clause 1.7 of the *Environmental Planning and Assessment Act 1979*, there is to be no significant adverse impact upon any threatened species, threatened populations or threatened ecological communities or their habitats, and

(b) consistency with the management actions and measures specified in all applicable recovery plans or priority action statements for relevant listed threatened species, threatened populations or TECs, and

(c) be sited outside of land containing the threatened species, threatened populations and threatened ecological communities and their habitat, and the ecological buffer areas required to protect them, unless no practical alternative is available, and

(d) maintain, or enhance where necessary, the natural environmental, hydrological and micro-climatic conditions in which that threatened species, threatened population or threatened ecological community exists, and

(e) protect, or enhance where necessary, the natural structure, species diversity and ecological processes of the native vegetation community, within which the threatened species or threatened population exists, or that are representative of the threatened ecological community.

C3. Prior to granting consent to development that has the potential to impact upon threatened species, threatened populations or threatened ecological communities, or their habitats, a Flora and Fauna Assessment prepared in accordance with Council's guideline may be required.

C4. A Biodiversity Development Assessment Report (BDAR) is required to be submitted with the application when the Biodiversity Offset Scheme (BOS) applies. The BOS applies when a development exceeds the BOS thresholds or is likely to have a significant impact on threatened species or threatened ecological communities according to the test of significance in section 7.3 of the Biodiversity Conservation Act 2016.

Note: Seeking preliminary advice from Council prior to the preparation of a BDAR is strongly recommended.

C5. A *Vegetation Management Plan* (VMP) prepared in accordance with Council's guideline may be required to guide the development of the site. The VMP is to demonstrate the appropriate protection, rehabilitation, restoration and management of threatened species, populations, and threatened ecological communities, and their habitat.

C1.3. Significant vegetation and rare flora species

Explanation

Native vegetation mapping in the Blue Mountains local government area has identified 31 significant vegetation communities (identified in LEP 2015 Schedule 6), including Blue Mountains Heath and Scrub, Blue Mountains Escarpment Complex, Megalong Granite Forest and Blue Gum River Flat Forest.

A number of these significant vegetation communities are also currently listed as critically endangered, endangered or vulnerable ecological communities in the *NSW Biodiversity Conservation Act 2016* and the *Commonwealth Environment Protection and Biodiversity Conservation Act 1999*.

Rare flora species as referred to in the LEP 2015 and this DCP means any flora species listed in the publication entitled "Rare or Threatened Australian Plants", Briggs & Leigh, 1995 Revised Edition, also referred to as ROTAP species.

Rare flora such as Eucalyptus burgessiana (Faulconbridge Mallee Ash), Lomandra brevis (Tufted Mat-rush) and Lissanthe sapida (Native Cranberry) and Almaleea incurvata are supported by unique habitats within the Blue Mountains LGA.

Significant vegetation communities and rare species are to be protected in accordance with the provisions of the LEP 2015 clauses 6.1 (Impact on environmentally sensitive land), 6.6 (Protected area-vegetation constraint area) and 6.7 (Protected area-ecological buffer area), and Schedule 6.



Part C1 - Figure 4: Blue Mountains Swamp Community



Part C1 - Figure 5: Eucalyptus deanei in Blue Mountains Shale Cap Forest

Objectives

O1. To protect, enhance and restore significant vegetation communities and rare flora species, and the ecological buffers required to protect them.

Controls

C1. Identify the location, extent and nature of all significant vegetation communities and rare species of flora that occur within the site, relative to the development proposal, on a *Site Analysis Plan*. Refer to B1 Site Analysis and Part I – Submission Requirements.

C2. The location, design and implementation of the development and any asset protection zone is to demonstrate achievement of the following:

(a) In accordance with the provisions of clause 6.1 (Impact on environmentally sensitive land) of LEP 2015, there is to be no adverse impact on any significant vegetation communities and rare flora species, and

(b) Be sited outside of land containing the significant vegetation communities and rare flora species, and the ecological buffer areas required to protect them, unless no practical alternative is available, and

(c) Maintain, or enhance where necessary, the natural environmental, hydrological and micro-climatic conditions in which the significant vegetation communities and rare flora species exists, and

(d) Protect, or enhance where necessary, the natural structure, species diversity and ecological processes of the native vegetation community within which rare flora species exists, or that are representative of the significant vegetation communities.

C3. Prior to granting consent to development that has the potential to impact upon significant vegetation communities and rare flora species, or their habitats, a *Flora and Fauna Assessment* prepared in accordance with Council's guideline may be required.

C4. A *Vegetation Management Plan* (VMP) prepared in accordance with Council's guideline may be required to guide the development of the site. The VMP is to demonstrate the appropriate protection, rehabilitation, restoration and management of threatened species, populations, and threatened ecological communities, and their habitat.

C1.4. Fauna corridors

Explanation

Within cleared or developed areas, remnant native bushland in the form of a corridor that provides a link between surrounding reserves is important to facilitate fauna movements and maintain connections between fauna populations.

Within the Blue Mountains, such linkages often occur along watercourse corridors and can also occur where bushland extends between townships and across ridgelines between adjoining catchments.

Smaller scale pockets or islands of native vegetation within urban areas are also important to maintain landscape quality, bushland character and provide fauna refuge areas.

Fauna corridors are to be protected in accordance with the provisions of LEP 2015, clause 6.3 (Terrestrial biodiversity) and where native vegetation or landscape features are otherwise identified as providing connectivity between vegetation remnants, habitats and fauna populations.

Objectives

O1. To protect, enhance and restore, and prevent the fragmentation of important landscape corridors and native vegetation linkages that provide for fauna movement and connectivity between habitats and fauna populations.

Controls

C1. Identify the location, extent and nature of all existing and potential fauna corridors that occur within the site, relative to the development proposal, on a *Site Analysis Plan*. Refer to B1 Site Analysis and Part I – Submission Requirements.

C2. If fauna corridors are not identified in accordance with clause 6.3 (Terrestrial biodiversity) of LEP 2015, but are otherwise confirmed to be present during the preparation or assessment of the development application, the site layout and design of development is to maximise the protection of those fauna corridors, unless no practical alternative is available.

C3. The location, design and implementation of the development and any asset protection zone is to demonstrate achievement of the following:

(a) That there is no adverse impact on any fauna corridor, unless no practicable alternative is available, and

(b) Be sited outside of land identified as an existing or potential fauna corridor unless no practical alternative is available, and

(c) Maintain, or enhance where necessary, the natural environmental, hydrological and micro-climatic conditions in which the native vegetation community comprising a fauna corridor exists, and

(d) Protect, or enhance where necessary, the natural structure, species diversity and ecological processes of the native vegetation community that comprises the fauna corridor, and

(e) Conserve an appropriate corridor width with regard to core habitat areas, potential edge effects and fragmentation, and

(f) Conserve natural habitat requirements and landscape features that are necessary for the fauna corridor to function effectively.

C4. Prior to granting consent to development that has the potential to impact upon a fauna corridor, a *Flora and Fauna Assessment* prepared in accordance with Council's guideline may be required. The assessment is to provide recommendations for an appropriate fauna corridor width to be protected or restored.

C5. A *Vegetation Management Plan* (VMP) prepared in accordance with Council's guideline may be required to guide the development of the site. The VMP is to demonstrate the appropriate protection, rehabilitation, restoration and management of fauna corridors and native vegetation connectivity.

C1.5. Ecological buffer areas

Explanation

Ecological buffer areas protect native vegetation from the adverse edge effects caused by adjacent development.

Ecological buffer areas are the optimal and necessary mitigating measure for development proposals on sites where significant vegetation such as LEP 2015 Schedule 6 Significant vegetation communities, or other flora determined to have regional, State or Commonwealth significance, is required to be protected.

An ecological buffer area comprising native vegetation helps to mitigate edge effects and limit long-term degradation of significant vegetation by regulating the microclimate, maintaining site hydrology, and buffering the impacts of nutrient enrichment, weed invasion, erosion, sedimentation, pollution and direct impacts from urban settlement.

An effective buffer area is sized and designed to maintain the significant vegetation to be protected in a natural self-sustaining condition.

An adequate buffer width should be determined with consideration of the existing or potential value of the vegetation being protected; site and buffer characteristics; the intensity of adjacent land use and the desired buffer performance outcomes.

Ecological buffer areas are to be protected and restored in accordance with the provisions of the LEP 2015 clauses 6.1 (Impact on environmentally sensitive land), clause 6.6 (Protected area–vegetation constraint area) and clause 6.7 (Protected area–ecological buffer area).

Appropriate vegetated buffer areas are also used as a mitigating measure to protect a range of other environmentally sensitive or significant natural features in the relevant sections of Part C1.

Objectives

O1. To protect, enhance and restore the ecological buffer areas necessary to protect and maintain in their natural condition:

(a) significant vegetation communities listed in Schedule 6 of LEP 2015, and

(b) rare species of flora, and

(c) flora of State or Commonwealth significance

Controls

C1. Identify the location, extent and nature of all ecological buffer areas that are required within the site, relative to the development proposal, on a Site Analysis Plan. Refer to B1 Site Analysis and Part I – Submission Requirements.

C2. If unmapped Significant vegetation communities are confirmed to be present during the preparation or assessment of the development application, then the development is to be sited to provide an adequately sized ecological buffer (presumed to be 60 metres until determined by environmental assessment) to protect that significant vegetation.

C3. For all ecological buffer areas, whether mapped or unmapped, the location, design and implementation of the development and any asset protection zone is to demonstrate achievement of the following:

(a) Be sited outside of land identified as an ecological buffer area unless no practical alternative is available, and

(b) Maintain, or enhance where necessary, the natural environmental, hydrological and micro-climatic conditions in which the native vegetation community comprising the ecological buffer area exists, and

(c) Protect, or enhance where necessary, the natural structure, species diversity and ecological processes of the native vegetation community that comprises the ecological buffer area, and

(d) Conserve an adequately sized buffer area with due regard for:

i. land use zoning and the relevant objectives of the LEP 2015 and this DCP, and

ii. achieving the effective protection of the significant vegetation being protected by the buffer, and

iii. the structure, composition and sensitivity of the significant vegetation being protected by the buffer, and

iv. effectively mitigating and absorbing the edge effects that would otherwise extend into and degrade the significant vegetation being protected by the buffer, and

v. adjacent land use intensity, and

vi. buffer area composition and site characteristics.

C4. Prior to granting consent to development that has the potential to impact upon an ecological buffer area, a *Flora and Fauna Assessment* prepared in accordance with Council's guideline may be required. The assessment is to identify the location, extent and nature of significant vegetation required to be protected, and provide recommendations for the ecological buffer areas necessary to protect that significant vegetation.

C5. A *Vegetation Management Plan* (VMP) prepared in accordance with Council's guideline may be required to guide the development of the site. The VMP is to demonstrate the appropriate protection, rehabilitation, restoration and management of ecological buffer areas.

C1.6. Riparian land and watercourses

Explanation

Watercourses and adjacent riparian land together form a riparian corridor that provides important ecosystem services and protects catchment health. Riparian land contains the transition zone between dry terrestrial and aquatic environments.

A healthy riparian corridor contributes significantly to the biodiversity value and stream health of a watercourse. They help sustain the food chain for wildlife and provide vegetation connectivity between wildlife habitats.

The protection and restoration of native vegetation on riparian land (within a riparian zone) is important for providing bed and bank stability, protecting water quality and providing food, shade, shelter and breeding sites for terrestrial, riparian and aquatic wildlife.

Vegetated buffer areas are needed adjacent to riparian zones to protect surface and groundwater regimes, minimise the entry of pollutants and mitigate adverse edge effects of development on riparian zone habitat areas.

Development applications for works in and near watercourses may also require State Government Approval under the *Water Management Act 2000* or *Fisheries Management Act 1994* and are to be lodged with Council as Integrated Development. These developments are to comply with State Government requirements and guidelines for in-stream works, riparian corridor protection and fish passage.

Riparian land, wetlands and watercourses are to be protected in accordance with the provisions of the LEP 2015 clause 6.1 (Impact on environmentally sensitive land) and clause 6.8 (Protected area – riparian lands and watercourses).



Part C1 - Figure 6: Riparian land and watercourse - Leura Falls Creek

Objectives

O1. To protect, enhance and restore watercourses, wetlands and riparian lands and their natural hydrologic, hydraulic, geomorphic and ecological functions.

O2. To protect and enhance the condition and function of riparian and aquatic ecosystems and the life forms they support.

O3. To protect and enhance surface and ground water quality and catchment health.

O4. To provide vegetated buffers between development, and watercourses and their riparian zones.

O5. To rehabilitate degraded watercourses, riparian corridors and riparian vegetation.

Controls

C1. Identify the location, extent and nature all watercourses, wetlands and riparian land that occur within the site, relative to the development proposal, on a *Site Analysis Plan*. Refer to B1 Site Analysis and Part I – Submission Requirements.

C1. If development is located adjacent to a watercourse, then that development is to comply with LEP 2015 clause 6.8 (Protected area – riparian lands and watercourses).

C2. The location, design and implementation of the development and any asset protection zone are to be sited outside of watercourses, wetlands and riparian land unless no practical alternative is available.

C3. Maintain and restore where necessary the health and integrity of watercourses and riparian land by:

(a) conserving and enhancing riparian corridors, and

(b) protecting the natural landforms and functions of water courses and riparian land, and

(c) avoiding modifications to natural watercourses, and

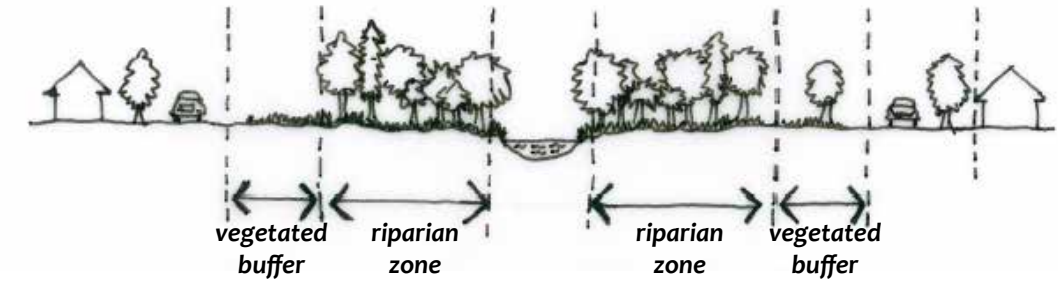
(d) facilitating the restoration of degraded watercourses, and

(e) applying best-practice design to in-stream works.

C4. Riparian corridors are to be conserved and enhanced by demonstrating the following:

(a) A native vegetation riparian zone is to be protected, and enhanced where necessary, on both sides of a watercourse as shown in *Part C1 - Figure 7.*

(b) To mitigate the adverse edge effects of adjacent development, particularly for intensive land uses, a vegetated buffer area is to be provided where ever possible between built or hard development surfaces and riparian zones as shown in *Part C1 - Figure 7.*



Part C1 - Figure 7: A native vegetation riparian zone is to be provided on both sides of a watercourse

(c) The riparian zone for a watercourse is to be in the range of 10 to 60m, provided on each side of the watercourse and measured from the top of each bank.

(d) The location, extent and nature of riparian zones and vegetated buffer areas required to protect the watercourse, are to be identified, relative to the development proposal, on a Site Analysis Plan.

(e) Council will assess the riparian corridor requirements of each development on its merits. In determining the actual width of the riparian zones and any vegetated buffer areas, the following factors are to be considered:

i. Land use zoning and the relevant objectives of the LEP 2015 and this DCP.

ii. Achievement of multiple outcomes including protection of biodiversity, riparian and aquatic systems, ecological processes, water quality and soil and stream stability.

iii. Maintaining or restoring riparian corridor connectivity up and downstream of the site.

iv. Protection of any significant vegetation communities or other flora determined to have regional, State or Commonwealth significance.

v. Stream order of the watercourse as defined by the Strahler system.

vi. Adjacent land use intensity.

vii. Buffer area composition and site characteristics.

(f) Native vegetation within the riparian zone is to be protected or regenerated/revegetated where necessary, to achieve, fully structured form (trees, shrubs and groundcover species) or other structure representative of the locally occurring riparian community.

(g) Riparian corridors are to be linked, with other areas of native vegetation to enhance fauna corridors and connectivity between significant natural features wherever possible.

C5. Natural landforms and functions of watercourses and riparian land are to be protected by ensuring any development:

(a) maintains the natural hydrologic regime and hydraulic conditions, and

(b) preserves the natural alignment of watercourses, and

(c) conserves natural habitat and landscape features that are necessary for the watercourses and riparian zones to function effectively, and

(d) maintains the natural geomorphic and ecological forms and functions of watercourses and riparian land, and

(e) maintains bed and bank stability, prevents erosion and sedimentation and preserves the diversity of in-stream aquatic habitats, and

(f) maintains environmental flow characteristics of the downstream watercourse as far as practicable, and

(g) protects the natural structure, species diversity and ecological processes of the native vegetation communities that occur on riparian land, and

(h) protects water quality and in-stream habitat with effective erosion and sedimentation controls and environmental protection works necessary to prevent erosion and pollution and maintain stream stability.

C6. Avoid modifications to natural watercourses by ensuring:

(a) watercourses are not piped, filled, excavated, re-located or straightened to reduce the natural meander or flow path, and

(b) overland flow paths are identified and accommodated in site planning and development design.

C7. Facilitate the restoration of degraded watercourses by ensuring consideration is given to the following:

(a) Where a development will have a direct impact on a watercourse that has been piped or modified with a constructed channel or gabion wall, this should be removed to restore a natural channel form except where: it is technically unfeasible to reverse the piping or channelization, or restoration of a natural channel form will create a hazard, and

(b) The design and location of any development is not to preclude the future retrofit of piped or otherwise modified watercourses into an open natural channel form.

C8. In-stream works are to achieve best practice design by:

(a) emulating the natural contours of the bed and banks of the watercourse, not obstructing or adversely affecting the flow of water and maintaining the ecological values and functions of the riparian corridor, and

(b) minimising the number of watercourse crossings, and

(c) providing stormwater outlets that are located to avoid impacts to existing native vegetation, directed downstream, graded at bed level or just below any permanent water level and providing adequate scour protection, and

(d) comprising a design and materials compatible with the natural environment e.g. an environmentally sensitive soft-engineered design, such as a rock-rip rap. Gabions or concrete are to be avoided, and

(e) where appropriate, bank or swamp stabilisation works are to preferably use biodegradable techniques such as brush or coir logs, and

(f) incorporating the re-establishment of native vegetation within disturbed areas utilising locally native sedges, rushes, groundcovers, shrubs and trees as appropriate for the location within the watercourse, riparian zone or floodplain. Revegetation is to emulate the naturally occurring community, species diversity and density.

C9. Prior to granting consent to development that has the potential to impact upon watercourses, wetlands and riparian land, a *Flora and Fauna Assessment* prepared in accordance with Council's guideline may be required. The assessment is to provide recommendations for the size and treatment of riparian zones and any vegetated buffer areas, necessary to protect watercourses and riparian land.

C10. A *Vegetation Management Plan* (VMP) prepared in accordance with Council's guideline may be required to guide the development of the site. The VMP is to demonstrate the appropriate protection, rehabilitation, restoration and management of fauna corridors and native vegetation connectivity.

C1.7. Escarpment areas and geological features

Explanation

The escarpment areas of the Blue Mountains comprise a range of unique landforms and features including moist sheltered rock faces, gorges, crevices, ledges, caves, bush rock, steep vegetated slopes and extensive outcroppings of sandstone, escarpments, geological formations and talus slopes.

These diverse escarpment landforms and features are associated with a complex variety of distinctive vegetation communities that can vary on a very small scale, ranging from moist cliff line vegetation, heath, swamp, rainforest, mallee and forest. Collectively, this vegetation is recognised as a significant vegetation community, Blue Mountains Escarpment Complex in LEP 2015 Schedule 6.

Escarpment areas support a great diversity of plant species, forms and habitat niches and provides unique habitat for many rare or threatened species of flora and fauna.

Escarpment areas and geological features are to be protected in accordance with the provisions of the LEP 2015 clause 6.1 (Impact on environmentally sensitive land), and clause 6.12 (Protected area - escarpment).



Part C1 - Figure 8 Blue Mountains escarpment

Objectives

O1. To protect the rock faces, crevices, ledges, caves, talus slopes, bush rock, rock outcrops, escarpments, geological formations, steep vegetated slopes and other significant landscape features that comprise the escarpment system of the Blue Mountains.

O2. To protect the Blue Mountains natural and cultural heritage and its distinctive character and amenity.

O3. To protect unique and rare flora and fauna habitat values and distinctive vegetation communities that the escarpment areas / system supports.

Controls

C1. Identify the location, extent and nature of all geological and landscape features comprising escarpment areas within the site, relative to the development proposal, on a Site Analysis Plan. Refer to B1 Site Analysis and Part I – Submission Requirements.

C2. Moist sheltered rock faces, crevices, ledges, caves, talus slopes, bush rock, rock outcrops, escarpments, geological formations, steep vegetated slopes and other significant landscape features that comprise the escarpment system of the Blue Mountains are to be protected and maintained as part of any development.

C3. The location, design and implementation of the development and any asset protection zone is to demonstrate achievement of the following:

(a) The integrity of geological and landscape features comprising the escarpment area are maintained, and

(b) The protection of vegetation features comprising the escarpment area is maximised, and

Note: The escarpment area provides natural habitat requirements for many threatened flora and fauna species, threatened ecological communities and significant vegetation communities and rare flora. The development is to comply with the provisions of Part C1.2 (Threatened Species, Threatened Populations and Threatened Ecological Communities) and Part C1.3 (Significant vegetation and rare flora species).

(c) Be sited outside of geological, landscape and vegetation features comprising the escarpment area unless no practicable alternative is available, and

(d) Canopy conservation within the escarpment area is maximised. Demonstrate the development and asset protection zones will not have an unacceptable impact with a view analysis from points where the site is visible.

C1.8. National Park and World Heritage Areas

Explanation

Blue Mountains National Park is part of the Greater Blue Mountains World Heritage Area; listed for its remarkable geographic, cultural and botanic values.

Development can result in the degradation of the National Park due to urban edge effects including nutrient enrichment, weed invasion and stormwater pollution and erosion.

Natural vegetated buffer areas are required to protect the natural, conservation, scientific, scenic, aesthetic, historical, and cultural and heritage values of the Blue Mountains National Park where adjoining development is proposed.

National Parks are included within LEP 2015 as Zone **C1 National Parks and Nature Reserves.**

National Parks are to be protected in accordance with the National Parks and Wildlife Act, 1974. The potential impacts of adjoining development are required to be managed in accordance with the provisions of the LEP 2015 clause 6.1(3) (Development near Blue Mountains National Park).



Part C1 - Figure 9: Blue Mountains National Park

Objectives

O1. To maintain the natural, conservation, scientific, scenic, aesthetic, historical, and cultural and heritage values of the Blue Mountains National Park (which forms part of the Greater Blue Mountains World Heritage Area).

O2. To recognise the importance of the Blue Mountains National Park as the setting of the City.

O3. To provide adequately sized native vegetation buffer areas to protect the natural environment of the Blue Mountains National Park.

Controls

C1. Development and any bush fire asset protection zone located on land adjacent to the Blue Mountains National Park is to protect the natural landscape and the aesthetic, historical, heritage, natural, scenic and scientific attributes of the land within the Blue Mountains National Park.

C2. Identify and incorporate all necessary measures to effectively avoid, remedy or mitigate any adverse environmental impact on land within the Blue Mountains National Park. Ensure all opportunities to avoid environmental impact are utilised.

C3. The location, design and implementation of the development and any asset protection zone is to provide an adequately sized native vegetation buffer area, to mitigate adverse environmental impacts of the development on the adjoining Blue Mountains National Park land.

C4. Identify the location, extent and nature of any native vegetation buffer areas required within the site to protect the Blue Mountains National Park, relative to the development proposal, on a *Site Analysis Plan*. Refer to B1 Site Analysis and Part I – Submission Requirements.

C1.9. Light Pollution

Explanation

Light pollution is the fastest growing pollutant around the globe, with scientific research showing an average increase year-on-year of more than 2%. This excess light or wasted energy results in a variety of negative environmental impacts including increasing greenhouse gases and carbon footprints, deleterious impacts on fauna and human health, and loss of cultural heritage connections associated with the night skies. Natural darkness has a conservation value in the same way that clean water, air and soil has intrinsic value. Animals perceive light differently from humans and artificial light can disrupt critical behaviour and cause physiological changes in wildlife.

Outdoor lighting should be designed to minimise light spill into neighbouring dwellings, adjacent bushland or into the night sky. Incorporation of good lighting design principles will improve neighbour amenity, reduce energy consumption, reduce impacts on biodiversity and maintain the quality of the night sky.

Objectives

O1. To ensure that outdoor lighting is designed in accordance with good lighting design principles to minimise the impacts of light spill/pollution on amenity, biodiversity and cultural values, the World Heritage listed National Park and the night sky.

Controls

This part applies to development on land that is adjacent to, directly opposite or separated only by a local road from land in the Blue Mountains National Park or zoned C2 Environmental Conservation.

C1. Outdoor lighting must be designed with consideration of the lighting design principles outlined in the National Light Pollution Guidelines for 2023.

C2. The use of outdoor lighting is to be minimised and limited to essential functional uses (e.g. pathways, security).

C3. Use of motion sensor lighting is encouraged to reduce overall light use.

C4. Outdoor lighting is to conform with Australian Standard 4282/2023 Control of Obtrusive Effects of Outdoor lighting.

PART C2

BUSHLAND AND WEED MANAGEMENT



Introduction

Among the principal aims of the LEP 2015 is the conservation and enhancement, for current and future generations, of the ecological integrity, environmental heritage and environmental significance of the Blue Mountains. Sensitive management of vegetation and natural features on private property will significantly increase the potential to achieve this goal, and will enable the ongoing persistence and evolution of the rich biodiversity contained within surrounding natural areas within Council Reserves, Crown land or the Greater Blue Mountains World Heritage areas.

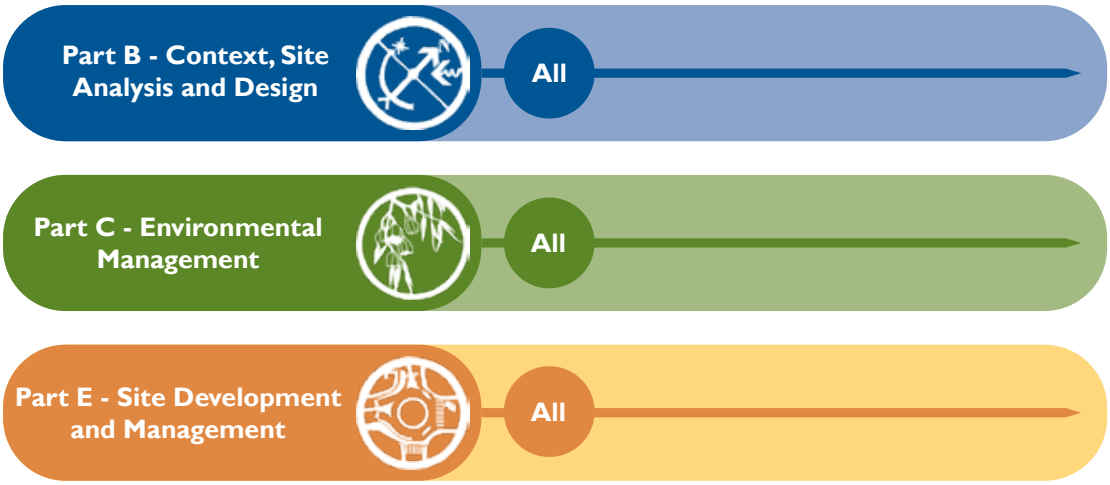
Damaging impacts upon natural areas may be cumulative, and each adverse influence may act together, compounding the effects and leading to degradation of ecological systems. Conversely, private landholders may create or enhance ecological diversity in the landscape at a localised scale, and may contribute to restoring and maintaining the biological values of the local area. Individual actions can improve ecological processes such as carbon capture and storage, groundwater recharge and nutrient cycling.

Council aims to provide residents with appropriate and adequate guidelines, to ensure that their development proposal does not have an unacceptable impact on the natural environment. This accords with the Key Directions of the Sustainable Blue Mountains 2025:

- *We value our surrounding bushland and the World Heritage National Park.*
- *Recognising that the Blue Mountains natural environment is dynamic and changing, we look after and enjoy the healthy creeks and waterways, diverse flora and fauna and clean air.*
- *Living in harmony with the environment, we care for the ecosystems and habitats that support life in the bush and in our backyards.*
- *We aim to conserve energy and the natural resources we use and reduce environmental impacts by living sustainably.*

This part provides direction and guidelines to enable applicants to manage bushland and control weeds through long-term preservation, restoration and management actions.

Read in conjunction with:



Submission Requirements:

One or more of the following types of reports may need to accompany a development application:

- *Landscape Plan*
- *Vegetation Management Plan*
- *Weed Management Strategy*

C2.1. Bushland management

Explanation

This Part applies to land that contains bushland and indigenous vegetation whether scheduled, or non-scheduled and in any condition.

Specific environmental protection works may be required as a component of development so as to achieve compliance with general or specific objectives within the LEP 2015. This may include measures to mitigate or offset likely impacts of the development or to repair existing damage to natural features on the development site.

Under Part 6 of LEP 2015, certain additional local provisions allow for the retention, restoration and management of areas of bushland or indigenous vegetation outside the area immediately required for the development. This may be required in order to minimise or mitigate the impact of development upon:

- Environmentally sensitive land, LEP 2015 clause 6.1 (Impact on environmentally sensitive land)
- Fauna Corridors, LEP 2015 clause 6.3 (Terrestrial biodiversity)
- Protected Areas; LEP 2015 clauses 6.4 (Protected area-slope constraint), 6.6 (Protected area-vegetation constraint area), LEP 2015 clause 6.7 (Protected area-ecological buffer area), 6.8 (Protected area-riparian land and watercourses), 6.12 (Protected area-escarpment), and 6.13 (Protected area-land between towns).

This part of the DCP provides guidelines to describe and direct the type of *environmental protection works* necessary under a number of different circumstances. This will ensure that these works are undertaken in a manner consistent with current best environmental practices, and in line with Council's requirements. This part applies to the following areas:

- parts of the development area which contain indigenous vegetation which is to be retained essentially intact, and
- areas of the development site which lie beyond those required for the development, and which are to be retained as predominantly natural areas, and
- parts of the development site which contain environmentally sensitive land or protected areas, and
- degraded land within Zones **C2 Environmental Conservation, C3 Environmental Management** and **C4 Environmental Living.**
- degraded land within Zone **SP3 Tourist** where adjoining land zoned **C2 Environmental Conservation.**

Qualifications and experience of workers may be required under certain circumstances.

Council may specify that, due to the sensitivity of the site, the presence of threatened species or communities, the complexity of the restoration tasks or the necessity for the works to be completed within a restricted time frame, suitably qualified personnel with specialist skills relevant to the situation be engaged to implement necessary Environmental Protection Works.

Note 1: As a guide, trained bushland regeneration practitioners should hold a minimum of the Bushland Regeneration Certificate II/Natural Area Restoration Certificate III, plus 500 hours of practical bushland regeneration under an experienced supervisor. Supervisors would be required to have at least some supervisory experience, preferably a higher qualification and a minimum of 700 hours bush regeneration.

Note 2: If bush regeneration practitioners are engaged to use pesticides (herbicides) for commercial and occupational purposes, they are to be trained to the appropriate level of competency in pesticide use in accordance with the provisions of the Pesticides Regulation 2017.

Qualifications and experience may not be required under other circumstances

It is often appropriate for private land owners to undertake the required environmental protection works themselves, particularly when an approved *Vegetation Management Plan* (VMP) establishes a strategy that runs over a number of years. Many professional restoration practitioners will provide training for landholders to facilitate this process. They may also provide a level of supervision of the works.

Note: Determination can be made on a case by case basis. If in doubt, ask the relevant Council staff to clarify this aspect.

Objectives

O1. To ensure that environmental protection works to conserve, maintain or restore the ecological, scientific, cultural, intrinsic and aesthetic values of native vegetation, waterways, habitat areas and natural features, are recognised and considered as a component of development.

O2. To ensure that the environmental values of the extensive urban/bushland interface in localities within the Blue Mountains are appropriately enhanced through responsive site management.

O3. To ensure that appropriate environmental protection works are identified during the development assessment process.

O4. To ensure that the primary objective for environmental protection works is to achieve restored bushland areas which are resilient, self-sustaining and suitably integrated into the larger ecological matrix, with which they may interact.

O5. To ensure that all environmental protection works necessary to protect biodiversity or reduce unacceptable impacts to the environment are implemented in a manner which achieves optimal restoration outcomes relevant to the site.

O6. To ensure that existing adverse impacts upon vegetation, including weed invasion and land degradation, are identified and reasonably managed through suitable environmental protection works as a component of development.

O7. To ensure that the conservation of the natural environment is balanced with the need to protect people and property from the risk of bushfire in the Blue Mountains.

Controls

C1. Environmental protection works necessary to achieve the conservation or restoration of identified natural features or ecological communities are to be identified and documented in an appropriately detailed Vegetation Management Plan (VMP), prepared in accordance with Council's guideline.

C2. The *ecological restoration* requirements for a development site are to be based on the site's existing condition and capacity to recover. The following hierarchy of methods, from least interventionist to most, is to form part of any proposed strategy:

(a) Conservation, which means all processes and actions of looking after a place so as to retain its natural significance and always includes protection, maintenance and monitoring.

(b) Bushland regeneration, which is the practice of restoring bushland by focusing on reinstating and reinforcing the systems' ongoing natural regeneration (recovery) processes for the maintenance of ecological integrity and the operation of dynamic processes.

(c) Assisted natural regeneration, which is a staged process which aims to stimulate or trigger the growth of native plant propagules (such as seed, tubers or rhizomes etc.) already present on site or having the ability to migrate onto the site. It may also involve low impact weed control and repair of damaged land surfaces.

(d) Reconstruction though revegetation, which involves the introduction of locally indigenous and preferably local provenance plant species. Planting patterns and species composition is modelled on the diversity and structural characteristics of a pre-existing vegetation community.

(e) Landscape fabrication refers to the creation, through land reshaping, soil amendment or replanting, of a vegetated area which is distinctly different in species diversity, composition, structure and appearance to the pre-existing bushland community on a site. This is only used on highly degraded sites where the restoration of a pre-existing vegetation community cannot be achieved due to the significant and permanent alteration of key abiotic habitat components.

C3. Within the areas subjected to ecological restoration works, the following actions are to be eliminated or restricted (unless they have been recommended within an approved VMP) as they may degrade existing bushland or inhibit or diminish the recovery potential of degraded areas of indigenous vegetation:

(a) Removal of indigenous species, or

(b) Mowing, slashing and trimming of vegetation, or

(c) Raking and / or removal of natural organic material, or

(d) Excavation or addition of soil material or compost, or

(e) Planting of non-endemic species, or

(f) Provision of deep organic mulch, or

(g) Vehicular movement or parking, or

(h) Storage of plant or equipment, or

(i) Stockpiling of materials or waste.

C4. Priority and environmental weed species are to be effectively and appropriately controlled in a timely manner, in all areas to which this part applies. A *Weed Management Strategy* may need to be submitted if weed infestation is substantial.

C5. All specified environmental protection works are to be consistent with industry recognised and recommended ecological restoration practices and techniques.

Note: These standards can be accessed through the Australian Association of Bush Regenerators (AABR).

C6. In providing an asset protection zone to satisfy Planning for Bushfire Protection 2019 requirements, the restoration and conservation of remnant vegetation is to be prioritised over landscape fabrication. Representative vegetation cover from all stratum (i.e. shrub, tree, groundcovers) is to be retained and restored to the maximum extent possible whilst still complying with the requirements of an asset protection zone.

C2.1.1. Environmentally Sensitive Land

C1. Where development is permitted in or adjacent to the habitat area of threatened species or threatened ecological communities, environmental protection works are to include management actions designed to limit the impact or operation of key threatening processes[1], including:

(a) the clearing of native vegetation, and

(b) removal of bushrock, and

(c) invasion and establishment of significant weed species, and

(d) the loss and degradation of native plant and animal habitat by invasion of escaped garden plants, including aquatic plants, and

(e) the removal of dead wood and dead trees, including those with hollows, and

(f) forest eucalypt dieback associated with over-abundant psyllids and Bell Miners.

Note: [1]A threatening process is something that threatens, or could potentially threaten, the survival or evolutionary development of a plant or animal species, population or ecological community. A threat can be listed under the *Biodiversity Conservation Act 2016* as a 'key threatening process' if it adversely affects threatened species, populations or ecological communities or if it could cause species, populations or ecological communities that are not threatened to become threatened. Current and updated listings may be accessed at the Office of Environment and Heritage website.

C2. Development of land containing *significant vegetation*, threatened flora species or threatened ecological communities is to include provisions for environmental protection works to conserve and enhance the operation of ecological functions and ecological connectivity which may include, but not be limited to the following:

(a) Retain, protect and manage any flora species which are considered to be characteristic of the threatened ecological community or significant vegetation community, and

(b) Improve resilience levels and retain genetic information by salvaging any viable indigenous plant material from the development footprint, for transplanting to suitable recipient sites, and

(c) Promote spontaneous recruitment and establishment of species representative of the threatened ecological community, or seedlings of the threatened species, by managing or eliminating degrading impacts or activities, and

(d) Improve the condition and stability of the land surface and reinstate or restore the natural hydrology and soil processes, and

(e) Re-establish suitable corridor links in fragmented habitat areas to improve recolonization opportunities for indigenous plant and animal species, enhance climate adaptation opportunities and improve the genetic health of small populations, and

(f) Undertake any other specific actions or measures recommended by any relevant recovery plan or priority action statement.

Note: Recovery plans and Priority Action Statements may be accessed via the Office of Environment and Heritage website.

C3. Development of any land adjoining natural waterways is to include any environmental protection works necessary to protect, manage and restore the natural ecological function, ecological processes and scenic value of the waterway. Refer to Part C1 Biodiversity and Natural Resources of this DCP.

C4. Development on land adjoining National Park is to implement any environmental protection works determined to be necessary to avoid, remedy or mitigate any adverse impact of development. This may include, but not be limited to, the maintenance, fabrication or enhancement planting of vegetated buffers and control of environmental weeds in all areas of the site.

C2.1.2. Land containing high quality remnant bushland

C1. Development of land containing high quality remnant native vegetation, or identified habitat areas for threatened species or threatened populations, is to include conservation works to maintain and enhance the endemic biodiversity and habitat areas of the Blue Mountains, so as to create breeding and local migration opportunities for indigenous wildlife. This may include, but not be limited to, any of the following:

(a) Salvage of suitable woody plant material removed as a component of development, and appropriate redistribution of the material into areas of suitable bushland to create habitat complexity ,and

(b) Retention of foraging opportunities for wildlife through the conservation and regeneration of high habitat value shrubs, and

(c) Retention of a healthy and diverse range of indigenous tree species, from a variety of age classes, which may represent components of wildlife corridors and which contain diverse habitat provisions, including hollows, rough bark, flowers and fruit, and

(d) Retention and management of dense ground layer and/or low shrub vegetation necessary as a key habitat component for a number of threatened fauna species, and

(e) Monitoring and protection from threats and disturbances, including recreational overuse, weed invasion, disturbance by domestic animals, nutrient enrichment or establishment of over abundant native species which may degrade the health and integrity of the bushland area.

C2.1.3. Land containing poor quality or weed affected native vegetation

C1. Existing threats to the health and integrity of the bushland area are to be identified.

C2. The means by which identified threats or disturbances may be eliminated or reduced is to be determined. This will inform the environmental protection works required on the site.

C3. Assisted natural regeneration is to be used manage weed species and improve the health and viability of existing indigenous vegetation.

C4. Resilience is to be enhanced in those areas displaying minimal to nil recruitment, through the use of reconstruction through revegetation, or reinstatement of habitat elements.

C5. Monitor and manage regeneration, and stage any weed control to conserve habitat. The short term habitat value of over-abundant colonising species or certain weed species is to be recognised, and these features retained or managed until other vegetation matures.

C2.1.4. Land containing highly degraded native vegetation, or mainly cleared areas

C1. Development of land containing disturbed native vegetation within slope constraint areas, escarpment conservation areas, land between towns and Zone **C2 Environmental Conservation** is to include provisions for environmental protection works to progressively restore, regenerate and maintain vegetation characteristic of the pre-existing locally indigenous vegetation community, or an appropriate reference community.

C2. Where land has been cleared and degraded through surface compaction and/or erosion, environmental protection works necessary to restore the functional capacity of the soil are to be undertaken. This may include any of the following measures:

(a) restore an appropriate level of organic matter, including light mulch, logs and seed bearing branches, to protect soil surfaces from rainsplash impacts, and

(b) to prevent erosion and allow for the retention and reuse of water and gradual accumulation of organic matter, use soft engineering solutions to stabilise and reinforce bare soil areas having a slope of 1:3 or steeper.

C3. Revegetation of locally indigenous species, characteristic of the reference ecosystem, is to be undertaken in a manner which will reinstate both resilience and restore ecological function to cleared or highly degraded areas.

C2.1.5. Environmental zones: C3 Environmental Management and C4 Environmental Living

C1. Development of any land within Zones **C3 Environmental Management** and **C4 Environmental Living**, regardless of the condition of the vegetation or the presence of any environmentally sensitive feature or area, is to include any environmental protection works necessary to achieve the following outcomes:

(a) the protection, restoration and conservation of areas of indigenous vegetation outside of the areas approved for development, including asset protection zones. These are to be retained as 'natural areas' for the life of the development, and

(b) the progressive restoration of cleared or degraded land which is outside of the areas approved for development, including asset protection zones, to an appropriate reference condition or pre-existing locally indigenous vegetation community.

C2.2. Weed Management

Note: For the purpose of this part, weed control and management is a class of Environmental Protection Works.

Explanation

Weeds are considered to be "a plant that requires some form of action to reduce its harmful effects on the economy, the environment, human health and amenity" (Commonwealth of Australia 2007).

In New South Wales, weeds pose the second greatest threat to the conservation of biodiversity and ecosystem function after land clearing and habitat loss. The loss and degradation of native plant and animal habitat by invasion of escaped garden plants, including aquatic plants, is listed as a key threatening process under the provisions of the *Biodiversity Conservation Act 2016*.

There are currently over 400 weed species known to occur in the Blue Mountains, and a substantial proportion of these have their origins as garden plants. Many are transported from development on ridge lines and upper catchment areas into surrounding bushland via water and wind movement, or via native and feral wildlife. The establishment of weeds into the natural environment is assisted by soil disturbance, clearing of native vegetation for urban development, nutrient enrichment of soils and water, and excessive stormwater runoff from urban areas. Once established, the weed populations themselves become a degrading factor which further alters environmental conditions, and accelerates the loss of native vegetation and habitat.

Weed management is an essential and integral part of the sustainable management of natural resources for the benefit of the economy, the environment, human health and amenity. Combating weed problems is a shared responsibility.

During the planning and implementation of development, measures need to be taken to prevent weeds being introduced to the natural environment, and ensure that effective control strategies for existing weed infestations are developed and implemented. This is consistent with targets and objectives identified in Blue Mountains City Councils Weed Management Strategic Plan and Sustainable Blue Mountains 2025. This section therefore identifies both control measures for existing weed infestations and restrictions and advice on the use of identified invasive species in new development.

Objectives

O1. To reduce the impact of urban development on the environment and catchment areas by requiring the control of existing priority and environmental weed infestations as an integral component of all private land development.

O2. To protect and enhance the habitat area of native flora and fauna on development sites through the restoration and rehabilitation of weed dominated flora.

O3. To ensure that the urban amenity of a locality is not eroded by the unregulated growth of weed species on private property.

O4. To enhance the ecological and amenity value of the urban/bushland interface by ensuring that these areas do not become dominated by weed invasion.

O5. To limit opportunities for the emergence of new weed infestations by restricting the use of recognised invasive species within landscaped areas on private and public property.

Controls

C1. Priority weeds are not to be planted in any land use zone.

Note: All current weeds relevant to the Blue Mountains are listed in Part C2.3 of this DCP.

C2. Existing infestations of priority weeds are to be systematically and effectively controlled.

Note: Information relating to current priority weed declarations and information may be accessed on the Blue Mountains City Council website and through the NSW Department of Primary Industries.

C3. To assist in offsetting the environmental impact of development, control of listed environmental weeds will generally be required to be undertaken over areas of the development site not required for building or asset protection.

C4. Control techniques employed are to be the most effective for the subject weed and the most suitable for use in the given situation, particularly within sensitive vegetation communities and adjacent to waterways.

Note: Weed management techniques for specific weeds may be accessed via the Blue Mountains City Council website.

C5. Weed control activities are not to result in environmental harm to the affected system, including harm to non-target species, destabilisation of steep slopes or pollution of water. All weed control is to be undertaken in accordance with current industry best practice techniques and practices.

Note: Guidelines for low impact control of weeds can be accessed via the Australian Association of Bush Regenerators website.

C6. Factors which are promoting weed growth are to be identified and reduced as an integrated component of weed control, and weed control is to be combined with measures designed to enhance the ability of the bushland to resist future weed invasion.

C7. Where herbicide is used in weed control, it is to be done so in accordance with the current regulations and permits designed to reduce the potential for environmental harm, (including to waterways and aquatic organisms) and to ensure the use of the most suitable and effective chemical against the target weed(s).

Note: Further information about the use of chemicals for weed control and precautions for their use can be accessed via the Australian Government website.

C8. Where weeds are present on land being subdivided, a detailed weed control strategy is to be devised and implemented, to ensure that all listed environmental weeds are effectively and systematically controlled in a timely manner. The weed control strategy will need to identify realistic milestones by which the success of the weed control may be determined prior to the release of a subdivision certificate.

C9. All weed control strategies are to be based on the existing site condition and capacity to recover. The following hierarchy of considerations and principles is to be included in any proposed strategy:

(a) Ensure that all weeds are correctly identified so that the appropriate control techniques may be determined, and

(b) Identify and mitigate site factors which have promoted the establishment of the weeds, and

(c) Identify the impact of each weed species, and determine which are causing harm to the affected community, and which may be providing benefit, and

(d) Determine if the weeds are providing wildlife habitat. If so, retain and control spread until compensatory plantings have matured, and

(e) Determine priorities for weed control, identifying high to low priority weeds and high to low priority areas on site, and

(f) Determine the level of control which will be realistically achieved for the weed(s), based on the current (localised) extent of the weed, its ecology and persistence and the likelihood of reinvasion within the short term, and

(g) Determine if other measures need to be undertaken in conjunction with the weed control, including revegetation, erosion control and waste management, and

(h) Determine the time frame for achieving control and / or containment of the weed(s), and

(i) Determine if the weed removal is likely to result in destabilisation of soil. If so, undertake weed removal in a gradual and staged manner ensuring that restoration (natural regeneration, planting) is successful at each stage.

C10. All weed control is to include any supplementary environmental protection works necessary to improve site resilience and increase resistance to weed reinvasion, including any of the following actions:

(a) Apply appropriate triggers to stimulate regeneration of soil stored weed seed to exhaust supply, and

(b) Apply triggers to stimulate regeneration of indigenous plant material, and

(c) Provide competitive plantings using provenance plant material, and

(d) Reduce light levels through canopy restoration, and

(e) Reduce any factors which may promote the growth of weeds and degrade the health of indigenous plant material, including nutrient enrichment or stormwater flows.

C2.2.1. Environmentally sensitive land

C1. Weed control in drainage lines, creek banks or riparian zones is not to result in soil erosion or other contamination of waterways.

Note 1: The manner in which weeds should be controlled in these areas may be accessed via the Australian Association of Bush Regenerators (AABR).

Note 2: Always observe restrictions on the use of chemicals and fertilisers near waterways imposed by *Pesticides Act 1999,* the *Pesticides Regulation 2017* and the *Protection of the Environment Operations Act 1997.*

C2. In order to protect and enhance the biodiversity value of the vegetation community, and reduce factors which may degrade the vegetation or the habitat area, listed weed species are to be systematically and effectively controlled over land identified as containing any Significant Vegetation Community listed in Schedule 6 of LEP 2015. A detailed Weed Management Strategy may need to be submitted.

C3. Where weed control is required within any vegetation community listed under the provisions of the Biodiversity Conservation Act 2016 (BC Act), or the *Environmental Protection and Biodiversity Conservation Act 1999* (EPBC Act), the preparation of a detailed *Weed Management Strategy* will generally be required. The strategy will need to be prepared in accordance with any best practice guidelines or principles relevant to the threatened community.

Note: Relevant guidelines include best practice guidelines: Sydney Turpentine Ironbark Forest available from the NSW Office of Environment and Heritage website.

C4. Weed control in *Threatened Ecological Communities* or within the habitat area of threatened species is to be undertaken by practitioners with adequate and appropriate qualifications in ecological restoration and pesticide application.

Note: Ongoing maintenance weeding and other restoration works undertaken in the habitat area of threatened species or within a threatened ecological community may require licensing under Part 2 of the *Biodiversity Conservation Act 2016* (unless the Director – General determines that a licence is not required in which case a certificate to that effect will be issued). The licence is to be held by the landowner.

C5. Weeds representing breeding or important foraging habitat of threatened fauna species are to be conserved, and controlled only following the establishment of suitable compensatory replanting or during non-breeding periods.

C2.2.2. Use of listed weed species in landscaped areas.

C1. In order to protect and enhance the scientific and recreational value of the natural environment, and avoid adverse environmental impact, properties which adjoin public reserves, Crown Land or National Park, are not to include listed weed species within new landscaped areas. Measures are to be implemented to ensure that propagules from existing established landscape species do not to escape beyond lot boundaries.

C2. So as to limit the potential for invasive species to result in deleterious impact upon the natural environment which may detract from the recreational experience, weed species listed within Part C2.3 of this DCP are to be excluded from landscaping works on land within Zone **RE1 Public Recreation** or **RE2 Private Recreation.**

C3. Landscaping undertaken for or on behalf of Council, or landscaping undertaken by private residents on Council owned land (including road reserves) is not to include the retention or planting of any weed species listed in Part C of this DCP.

C4. So as to protect and conserve the ecological integrity of sensitive environments within the Blue Mountains, weed species listed within Part C2.3 of this DCP are not to be used within landscape areas in any of the following zones or areas:

(a) on any Environmentally Sensitive Land, or

(b) within any asset protection zone (APZ), or

(c) in any land use zone, unless it can be demonstrated to Councils satisfaction, that no alternative species is capable of achieving the heritage or character objectives relevant to the landscaping of the site.

C2.3. Weeds of the Blue Mountains

Explanation

The following is a list of priority and other environmental weeds which have been evaluated as posing a potential weed risk in natural environments in the Blue Mountains Local Government Area (LGA). This assessment has been made using the NSW Weed Risk Management System (Johnson 2009*).

These weed risk assessments have been based on current knowledge and research, and are not based solely on current distribution in the local government area: the evaluation also takes into account the weed's potential distribution and impacts in the specific climatic, geological and biological conditions occurring in the region.

The list is not exhaustive. The selection of species to be assessed was based on plants or groups of plants known to have weed potential, which are commonly used in landscaping, gardening or agricultural situations and are generally commercially available.

This list will be periodically reviewed and updated. As new species become commercially available, or new information on the weed risk of some plants emerges, species may be reassessed and added or removed from the list.

Disclaimer: *Council recognises that some weed species pose a lesser or higher risk in specific climatic and geological regions within the LGA. The current weed risk assessments have been conducted based on current knowledge and broad vegetation and risk categories. It should be noted, however that there are limitations with the assessment process for weeds which are only invasive in a small percentage of habitats, but these habitats have high conservation significance (e.g. swamps/shale based communities). In these instances, the weed risk may be assessed as lower than might truly be the case for that specific weed where it occurs in these specific habitats. These plants have been identified by an asterisk against their Weed Risk Category.*

Use of this list

This list is to be read in conjunction with Part C2: Bushland and Weed Management, Part C3: Landscaping, and Part C5: Tree and Vegetation Removal, of this DCP, where general and specific controls on the removal, control and landscape use of these plant species is provided.

This list is also a useful reference for land management agencies, consultants, landowners and residents operating in the Blue Mountains LGA as a tool for identifying weeds in their area and species to be avoided when selecting plants for landscape purposes.

* Dr Stephen Johnson, Industry and Investment New South Wales, ORANGE NSW 2800, September 2009/ © State of New South Wales through Industry and Investment NSW 2009.

C2.3.1. Categories of weeds listed

Priority weeds are those that have been identified as posing a significant threat to human health, agriculture or natural resources and which legally require some measure of control.

Further information on priority weeds can be sourced from Council's website at:

www.bmcc.nsw.gov.au/weeds

Priority weeds are classified as follows:

- Prohibited Matters
- State Priority Weeds
- Regional Priority Weeds
- Local Priority Weeds

Local, regional and state Priority weeds are listed in the Blue Mountains Weed Management Strategic Plan. State priority weeds are high risk weed species across the entire state of NSW, as determined by the Department of Primary Industries. Regional priority weeds are high risk weed species within the Greater Sydney region, as determined by the Greater Sydney Regional Weeds Committee. Local priority weeds are high risk weed species within the Blue Mountains Local Control area, as determined by a professional panel within BMCC.

Environmental Weeds include, but are not restricted to species classified priority weeds. The nomination of species eligible for listing as environmental weeds is generally a matter for local agencies and organisations including Local Councils, Local Land Services, Community groups such as Landcare, Conservation societies and organisations, Research Centres and other members of the scientific community.

Further information can be sourced from the Department of Primary Industries and the Council's website through the following links:

www.dpi.nsw.gov.au/biosecurity_old/weeds

www.bmcc.nsw.gov.au/weeds

Common Name	Scientific Name	Risk Category	Note
Trees			
African Olive	*Olea europaea* spp.*africana*	Regional Priority Weed	
Albizia/Cape Leeuwin Wattle	*Paraserianthes lophantha*	Very High	
Athel Pine	*Tamarix aphylla*	State Priority Weed	
Black Locust*	*Robinia pseudoacacia*	Local Priority Weed	
Black Willow	*Salix nigra*	Regional Priority Weed	
Blackthorn	*Prunus spinosa*	High	
Box Elder	*Acer negundo*	Local Priority Weed	
Broad leaf Pepper tree	*Schinum terebinthifolius*	Medium	
Camphor Laurel	*Cinnamomum camphora*	Local Priority Weed	
Canadian Poplar	*Populus x canadensis*	High	
Canary island Date Palm	*Phoenix canariensis*	Medium	Removal is only exempt if the tree is <6m high
Cherry Plum	*Prunus cerasifera*	Medium	
Chinese Celtis	*Celtis sinense*	Medium	
Chinese Pistachio	*Pistachia chinensis*	High	
Chinese Tallow	*Triadica sebifera*	High	
Coastal Lily Pily	*Syzigium paniculatum*	Medium	Removal is only exempt if the tree is <6m high
Common Alder	*Alnus glutinosa*	Very High	
Cootamundra Wattle	*Acacia baileyana*	Local Priority Weed	
Coral Tree	*Erythrina crista-galli, E. x sykesii*	High	
European Nettle Tree	*Celtis australis*	Medium	Removal is only exempt if the tree is <6m high

Common Name	Scientific Name	Risk Category	Note
Trees			
European Olive	*Olea europaea europaea*	High	
Evergreen Alder	*Alnus jorullensis*	Local Priority Weed	
Evergreen Dogwood & Himalayan Strawberry Tree	*Cornus capitata*	Local Priority Weed	
Flame Tree	*Brachychiton acerifolius*	Very High	Removal is only exempt if the tree is <10m high
Honey Locust	*Gleditsia triacanthos*	Local Priority Weed	
Hoop Pine	*Araucaria cunninghamia*	Medium	Removal is only exempt if the tree is <10m high
Irish Strawberry Tree	*Arbutus unedo*	Low	Removal is only exempt if the tree is <3m high
Jacaranda	*Jacaranda mimosifolia*	Medium	Removal is only exempt if the tree is <6m high
Kei Apple	*Dovyalis caffra*	Regional Priority Weed	
Kohuhu	*Pittosporum tenuifolium*	High	
Lemon scented Gum	*Corymbia citriodora*	Very High	
Loquat	*Eriobotrya japonica*	Medium	
Mesquite	*Prosopis spp.*	State Priority Weed	
Mexican Pine	*Pinus patula*	Very High	
Miconia	*Miconia calvescens*	State Priority Weed	
Mt Morgan Wattle/ Queensland Silver Wattle	*Acacia podalyriifolia*	Local Priority Weed	
New Zealand Cabbage Gum	*Cordyline australis*	Medium	

Common Name	Scientific Name	Risk Category	Note
Trees			
New Zealand Pittosporum	*Pittosporum eugenioides*	Medium	
Persian silk tree	*Albizia julibrissin*	Medium	Removal is only exempt if the tree is <6m high
Portuguese Laurel	*Prunus lusitanica*	Local Priority Weed	
Privet Large-leaf	*Ligustrum lucidum*	Local Priority Weed	
Radiata Pine,Monterey Pine	*Pinus radiata*	Very High	Removal is only exempt if the tree is <10m high
Rhus Tree	*Toxicodendron succedaneum*	Local Priority Weed	
Rubber Tree	*Ficus elastica*	Medium	
Silky Oak	*Grevillea robusta*	High	Removal is only exempt if the tree is <10m high
Spotted Gum	*Corymbia maculata*	High	Removal is only exempt if the tree is <10m high
Sycamore Maple	*Acer pseudoplatanus*	High	
Sycamore	*Platanus orientalis*	Local Priority Weed	
Tree Lucerne/ Tagasaste	*Chamaecytisus palmensis*	Local Priority Weed	
Tree-of-Heaven	*Ailanthus altissima*	Local Priority Weed	
Umbrella Tree	*Schefflera actinophylla*	Medium	
White Cedar	*Melia azedarach*	Very High	
White Poplar	*Populus alba*	Local Priority Weed	
Wild Black Cherry/ Bird Cherry	*Prunus serotina*	Local Priority Weed	

Common Name	Scientific Name	Risk Category	Note
Trees			
Willow Spp	*Salix babylonica, Salix x reichardii, Salix calodendron*	Very High	
Willows	*Salix spp.*	State Priority Weed	

Common Name	Scientific Name	Risk Category
Shrubs		
African Boxthorn	*Lycium ferocissimum*	State Priority Weed
Barberry	*Berberis aristata*	Local Priority Weed
Bellyache Bush	*Jatropha gossypiifolia*	State Priority Weed
Berberis	*Berberis thunbergia, B. vulgaris, B. darwinii*	Very High
Bitou Bush	*Chrysanthemoides monilifera subsp. rotunda*	State Priority Weed
Blackberry	*Rubus fruticosus spp. agg*	State Priority Weed
Boneseed	*Chrysanthemoides monilifera ssp. monilif*	State Priority Weed
Butterfly Bush	*Buddleja davidii*	Local Priority Weed
Cape/Montpellier Broom	*Genista monspessulana*	State Priority Weed
Cassia, Arsenic Bush	*Senna Septemtrionalis*	High
Cassia/Senna	*Senna pendula*	Local Priority Weed
Castor Oil Plant	*Ricinus communis*	Local Priority Weed
Cestrum-Red Flowering	*Cestrum elegans*	High
Cherry Laurel	*Prunus laurocerasus*	Local Priority Weed
Chinese Holly	*Mahonia aquifolium* (syn. *Berberis aquifolium*)	High
Chinese Lantern	*Abutilon pictum*	High

Common Name	Scientific Name	Risk Category
Shrubs		
Cotoneaster	*Cotoneaster spp.*	Local Priority Weed
English Holly	*Ilex aquifolium*	Local Priority Weed
Firethorn	*Pyracantha spp.*	Local Priority Weed
Flax-leaf Broom	*Genista linifolia*	State Priority Weed
Gold Flower	*Hypericum kouytchense*	High
Golden Wreath Wattle	*Acacia saligna*	Very High
Gorse	*Ulex europaeus*	State Priority Weed
Green Cestrum	*Cestrum parqui*	Regional Priority Weed
Hawthorn	*Crataegus monogyna*	Very High
Indian Hawthorn	*Raphiolepis indica*	Medium
Karo	*Pittosporum crassifolium*	Very High
Karoo Thorn	*Acacia karoo*	Local Environmental Weed
Lantana	*Lantana camara*	State Priority Weed
Lions Tail	*Leonitus leonurus*	Medium
Looking Glass Plant	*Coprosma spp.*	Very High
Mickey Mouse Plant/ Ochna	*Ochna serrulata*	Local Priority Weed
Milkwort/Polygala	*Polygala paniculata; P. virgata*	Very High
Mysore Thorn	*Caesalpinia decapetala*	Regional Priority Weed
Oleander	*Nerium oleander*	Low
Plectranthus shrubs - Blue spur flower	*Plectranthus ecklonii, P grandis,*	High
Privet European	*Ligustrum vulgare*	Local Priority Weed

Common Name	Scientific Name	Risk Category
Shrubs		
Privet Small-leaf	*Ligustrum sinense*	Local Priority Weed
Rhizomatous Bamboo	*Phyllostachys aurea*	Local Priority Weed
Rhizomatous Bamboo (Black)	*Phyllostachys nigra*	Local Priority Weed
Sacred bamboo	*Nandina domestica*	Low
Scotch/English Broom	*Cytisus scoparius*	State Priority Weed
Scurf Pea	*Psoralea pinnata*	Very High
Siam Weed	*Chromolaena odorata*	Very High
Spanish/Portuguese Heath	*Erica lusitanica*	Local Priority Weed
Sweet Pea Shrub	*Polygala myrtifolia*	Very High
Thorny olive/Evil Agnes	*Elaeagnus pungens*	Very High
Tree Heath	*Erica arborea*	Very High
Tropical Soda Apple	*Solanum viarum*	State Priority Weed
Tutsan	*Hypericum androsaemum*	Local Priority Weed
Wild Tobacco Bush	*Solanum mauritianum*	Local Priority Weed
Yellow Bells	*Tecoma stans*	Local Priority Weed

Common Name	Scientific Name	Risk Category
Perennials and Ground Covers		
Agapanthus	*Agapanthus praecox ssp. orientalis*	Local Priority Weed
Alligator Weed	*Alternanthera philoxeroides*	State Priority Weed
Arum Lily	*Zantedeschia aethiopica*	Local Priority Weed
Asparagus Weeds	*Asparagus aethiopicus, A. africanus, A. asparagoides, A. pumosus & A. scandens spp.*	State Priority Weed
Asthma Weed	*Parietaria judaica*	Low

Common Name	Scientific Name	Risk Category
Perennials and Ground Covers		
Bathurst/Noogoora/ Californian Burr	*Xanthium* spp.	High
Blue Periwinkle	*Vinca major; V. minor*	Local Priority Weed
Broomrapes	*Orobanche* spp.	Local Environmental Weed
Canna Lily	*Canna indica*	Medium
Chinese knotweed	*Persicaria chinensis*	Regional Priority Weed
Coreopsis	*Coreopsis lanceolata*	Very High
Creeping Buttercup	*Ranunculus repens*	*Low
Creeping Lantana	*Lantana montevidensis*	Medium
Crofton Weed	*Ageratina adenophora*	Local Priority Weed
Day Lily	*Hemerocallis* spp.	*Low
Fennel	*Foeniculum vulgare*	*Low
Fireweed	*Senecio madagascariensis*	State Priority Weed
Forget-Me-Not	*Myosotis sylvatica*	Very High
Formosan Lily	*Lilium formosanum*	Very High
Foxtail Asparagus	*Asparagus meyeri*	Medium
Gaura, Clock weed; Butterfly plant	*Oenothera curtiflora (syn Gaura parviflora),*	High
Ginger Lily	*Hedychium gardnerianum*	Very High
Glory Lily	*Gloriosa superba*	Regional Priority Weed
Holly-leaved Senecio	*Senecio glastifolius*	Regional Priority Weed
Horsetail	*Equisetum arvense*	Regional Priority Weed
Impatiens/Busy Lizzy	*Impatiens balsamina/walleriana*	Medium
Japanese Knotweed	*Persicaria capitata*	Medium
Japanese Windflower	*Anemone x hybrida*	Medium

Common Name	Scientific Name	Risk Category
Perennials and Ground Covers		
Kochia	*Kochia scoparia*	Local Environmental Weed
Mistflower	*Ageratina riparia*	Local Priority Weed
Montbretia	*Crocosmia x crocosmiiflora*	Local Priority Weed
Mouse-eared Hawkweed	*Hieracium pilosella*	State Priority Weed
Nasturtium	*Tropoleum majus*	Low
New Zealand Flax	*Phormium sp.*	*Low
Orange Hawkweed	*Hieracium aurantiacum.*	State Priority Weed
Ox-eyed Daisy	*Leucanthemum vulgare*	Very High
Oyster Plant	*Acanthus mollis*	Very high
Parthenium Weed	*Parthenium hysterophorus*	Very High
Patersons Curse	*Echium spp.*	Local Priority Weed
Peruvian Lily	*Alstromeria aurea*	High
Plectranthus ground covers	*Plectrantus ciliatus, P. verticillatus, P. amboinicus, P. ornatus, P.oertendahlii*	High
Prostrate Cotoneaster	*Cotoneaster microphyllus / horizontalis*	Very high
Red Hot Poker	*Kniphofia sp.*	High
Scotch Thistle	*Onoporum acanthium*	Local Priority Weed
Seaside Daisy	*Erigeron karvinskianus*	Medium
Self Heal	*Prunella vulgaris*	Medium
Shasta Daisy	*Leucantheum sp.*	Low
Sicilian Sea Lavender	*Limonium hyblaeum*	Regional Priority Weed
Silver-leaf nightshade	*Solanum eleagnifolium*	State Priority Weed

Common Name	Scientific Name	Risk Category
Perennials and Ground Covers		
Singapore Daisy	*Sphagneticola trilobata*	Regional Priority Weed
Spotted Knapweed	*Centaurea maculosa*	Medium
St Johns Wort	*Hypericum perforatum*	Local Priority Weed
Topped lavender, French/ Italian	*Lavandula stoechas*	Low
Trad	*Tradescantia fluminensis*	Local Priority Weed
Veldt Daisy	*Osteospermum* (syn. *Dimophotheca*) *ecklonis*	Very High
Viper's Bugloss	*Echium vulgare*	Local Priority Weed
Watsonia	*Watsonia meriana*	Local Priority Weed
Watsonia	*Watsonia borbonica*	High
Wild Iris	*Dietes bicolor, D. iridioides*	High

Common Name	Scientific Name	Risk Category
Grasses and Grass-Like Plants		
African Feathergrass	*Pennisetum macrourum*	High
African Lovegrass	*Eragrostis curvula*	Local Priority Weed
Brown Top Bent	*Agrostis cappillaris*	Low
Chilean Needle Grass	*Nassella neesiana*	State Priority Weed
Cocksfoot	*Dactylis glomerata*	**High
Columbus Grass	*Sorghum x almum*	High
Common/Soft Rush	*Juncus effusus*	High
Coolatai Grass	*Hyparrhenia hirta*	Local Priority Weed
Creeping Bent	*Agrostis stolonifera*	**High
Ehrharta	*Ehrharta erecta*	Very High
Giant Coolatai/Thatch Grass	*Hyparrhenia rufa*	High

Common Name	Scientific Name	Risk Category
Grasses and Grass-Like Plants		
Giant Parramatta Grass	*Sporobolus fertilis* (syn.*indicus*)	High
Giant Reed	*Arundo donax*	Regional Priority Weed
Johnson Grass	*Sorghum halepense*	High
Mexican Feather Grass	*Nassella tenuissima*	Local Environmental Weed
Pampas Grass	*Cortaderia selloana*	Regional Priority Weed
Parramatta Grass	*Sporobolus indica*	Medium
Paspalum	*Paspalum dilatatum*	**Very High
Plume/ Fountain Grass	*Pennisetum setaceum, P. alapecaroides*	High
Prairie Grass	*Bromus catharticus*	**Medium
Red Natal Grass	*Rhynchelytrum repens* (syn.*Melinis repens*)	**Medium
Reed Canary Grass	*Phalaris urundinaceae*	**Very High
Rhodes Grass	*Chloris gayana*	**High
Sedge	*Cyperus congestus*	Medium
Serrated Tussock	*Nassella trichotoma*	State Priority Weed
Spider Plant/Ribbon Grass	*Chlorophytum comosum*	Medium
Sweet Vernal Grass	*Anthoxanthum odoratum*	Local Priority Weed
Tall Fescue	*Festuca arundinacea* (syn. *F. elatior*)	**High
Tiny-headed Rush	*Juncus microcephalus*	Very High
Umbrella Sedge	*Cyperus eragrostis*	*Medium
Vasey Grass/Giant Paspalum	*Paspalum urvillei*	High
Whiskey Grass	*Andropogon virginicus*	High
Yorkshire Fog	*Holcus lanatus*	Local Priority Weed
Yorkshire/Creeping Fog	*Holcus lanatus, H. mollis*	Very High
Zebra Grass	*Miscanthus sinensis* and varieties	Medium

Common Name	Scientific Name	Risk Category
Vines		
Balloon Vine	*Cardiospermum grandiflorum*	Local Priority Weed
Banana Passionfruit	*Passiflora tarminiana*	Local Priority Weed
Black-eyed Susan	*Thunbergia alata*	Medium
Blue Morning Glory	*Ipomoea indica*	Local Priority Weed
Blue-bell Creeper	*Billardiera* (syn. *Sollya*) *heterophylla*	Very High
Bridal Creeper	*Asparagus asparagoides*	State Priority Weed
Canary Creeper	*Senecio tamoides*	Medium
Cape Honeysuckle	*Tecomaria capensis*	Very High
Cape Honeysuckle	*Tecoma capensis*	Very High
Cape Ivy	*Delairea odorata*	Local Priority Weed
Cats Claw Creeper	*Dolichandra* (syn. *Macfadyena*) *unguis-cati*	State Priority Weed
Chilean Jasmine	*Mandevilla laxa*	High
Clematis cultivars	*Clematis spp.*	Very High
Climbing Asparagus	*Asparagus africanus*	Regional Priority Weed
Climbing Groundsel	*Senecio angulatus*	Very High
Dodder	*Cuscuta campestris*	Low
Dog rose	*Rosa canina*	Very High
Dolichos pea	*Dipogon lignosus*	Very High
English Ivy	*Hedera helix*	Local Priority Weed
Field Bindweed	*Convovulus arvensis*	Medium
Greater Bindweed	*Calystegia sylvatica*	Medium
Himalayan Honeysuckle	*Leycesteria formosa*	Local Priority Weed
Japanese Honeysuckle	*Lonicera japonica*	Local Priority Weed

Common Name	Scientific Name	Risk Category
Vines		
Kudzu	*Pueraria lobate*	Regional Priority Weed
Leaf cactus	*Pereskia aculeate*	Regional Priority Weed
Madeira Vine	*Anredera cordifolia*	State Priority Weed
Maidenhair Vine	*Muehlenbeckia complexa*	Low
Morning Glory	*Ipomoea indica, I. cairica*	Local Priority Weed
Moth Vine	*Araujia sericifera*	Local Priority Weed
Passionfruit (Common Banana, White, Corky)	*Passiflora edulis, P. mollisina P. subpeltata, P. cinnabarina, P. suberosa*	High
Pink Trumpet Vine	*Pondranea ricasoliana*	High
Sicklethorn	*Asparagus falcatus*	Regional Priority Weed
Skunk Vine	*Paederia foetida*	Regional Priority Weed
Sweet Briar	*Rosa rubiginosa*	Local Priority Weed
Turkey Rhubarb	*Acetosa sagittata*	Local Priority Weed
White Jasmine	*Jasminum polyanthum*	Local Priority Weed

Common Name	Scientific Name	Risk Category
Ferns		
Sword Fern/Fishbone Fern	*Nephrolepis cordifolia*	Low
Coin spot/Scaly tree fern	*Cyathea cooperi*	Medium

Common Name	Scientific Name	Risk Category
Cacti and Succulents		
Agave	*Agave americana*	Medium
Aloe	*Aloe arborescens*	Medium
Fairy Crassula	*Crassula multicava*	Medium

Common Name	Scientific Name	Risk Category
Cacti and Succulents		
Harrissia Cactus	*Harrissia spp*	Low
London's Pride	*Saxifraga umbrosa*	Low
Mother of Millions	*Bryophyllum spp.*	Local Priority Weed
Prickly Pear	*Opuntia spp.*	State Priority Weed
Sea Spurge	*Euphorbia paralias*	Regional Priority Weed
Swedish Ivy / Creeping Charlie	*Plectranthus verticelatus*	High
Yucca	*Yucca aloifolia*	High

Common Name	Scientific Name	Risk Category
Aquatic Plants		
Cabomba	*Cabomba caroliana*	State Priority Weed
Dwarf Rotala	*Rotala rotundifolia*	High
Egeria	*Egeria densa*	DECLARED
Frogbit	*Limnobium laevigatum*	State Priority Weed
Hygrophilla	*Hygrophilla costata*	Regional Priority Weed
Kidney leaf mud plantain	*Heteranthera reniformis*	Regional Priority Weed
Lagarosiphon	*Lagarosiphon major*	Local Environmental Weed
Mexican/Yellow Water Lily	*Nymphaea mexicana*	High
Sagittaria	*Sagittaria platyphylla*	State Priority Weed
Salvinia	*Salvinia molesta*	State Priority Weed
Senegal Tea	*Gymnocoronis spilanthoides*	Regional Priority Weed
Water Hyacinth	*Eichhornia crassipes*	State Priority Weed

Common Name	Scientific Name	Risk Category
Aquatic Plants		
Water Lettuce	*Pistia stratiotes*	Regional Priority Weed
Water Poppy	*Hydrocleys nymphoides*	Regional Priority Weed
Willow-leaf Primrose/ Ludwigia	*Ludwigia peruviana*	Regional Priority Weed

*These species are identified as high risk in swamps and riparian (creekline) habitats

** These species are identified as low risk in pasture situations in rural land use areas

PART C3
LANDSCAPING



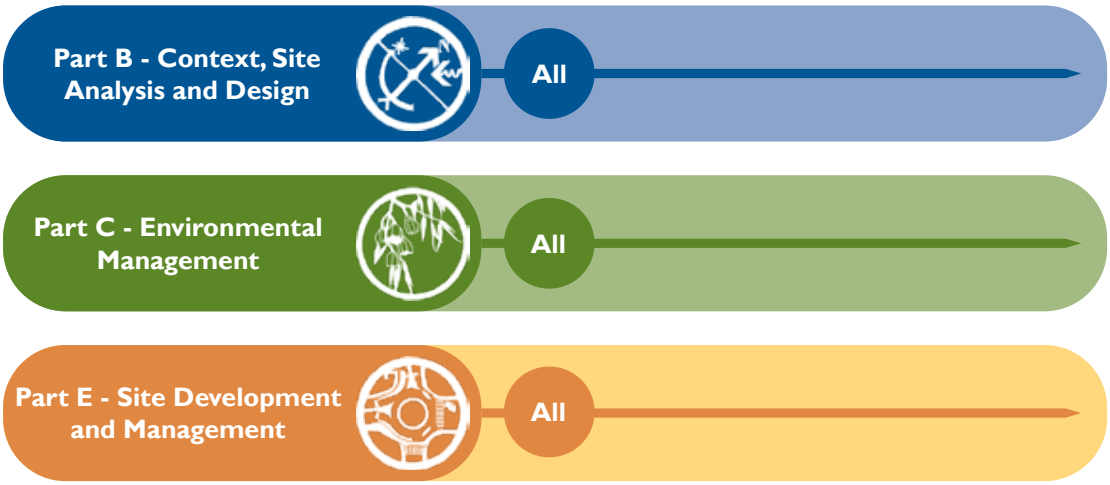
Introduction

Well designed, constructed and maintained landscapes are an asset to the community, and provide numerous positive outcomes.

Landscaping contributes to the attractiveness and liveability of our outdoor spaces, both public and private, and to the health and wellbeing of the community. Landscape plantings may assist in maintaining privacy between adjoining properties and between private and public spaces, contribute to energy efficiency and help to establish a character setting for development in a manner that defines the locality. Many landscaped areas also contribute to the habitat requirements of indigenous wildlife, and much of the Blue Mountains rich flora is well suited to landscape applications. Appropriately landscaped areas also help to conserve and enrich the significant cultural heritage of the village setting in the older areas of the Blue Mountains, in a manner sensitive to surrounding natural areas, including the surrounding World Heritage Blue Mountains National Park, the Sydney drinking water catchment and the Hawkesbury Nepean River System.

In order to ensure that landscaped areas and the plants selected for use on private property achieve these positive outcomes, it is important that the landscaping of the site is considered in the early planning stage of a development. This includes careful consideration of existing vegetation on the site, the potential impact of the development on this vegetation and if it may be conserved and incorporated into the landscape, and the requirement for new landscape areas and treatments that are appropriate to design and siting of the development. This part of the DCP provides both general and specific objectives, controls and guidelines relating to the landscape design, plant selection and outcomes for each land use zone within the Blue Mountains.

Read in conjunction with:



Submission Requirements

One or more of the following types of reports may need to accompany a development application:

- *Tree Removal and Retention Plan or Arboricultural Survey Report*
- *Arboricultural Impact Assessment*
- *Vegetation Survey and Species Inventory*
- *Vegetation Management Plan and/or Weed Management Strategy*
- *Landscape Plan*

C3.1. Landscaping Objectives

O1. To ensure that landscaping preserves and contributes to the visual amenity of the Blue Mountains and the existing and past cultural landscape.

O2. To promote the use of a landscaping style which reflects and reinforces the character of the locality and the maintenance of any aesthetic, cultural, scientific and biodiversity values which apply to the site.

O3. To encourage the retention of trees and other vegetation which are of ecological, aesthetic and cultural significance, through the integration of these features into the landscape.

O4. To provide adequate landscaping in accordance with the type, scale, location, climatic conditions and land use zone of the proposed development.

O5. To ensure that landscaping is designed, constructed and maintained to appropriately manage the interface between the natural landscape and the urban environment, in a manner consistent with the sustainability of the Blue Mountains ecological systems and biodiversity and the need to protect life and property from bush fire.

O6. To minimise the potential harm to fauna and flora habitat areas and ecological processes through the selection of non-invasive plant material as a first choice for landscaping in all land use zones.

O7. To ensure that landscaping does not unreasonably interfere with the liveability and/or solar access to buildings or open space areas on adjoining properties.

O8. To promote climate change adaptation through landscape design which minimises water use, provides for microclimate modification, reduces urban heat, consolidates and interconnects vegetation, habitat and waterways, is resilient to storms and drought and minimises bushfire risk.

O9. To achieve a neutral or beneficial effect on the urban forest tree cover, to mitigate urban heat island effect and maintain biodiversity, landscape amenity and liveability.



Part C3 - Figure 1: "It is possible to beautify almost any area, whatever shape or size, with a garden". Ellis Stones

C3.2. Demonstration of landscaping objectives

Explanation

Landscape Plans assist Council in determining the manner in which the proposed development will retain and manage existing vegetation, provide adequate and appropriate new plantings and mitigate the visual and environmental impact of the development. Landscape plans need to demonstrate that both general and zone specific landscaping objectives will be met.

Controls

C1. A landscape plan, drawn at an appropriate scale and including all relevant site and landscaping information as detailed in this Part, is to be submitted in support of all development applications (including subdivisions) which involve site works and land disturbance of an area >50m² within the following land use zones:

- R1 General Residential,
- R2 Low Density Residential,
- R3 Medium Density Residential;
- E3 Productivity Support;
- E4 General Industrial;
- C3 Environmental Management,
- C4 Environmental Living;
- W1 Natural Waterways, and
- SP3 Tourist

Note: The content and format of the landscape plan is detailed in Part I2.1.9 of this DCP.

C3.3. Retention and protection of trees and other vegetation

Explanation

The retention of established trees and other vegetation on development sites contributes to the conservation of existing landscape amenity, the retention of habitat components within the urban setting, and may reduce the overall environmental impact of new development.

Controls

C1. Development should minimise the removal of mature vegetation and site disturbance, and wherever practicable, existing established indigenous or other desirable landscape trees and other vegetation are to be prioritised for retention and incorporation into the landscape design.

Note: Trees and landscape settings within conservation areas or heritage items may require special consideration for retention. Refer to D9.4 Curtilages, settings, gardens and landscape settings, and C5.3 .

C2. Vegetation retention is to consider resilience. Healthy, structurally sound and undamaged trees and other vegetation are to be the first priority for conservation within proposed landscape areas.

C3. Consideration is to be given to the retention of trees in groups wherever it is possible to do so, as this generally provides increased resistance to storm events.

C4. Existing street front and road reserve plantings and remnant vegetation are to be conserved except where their removal is necessary for site access via a single road opening, which is to be located so as to minimise vegetation loss.

C5. Where the removal of landscape features which contribute to the streetscape or character of the locality is required to enable development, their loss is to be adequately compensated by replanting, subject to:

(a) whether site conditions permit replanting, and

(b) the necessity for further planting, having regard to the overall retention of trees and vegetation on the development site.

Note: Species selected to meet compensatory tree planting requirements should generally be of an equivalent stature (particularly height at maturity) to the landscape elements removed.



Part C3 - Figure 2: Streetscape trees: Tall streetfront trees with strong architectural form may enhance the amenity of the locality and contribute significantly to the neighbourhood character.

C6. Cut and fill and other soil modification is to be avoided in locations where vegetation and trees are to be retained. Where this cannot be avoided, an *Arboricultural Impact Assessment* is to be submitted which provides an assessment of the safe useful life expectancy (SULE) of that vegetation as it is affected by the proposal.

C7. Provision is to be made for the adequate protection of existing trees and vegetation to be retained. Exclusion zone fencing and any other measures determined to be necessary to ensure their protection during construction, are to be detailed on a landscape plan or similar documentation.

Note: A *Tree Removal and Retention Plan* or an *Arboricultural Survey Report* of existing trees on the development site and/or on lands which immediately adjoin the development area may be required. This may be necessary to determine the viability and suitability of trees for retention in the landscape, and in order to determine the extent of tree protection required.

C3.4. Landscape design, scale and amenity

Explanation

Landscaped areas perform a multiplicity of functions; some primary (such as screening and privacy or conservation of local amenity) and others secondary (including micro-climate modification). In the development of the landscape design, it is therefore important to identify all of the functional requirements that relate to a development, so that these can inform your design requirements.

Controls

C1. Landscape design should be responsive and complementary to the bulk and scale of the proposed development. It is to include plant material capable of achieving sufficient mature height to achieve a balance between the built form and the landscape elements.

C2. Sufficient space is to be reserved for the installation of appropriate plantings within parts of the site proposed for private open space areas, deep soil zones and permeable surfaces. These areas are to be effectively distributed on the site to achieve a visual balance between the built form and open space.

C3. Landscape planting is to assist in minimising the visual impact of the development from public areas and from adjoining properties, whilst also maintaining opportunities for natural surveillance. To achieve this outcome, screen planting should generally be provided at the following densities within garden bed areas:

(a) trees at 8m centres, and

(b) larger shrubs at 3m centres, and

(c) medium shrubs at 1.5m - 2m centres, and

(d) groundcovers and small shrubs at 0.5m – 1.0m centres.

Note 1: The outcome can be adjusted by varying the planting density of large to medium shrubs.

Note 2: The overall depth and continuity of the screen planting area may need to be adjusted on bushfire prone land.

C4. A proportion of the front building setback on land within **Residential Zones** and within Zones **C3 Environmental Management** and **C4 Environmental Living** is to be maintained as landscaped area (comprised of garden beds containing trees, shrubs and groundcovers or areas of retained indigenous vegetation). As a general guide:

(a) 25% for lots less than 18m wide, and

(b) 50% for lots greater than 18m wide.

Note: Provision for the planting of trees is encouraged in the property front setback area unless the width of the block or the dimensions of the existing front setback area are inappropriate for tree planting.



Part C3 - Figure 3: Landscape scale: Tall mid-block canopy plantings form a visual backdrop for multiple layers of more complex and smaller-scale plantings in proximity to the dwelling.

C5. The landscape is to incorporate garden bed areas of appropriate dimension and sufficient soil volume, to enable the successful establishment of landscape elements. The following minimum garden bed widths are generally required:

(a) garden beds without trees: 1.5m, and

(b) garden beds with trees: 5m.

Note: The soil volume required for tree establishment may, under certain circumstances, be partially met by any pervious area devoid of structures, which adjoins a narrower garden bed. Also see requirements for deep soil zones.

C6. Deep Soil Zones, retained as areas of natural ground, free of artificial structures and with a relatively unmodified soil profile are to be provided within the development site. *Deep Soil Zones* are to have minimum dimensions of 3m x 3m in order to permit the establishment and growth of large trees, and to promote rainwater infiltration. The following minimum areas are to be provided:

(a) single dwelling development within Zones **C3 Environmental Management** and **C4 Environmental Living**: 40% of the site area, and

(b) within Zone **R1 General Residential, R2 Low Density Residential** and **R3 Medium Density Residential**: 15% of the site area, and

(c) within Zone **E3 Productivity Support** and **E4 General Industrial**: 10% of the site area.

(d) within Zone **SP3 Tourist**: 20% of site area.

C7. The principles of passive solar design are to be incorporated into landscape design and the selection and location of tree and hedge species. There is to be consideration of the current and future amenity of residents and neighbours in terms of solar access, including ensuring that existing roof mounted solar energy systems are taken into account and not obstructed by vegetation.

C8. The location of structures, services (overhead and underground) and footpaths is to be adequately considered during the design of the landscape and in the selection of plant material. This will reduce potential future damage to these features.

C9. Stormwater management, including improved infiltration and the reduction of overland flow, is to be promoted within the landscape design and in the selection of plant material. The inclusion of Water Sensitive Urban Design features is to be considered.

Note: Refer to Part C6 Water Management of this DCP, with particular reference to Water Sensitive Urban Design (WSUD).

C10. Landscaping should contribute in a positive manner to bushfire protection strategies. The design is to ensure that landscaped areas are established and maintained in accordance with any Asset Protection Zone (APZ) requirements that may apply to the site, particularly in relation to building setbacks, vegetation density and canopy continuity.

Note: Refer to Part C4 Bushfire of this DCP.

C11. Landscaping in the vicinity of a heritage item is not to detract from the heritage value of the place. Careful consideration is to be given to plant selection and placement to ensure the potential for overshadowing or visual obstruction of heritage features is minimised.

Note: Refer to D5 Development in the vicinity of heritage items or conservation areas.

C12. Constructed elements within the landscape, including retaining walls, excavated and filled areas and ponds, are to be designed and located so as not to adversely impact on retained trees, natural features or an adjoining property.

Note 1: Adverse impacts may include damage to plant roots, diversion of stormwater runoff, modification or interference with ground water or surface aquifers, and any changes to the structural integrity of existing fencing or privacy screening.

Note 2: If the ground level is modified within the canopy spread of a tree to be retained, an arborist's report may be required to assess the impact of the proposed works. Refer to AS4970 Protection of Trees On Development Sites.



C3.5. Plant Material Selection

Explanation

Plant material is one of the key elements to consider within the landscape design. Living in the Blue Mountains does not limit plant selection to the use of only indigenous species. Although this is an objective of some land use zones (**C3 Environmental Management** and **C4 Environmental Living**), within other residential zones (**R1 General Residential, R2 Low Density Residential** and **R3 Medium Density Residential**), the use of exotic species may be more appropriate, depending upon the character and setting of the development site. However, when using non-indigenous species in any zone, whether these are native or exotic, it is important to ensure that these are not known to be invasive in bushland areas of the Blue Mountains.

Controls

C1. Plant material selection is to consider the land use zone within which the development is located, and compatibility with the specific objectives of that zone.

C2. Plant selection for all landscaping work is to consider its suitability to existing site conditions such as soils, aspect, drainage, elevation above sea level and prevailing climatic conditions.

C3. Plant material selected for use within each area of the landscape is to include an appropriate range of height and foliage density to meet specific landscape functional requirements identified for the landscape.

C4. Plant material with high aesthetic appeal, longevity, water use efficiency and known or predictable maintenance requirements should be prioritised for use in the landscape.



Part C3 - Figure 4: Blended landscapes: Indigenous canopy trees can be combined with traditional elements such as dry stone walls and exotic understorey plantings.

C5. Landscape planting proposed along front and side boundary alignments is to consider the plants mature height, width and root architecture to ensure that they will not unreasonably impact on adjoining land.

C6. Plant species listed within Part C2.3 Weeds of the Blue Mountains of this DCP, are not to be included in any new landscaped areas.

C7. Where development is located in or adjoining environmentally sensitive areas, the selection of plant material is to be restricted to appropriate locally indigenous vegetation.

C8. Within Asset Protection Zones, plant material selected for use is to contain low levels of volatile oils, possess high leaf moisture content and retain low levels of dead material, flaky or stringy bark.

Note: Refer to Part C4 Bushfire of this DCP.

C9. Landscape planting and plant selection within any road verge area is to be in accordance with the Blue Mountains City Council Street Tree Masterplan guidelines.

Note: Refer to Part E8 Public Domain of this DCP.

C10. Deciduous species are to be located away from stormwater flow paths and roadways to limit the pollution and obstruction of waterways from leaf material.

C3.6. Soil management

Explanation

Soil is a living, dynamic resource that supports plant life. It is a source of water, minerals, nutrients and symbiotic or partnership organisms. The soils of the Blue Mountains are ancient and fragile, and have taken millions of years to form. Their conservation should be a priority during development and construction of the landscape, as soils in good condition will grow healthy plants, maintain or improve the quality of water, reduce the impact of stormwater runoff on catchment areas and help remove pollution and waste from the environment.

Controls

C1. The retention and appropriate management of existing site soils for reuse is to be prioritised and recommended where appropriate on landscape plans. Any imported soil is to comply with *AS4419 Soils for landscaping and garden use.*

C2. Native topsoils and organic layers represent a component of the Blue Mountains biodiversity. Their conservation in situ and protection from inversion, burial or other loss is to be prioritised in all areas beyond the approved construction zone.

C3. All landscaped embankments having a slope of 1 horizontal to 3 vertical, or steeper, are to be stabilised to prevent erosion or slumping. Stabilisation techniques may include the use of dense ground covers, erosion control netting, mesh or fabric, coir logs or rock stabilisation.

C3.7. Plant material establishment and maintenance

Explanation

It is necessary to support the establishment of plant material. It is equally important to ensure that planted areas are adequately maintained if the approved landscape is to mature and satisfy the expected landscape outcomes.

C1. In order to ensure the establishment and growth of all plantings, the landscape plan is to include recommendations for the management or improvement of soil prior to planting, and the type of fertilisers which may be necessary to ensure optimal growth of species within the planting schedule.

C2. Following successful installation and establishment of approved landscape plantings, landscaped areas are to be adequately maintained (weeding, watering, spraying, pruning, re-fertilising) for the life of the development. Plants that die or are removed are to be replaced with the same species or similar species in an equivalent stage of growth, unless remaining plantings satisfactorily achieve the identified landscape objectives.

C3. Excessive use of fertilisers or pesticides in landscaped areas is to be avoided as it may lead to undesirable impacts including eutrophication of waterways, contamination of groundwater, release of greenhouse gases or unintentional loss of biodiversity.

C4. Netting and wire fencing used around any landscape elements or area is not to result in entanglement, entrapment or injury to wildlife. These measures are to be installed in accordance with current best practice guidelines.

Note: The Wildlife Friendly Fencing Project provides clear guidelines.

C3.8. Additional Zone Specific Controls

Explanation

Each land use zone listed under LEP 2015 has specific objectives for development, and a number of these relate to character, landscape and environmental outcomes. This section provides guidance and controls relating to the landscape character for specific land use zones, and identifies the level of detail required within the landscape plan.

Controls

C1. A landscape plan is required to support any development application within Zones **R1 General Residential, R2 Low Density Residential, R3 Medium Density Residential**; **E3 Productivity Support**; **E4 General Industrial**, **C3 Environmental Management**, **C4 Environmental Living**; **W1 Natural Waterways** and **SP3 Tourist** except for those developments which do not include external works, or are for minor additions and alterations.

Note 1: The following zone specific controls, in combination with the former general controls, and any development specific controls which are nominated elsewhere in this DCP, provide guidance relevant to the character, style and spatial extent of landscaping required within each land use area.

R1 General Residential, **R2 Low Density Residential** and **R3 Medium Density Residential**

C2. Sufficient landscape planting is to be included as an integral component of any development proposal, except for minor additions and alterations.

C3. Landscape style and plant selection is to recognise and maintain the distinct character of the relevant village location and is to maintain appropriate transitions to adjoining land uses.

C4. The established traditional streetscape character and heritage values are to be retained and enhanced by the use of appropriate landscape design and thoughtful species selection, whilst being mindful of issues such as solar access and the location of overhead and underground services.

C5. Given the invasive potential of certain traditional plant species, the use of alternative, non-invasive or low weed risk species capable of achieving the desired landscape outcomes is to be investigated for all new landscaping works.



Part C3 - Figure 5: Enhanced cultural landscape: Landscaping within heritage precincts and along major tourist routes should seek to retain and enhance traditional garden settings and promote interactive street frontages.

C6. Interactive street frontages are to be promoted through the use of tiered planting in preference to tall mono-specific hedges within the front setback. The planting palette in these areas should contain a suitable selection of tree, shrub and groundcover species of mixed form and colour to enhance the street front presentation, visual accessibility and local amenity.

C7. Where development is located on streets which represent major tourist routes, priority is to be given to the retention of healthy and structurally sound indigenous canopy trees or mature landscape trees which retain and frame view lines. The planting of tall hedges which may impede views is not supported.

C8. Street trees are to be provided at a ratio of 1 per 9 linear metres of site frontage.

Note: Species selection is to be in accordance with the guidelines provided within the Street Tree Master Plan. Refer to Part E8 Public Domain, of this DCP.

E3 Productivity Support and E4 General Industrial

C9. A landscape setting for industrial land uses is to be promoted, and wherever possible, existing bushland, environmentally sensitive features or other local character elements are to be retained and incorporated into the landscape setting.

C10. Dominant plant species selected for use should be non-invasive and long lived exotic or native species of moderate diversity, with low maintenance requirements. Plant material selected needs to be tolerant of higher intensity site use.

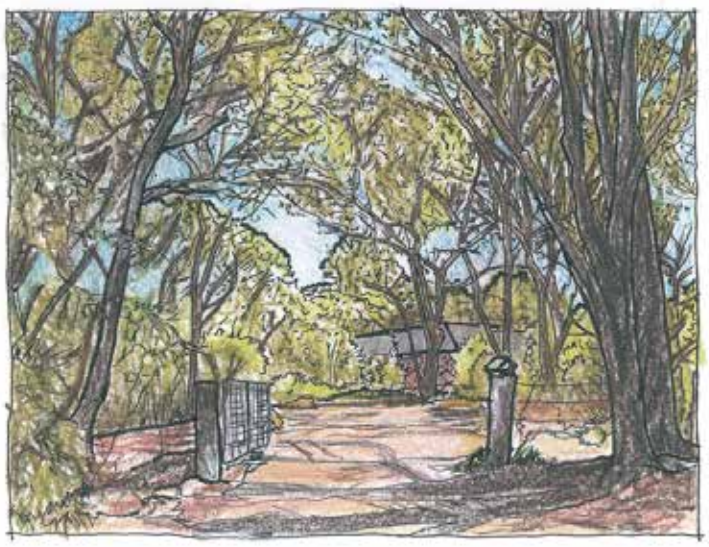
C11. Landscape elements proposed are to be consistent with any provided or required Water Sensitive Urban Design (WSUD) elements, and are not to result in any conflict in achieving water quality outcomes.

C12. Street trees are to be provided at a ratio of 1 per 9 linear metres of site frontage.

Note: Species selection is to be in accordance with the guidelines provided within the Street Tree Master Plan. Refer to Part E8 Public Domain, of this DCP.

C3 Environmental Management and **C4 Environmental Living**

C13. Landscape style is to reflect the characteristics of the locally indigenous vegetation and reinforce the bushland character and natural qualities of the environment. The use of single species or formal style hedges and other formal planting styles is discouraged.



Part C3 - Figure 6: Landscape settings in zone C3 Environmental Management: Landscape settings feature retained indigenous vegetation as a dominant element to minimise the visual impact of development.

C14. Within Zone **C3 Environmental Management** existing locally indigenous plant species are to be retained within the front setback to the maximum practical extent so as to maintain the bushland character and provide mitigation of the visual impact of the proposed development, subject to protecting life and property from bushfire.

C15. Within Zone **C4 Environmental Living**, existing locally indigenous plant species, where present, are to be retained and incorporated into site landscaping.

C16. Locally indigenous or other non-invasive native species are to be provided as the dominant component of new planting. As a guide, the following proportions are recommended:

(a) within Zone **C3 Environment Management**: 80%, and

(b) within Zone **C4 Environmental Living**: 60%.

Plant selection should prioritise species of high habitat value and low flammability.

C17. Within areas adjoining or overlooking the Blue Mountains National Park, existing locally indigenous vegetation and/or new landscaping is to be provided between the built form and the Park boundary, subject to protecting life and property from bushfire, so as to minimise the visual impact of development.

C18. Where natural features such as rock outcrops occur on site, these are to be protected and conserved for their amenity and habitat value.

C19. Where indigenous vegetation has been previously cleared or damaged, replanting of locally indigenous species is to be provided so as to emulate a bushland setting for the development, subject to the provision of any required Asset Protection Zone (APZ). New planting is to include species from all stratum (tree, shrub and ground layer) which may have naturally occurred in the locality.

Note: Additional environmental protection works may be required to facilitate the establishment of plantings. Refer to Part C2 Bushland and Weed Management.

C20. Should additional or complementary planting be required in areas identified as containing a *Significant Vegetation Community* or a threatened ecological community, only local provenance plant material is to be used.

C21. Plant species listed within Part C2.3 Weeds of the Blue Mountains of this DCP are not to be retained or used in any new landscaped areas. Existing weed infestations are to be systematically and progressively controlled.

Note: Refer to Part C2.2 for weed control requirements

C22. Landscaped areas are to contain good structural and plant diversity to provide a variety of habitat niches for a broad range of indigenous wildlife. Excavated bush rock should be reused in the landscape to create habitat complexity and assist in soil conservation.

C23. Within zone **C3 Environmental Management**, development for the purposes of horticulture is to be designed to minimise adverse impacts on adjoining natural areas through implementation of the following elements:

(a) ensuring that the planted area is adequately bunded to minimise impacts from fertilizers and irrigation, and

(b) providing buffer plantings which are nutrient tolerant and non-invasive around the perimeter of horticultural areas to limit nutrient leachate.

W1 Natural Waterways

C24. Only locally indigenous or where possible, local provenance plant material is to be used. Plant species are to be from all stratum which might naturally occur and are to be planted in a manner which protects and enhances the ecological function and value of the land and waterway.

C25. Existing environmental and priority weed infestations present within the riparian areas are to be systematically and progressively controlled using appropriate low impact techniques suited to the riparian environment.

Note: Refer to Part C2.2.1 for weed control requirements in environmentally sensitive land.

SP3 Tourist

C26. Within areas adjoining or overlooking Reserves or the Blue Mountains National Park, existing locally indigenous vegetation and/or new landscaping is to be provided between the built form and the Reserve or Park boundary, subject to protecting life and property from bushfire, so as to minimise the visual impact of development.

C27. The established traditional streetscape character and heritage values are to be retained and enhanced by the use of appropriate landscape design and thoughtful species selection, whilst being mindful of issues such as solar access and the location of overhead and underground services.

C28. Interactive street frontages are to be promoted through the use of tiered planting in preference to tall mono-specific hedges within the front setback. The planting palette in these areas should contain a suitable selection of tree, shrub and groundcover species of mixed form and colour to enhance the street front presentation, visual accessibility and local amenity.

C29. Where development is located on streets which represent major tourist routes, priority is to be given to the retention of healthy and structurally sound indigenous canopy trees or mature landscape trees which retain and frame view lines. The planting of tall hedges which may impede views is not supported.

C30. Dominant plant species selected for use should be non-invasive and long lived exotic or native species of moderate diversity, with low maintenance requirements. Plant material selected needs to be tolerant of higher intensity site use.

C31. Landscape elements proposed are to be consistent with any provided or required Water Sensitive Urban Design (WSUD) elements, and are not to result in any conflict in achieving water quality outcomes.

C32. Street trees are to be provided at a ratio of 1 per 9 linear metres of site frontage.

Note: Species selection is to be in accordance with the guidelines provided within the Street Tree Master Plan. Refer to Part E8 Public Domain, of this DCP.

PART C4
BUSHFIRE



Introduction

The Blue Mountains is highly bush fire prone due to a combination of climatic, vegetation and geographic factors. The City is also located within a World Heritage Estate, recognised for its unique landforms and ecology.

This combination of factors creates a challenging environment where the measures to address bush fire threat need to be balanced with the protection of a highly sensitive natural environment.

One of the core aims of the LEP 2015 is to ensure that development of bush fire prone land incorporates effective measures that protect human life, property and highly valued environmental and other assets from bush fire, without unacceptable environmental impacts.

It is not possible to prevent all damage and loss due to the unpredictable nature of bush fires. However the objectives and controls within this section of the DCP are intended to guide development towards measures which, in compliance with the relevant legislation, employ ecologically sustainable methods to lessen bush fire risk to that development.

This section applies to land identified on the Bush Fire Prone Land Map.

Submission requirements

For all applications over land mapped as bush fire prone, one or more of the following types of documents may need to accompany a development application:

- *Bushfire Threat Assessment:*
  - *For dwellings in accordance with section 4.14 of the Environmental Planning and Assessment Act 1979;*
  - *For subdivision and Special Fire Protection Purposes in accordance with section 100b of the Rural Fires Act 1997.*
- *Vegetation Management Plan*

C4.1. Legislative requirements for bush fire prone land

The following state level legislation and policy provides the requirements and provisions for the implementation of bush fire protection measures on bush fire prone land:

Environmental Planning and Assessment Act 1979 (EP&A Act)

Clause 4.14 of the EP&A Act requires that Council as the consent authority does not grant consent to any development, other than for the purposes of subdivision or development for a special fire protection purpose, unless it is satisfied that the development conforms to the specifications and requirements of *Planning for Bush fire Protection* 2019 (PBP) that are relevant to the development. The Council is also required to consult with the NSW Rural Fire Service concerning matters to be considered, with respect to the development, to protect persons, property and the environment from bushfire.

Development on bush fire prone land for the purposes of residential or rural residential subdivision or a special fire protection purpose is considered to be Integrated Development under clasue 4.46 of the EP&A Act. Council is required to refer these development applications to the NSW Rural Fire Service for their assessment and issue of a Bush Fire Safety Authority under Section 100B of the Rural Fires Act prior to the issue of consent.

Rural Fires Act 1997

Section 100B of the Rural Fires Act requires the issue of a Bush Fire Safety Authority by the NSW Rural Fire Service before development for the purposes of residential or rural residential subdivision, or development for special fire protection purposes can be consented to.

Special Fire Protection Purpose is defined under the Rural Fires Act 1997 and includes land uses such as a school, child care centre, hospital (including a hospital for the mentally ill or mentally disordered) and hotel, motel or other tourist accommodation. The definition also includes a building wholly or principally used as a home or other establishment for mentally incapacitated persons, seniors housing and a group home within the meaning of State Environmental Planning Policy (Housing) 2021, a retirement village and any other purpose prescribed by the regulations.

Note: Additional fees will also be required for referral to the RFS when lodging a development application.

Planning for Bushfire Protection 2019

The development controls and acceptable solutions for development on busfire prone land are contained within *Planning for Bushfire Protection 2019* (PBP). This guide is prepared by the NSW Rural Fire Service (RFS).

Compliance with PBP is required under section 4.14 of the EP&A Act, except for development for either residential or rural residential subdivision or development defined as a special fire protection purpose. These types of development are referred to the Rural Fire Service for assessment under the provisions of Planning for Bush Fire Protection.

10/50 Vegetation Clearing Code of Practice

The 10/50 Vegetation Clearing Code of Practice prepared by the Commissioner of the NSW Rural Fire Service, was released on 1 August 2014. The Code allows for the clearing on properties within the entitlement area, of trees within 10m and vegetation within 50m of an external wall of a building containing habitable rooms that comprises or is part of residential accommodation or a high-risk facility. There are certain limitations to this entitlement. Details on these limitations and the mapped entitlement area can be sourced from the Rural Fire Service website www.rfs.nsw.gov.au.

C4.2. Bush Fire Prone Land Map and Vegetation Categories

Bushfire prone land is land that is likely to be subject to bushfire attack. The Blue Mountains Bush Fire Prone Land Map, certified by the Rural Fire Service (RFS) in accordance with clause 10.3 of the *Environmental Planning and Assessment Act*, 1979, shows all land considered at risk from bushfire. The RFS identifies three vegetation types indicative of bushfire threat:

"Vegetation Category 1' is the most hazardous vegetation category, and refers to

forest, woodlands, heath and wetlands greater than 1 hectare in size.

'Vegetation Category 2' refers to moist forests, shrublands, open woodlands, mallee, grasslands, and pockets of Category 1 vegetation less than 1 hectare in size. 'Category 2' vegetation is at less risk from bush fire than Category 1 vegetation.

'Vegetation Category 3' refers to grasslands, freshwater wetlands, semi-arid woodlands, alpine complex and arid shrublands. 'Category 3' vegetation is a medium bushfire risk and higher risk than 'Category 2' vegetation but lower than 'Category 1' vegetation.

Land that directly adjoins bushland is also classified as 'Vegetation Buffer 100m & 30m'. These are the areas in which developments and people are most likely to be affected by a bushfire burning in the adjacent land.

The buffer area extends for a distance of 100m from the vegetation Category 1 areas and for a distance of 30m from the vegetation Category 2 and vegetation Category 3 areas.

C4.3. Development on Bush Fire Prone Land

Explanation

Large portions of the Blue Mountains are mapped as bush fire prone land. Therefore, development in these areas needs to take bush fire hazard into account when designing development, but balance this with the environmental constraints of the land. This could mean that the size, scale or location of development on the site may need to be adjusted to ensure that bush fire, environmental and cultural values are not compromised. The importance of site analysis and the careful planning of building location and design, landscaping, access and services is therefore underscored.

Objectives

O1. To balance conservation of the natural environment, with the protection of life and property from bushfire.

O2. To protect habitable buildings during the passage of any fire front by appropriate siting, design, choice of materials and construction.

O3. To contain asset protection zones and other bush fire protection measures within the boundaries of the site of the development that they are designed to protect, wherever possible.

O4. To enable the maintenance of bushfire protection measures and ongoing vegetation management for the life of the development by the owner or occupier.

O5. To ensure the protection and maintenance of biodiversity values, and the avoidance of unnecessary vegetation removal.

O6. To ensure that retained vegetation and introduced plantings are appropriate in bush fire prone areas.

O7. To place restrictions on the erection of buildings requiring special fire protection measures, including buildings whose occupants could be highly vulnerable to bush fires, buildings in which people work or assemble and buildings situated on sites where the proposed intensity of land use provides an unacceptable level of bushfire risk.

O8. To enable infill development that provides effective asset protection and that responds to the pattern of development within existing urban areas that are bushfire prone.

O9. To enable the defence of habitable buildings against bushfire attack by providing adequate water supplies, on-site access and safe access networks for firefighting purposes.

C4.4. Asset Protection Zones

Explanation

Asset Protection Zones (APZs) are one of the key *bushfire protection measures* outlined in *Planning for Bushfire Protection 2019* (PBP) and further detailed in the document Standards for Asset Protection Zones. The implementation of APZs, in combination with measures such as the construction standard and design of a building, the provision of adequate water supply and suitable landscaping, reduce the bushfire risk to life and property.

Asset Protection Zones generally fall into the following two categories:

- Inner Protection Areas (IPA)

  The IPA is located closest to the building or asset, is intended to provide defendable space and to manage heat intensities at the surface of the building.
- Outer Protection Areas (OPA)

  The OPA is located beyond the Inner Protection Area, and is intended to slow the fire, filter embers and reduce the likelihood of a crown fire.

The requirements and specifications for these types of APZs are contained in PBP.

Where APZs are required within bushland areas, it is important that vegetation modification (i.e. slashing, pruning, thinning and removal) is undertaken in a way which will comply with the relevant bushfire legislation, as well as the requirements for environmental protection under LEP 2015.

Controls

C1. Development is to balance the conservation of the natural environment with the protection of life and property from bushfire by:

(a) locating and designing developments and asset protection zones that minimise impacts on native flora and fauna and where possible, are located outside of environmentally sensitive areas, and

(b) considering measures to avoid and minimise environmental impact when implementing all appropriate *bushfire protection measures*, and

(c) utilising environmentally sensitive methods for fuel reduction in bushland areas.

Note: With reference to C1(b), these measures may include radiant heat shields; sprinklers; increased building construction Bushfire Attack Levels (BAL) or alternative development location

C2. The extent, nature, method and timeframes for vegetation modification required to establish and/or maintain the required asset protection zones is to be detailed and implemented in accordance with a *Vegetation Management Plan* (VMP) or as otherwise conditioned by Council in the development consent.

C3. Vegetation modification is to involve the minimum level of disturbance (i.e. slashing, pruning, thinning or removal) to existing vegetation required to comply with PBP and *Standards for Asset Protection Zones.*

C4. Vegetation modification is to maintain the natural vegetation character and habitat values of the site by retaining and/or allowing the regeneration of elements of each stratum (groundcover/ shrub/ canopy).

C5. To achieve the reduced fuel levels, any indigenous vegetation removal is to selectively target the removal of more flammable species and prioritise the retention of those species with lower flammability and high habitat value.

C6. Undertake vegetation modification using methods which result in minimal soil disturbance and minimal damage to adjacent bushland to be retained.

C7. Ensure retained shrub and tree clumps do not form a continuous canopy across the APZ and prune selected limbs to achieve both vertical and horizontal separation in preference to complete tree removal.

C8. Maintain at least 75% cover of appropriate ground layer vegetation, and up to 90% on steep slopes to reduce the potential for soil erosion and weed invasion.

C9. Where tree removal is required to achieve canopy separation:

(a) Select trees for removal based on their type, health and structure, and prioritise trees of poor quality and form, and

(b) Only remove trees where necessary to achieve a 15% canopy cover for an IPA or a 30% canopy cover for OPA, and

(c) Only remove trees where necessary to achieve a maximum canopy separation of two to five metres between clusters or islands of vegetation or between retained vegetation and built structures, and

(d) Aim to retain a variety of age classes of trees within the APZ to ensure sustainability of the canopy, and

(e) Retain tall, forest form trees with light or open canopies in small clumps or islands, where they are better able to resist wind , and

(f) Stumps may be ground or retained in situ to maintain soil stability, and

(g) Prioritise species with smooth or hard, persistent bark for retention over rough or ribbon barked species within the IPA, and

(h) To retain greater habitat complexity, species with rough, flaky or ribbon bark may be retained in small numbers within the APZ, but not directly adjacent to any asset, and

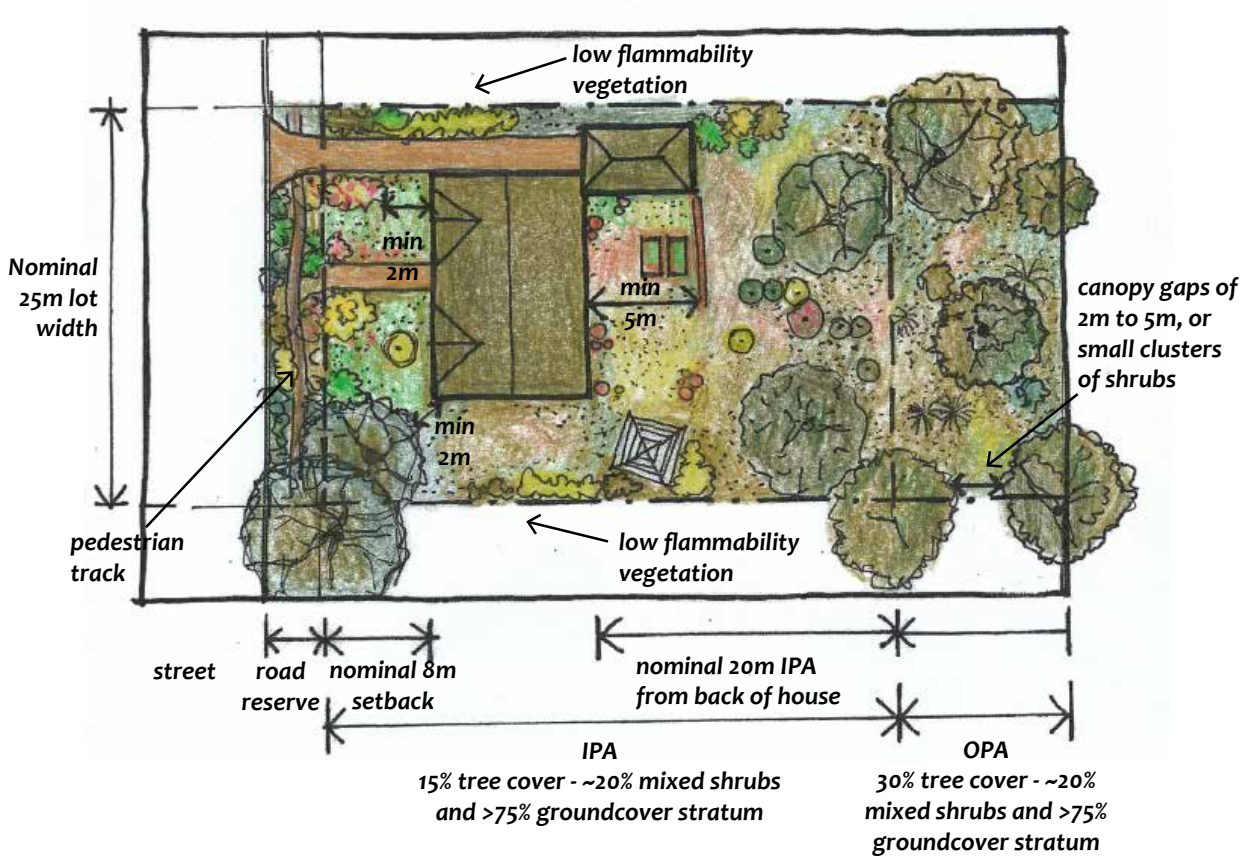
(i) Avoid removal of trees with hollows, or where there is evidence of fauna occupation. Where removal of an occupied tree is unavoidable, safe removal and relocation of fauna is to be undertaken by a licensed wildlife handler prior to carrying out any works.

C10. Weed removal is to be prioritised over indigenous vegetation removal within any APZ. Any priority or environmental weeds are to be removed to ensure regeneration of these weeds does not occur.

Note: For a list priority and environmental weeds refer to Part C2.3 Weeds of the Blue Mountains of this DCP.

C11. Where the APZ extends into degraded bushland, facilitate the restoration of cleared or degraded areas, using low impact bush regeneration methods or native revegetation using native species local to the area, within the limits of maintaining the required fuel levels.

C12. Employ strategies to ensure the maintenance of soil and slope stability on land with slope in excess of 20%.



Part C4 - Figure 1: Indicative diagram of the appropriate native vegetation retention, tree layout and plantings for a dwellings house with an Inner Protection Area (IPA) and an Outer Protection Area (OPA) in bushfire prone area.

C4.5. Plant species selection and landscape design in Bush Fire Prone Areas

Explanation

Bush fire protection often incorporates a number of different measures for any single property, including the implementation of Asset Protection Zones (APZs) and the construction standards of a building. Appropriate species selection for retention and planting on bush fire prone land can also contribute to lessening the bushfire risk to a property.

Two terms are often used when talking about the flammability characteristics of a plant, and these have very specific and quite different meanings.

Fire resistant is a term that describes plant species that survive being burnt and will regrow after a bushfire. They are resistant to being killed by a bushfire, but not to being burnt. Therefore, they may be highly flammable and inappropriate for a garden in a high bushfire risk area.

Fire retardant can also be misleading when referring to plants. It implies that a plant will not burn readily or may slow the passage of a fire. All plants will burn under the right conditions, however, certain species are less likely to ignite or burn as intensely as others, and many have growth characteristics which make them less likely to promote the spread of fire.

Poorly located vegetation that burns readily may expose a house to increased levels of radiant heat and flame contact.

Well-placed vegetation with low flammability may help protect houses by:

- reducing the amount of radiant heat received by a house, and
- reducing the chance of direct flame contact on a house, and
- reducing wind speed around a house, and
- deflecting and filtering embers, and
- reducing flammable landscaping materials within the defendable space.

This section provides guidelines for the planting and retention of both native and non-invasive exotic species on bush fire prone land based on current understanding, and emphasises basic garden design principles suitable for application in these areas.

The species selection provided is indicative only, and based on relevant guidelines and current theory. There are always exceptions to general plant selection guidelines. For example, Cherry Laurel (*Prunus laurocerasus*) a commonly used hedge plant in the Blue Mountains, and a environmental weed, is highly flammable, despite possessing large, dark green leaves with apparently low oil content.

Differences of opinion also surround the use of conifers in bushfire prone locations. Most contain high levels of resin, and have leaf and architectural characteristics which would indicate their unsuitability in bushfire prone locations, but there is both anecdotal evidence and research to suggest that this may not always be the case. The advice provided herein therefore provides guidance rather than guarantees, and ongoing research will continue to inform such advice.

C4.5.1. Location and Maintenance of Vegetation

The location, type and ongoing maintenance of vegetation within a property have a significant impact on the bushfire risk to any house. These factors can prevent the accumulation of debris and prevent the spread of fire towards a building.

When designing a new garden or modifying an existing one, carefully consider the placement of garden beds, trees and other vegetation to reduce the bushfire risk to the house.

When selected and located correctly, plants can contribute positively to the protection of assets. However, plants can also contribute to house loss by:

- providing a continual fuel path to the house, allowing direct flame contact, and
- dropping leaf litter on the ground, which readily ignites leading to ember attack, and
- dropping limbs or tree branches onto the house, and
- adding to fuel loads on or near the house, such as creepers over pergolas, fences or verandahs, and
- if located too close, producing radiant heat that may ignite the house or cause windows to break, allowing embers into the house, and
- acting as ladder fuel from the ground into tree canopies, increasing the intensity and potential spread of the fire.

Vegetation should always be kept clear of access to and from the house and property, and use driveways and paths to create separation between vegetation and the house.

Pathways, either formal or informal throughout the asset protection zone, also provide easy access to the area for regular maintenance and they provide separation between clusters of plantings or retained vegetation.

C4.5.2. Principles for suitable plant material selection

The following characteristics are important considerations when selecting trees or vegetation for retention in bushland areas of an asset protection zone (APZ), or for new plantings on bush fire prone land. Select plant material which has all or most of the following characteristics:

P1. Relatively high moisture content of leaves

However, this will vary in most plants according to the time of day, season, age of the leaf and the plant itself, location of the plant and exposure levels and environmental conditions, including soil moisture levels and air temperature. Some plants also loose moisture more readily than others under pre-ignition conditions.

P2. Low levels of volatile oils, waxes and resins in leaves

The leaves of plants containing significant amounts of oils, waxes and resins will often have a strong scent when crushed. For example Rosemary, Lavender and Tea Trees have oil in their foliage and pine trees can have high resin content.

P3. Large or hard leaves with simple margins

Small, thin and narrow leaves have a high surface-area-to-volume ratio, which may make them more susceptible to drying out, and more flammable. However, some small leaved native species have strategies to conserve moisture. The shape of leaves also influences how easily they are caught in vegetation when they fall off the plant.

P4. Plants that do not accumulate dead leaf and stem material below them or suspended in the foliage

If leaves are caught within plants it will increase that plant's flammability as leaf litter dries out and ignites readily. Dead pine needles are a good example of leaves that readily catch in other plants.

P5. Smooth, hard, or persistent bark, rather than flaky, papery or ribbon bark

Bark that is loose, stringy or fibrous will ignite easily and can break off to create burning embers that are carried ahead of a bushfire. These types of bark can also act as ladder fuels that carry fire into the canopy of a tree, increasing the intensity of the fire.

P6. Plants which have a reliable and fairly predictable size and shape

Alternatively, choose species which can easily be pruned to restrict their spread or ultimate size.

P7. Plants which may be periodically heavily pruned to reduce biomass ahead of the fire season, without needing to remove the whole plant

Many plants respond positively to regular pruning, and may flower more prolifically in subsequent years as a result.

P8. Avoid the use of vines

These typically add fuel directly to a structure or to supporting vegetation. They act as ladder fuels, bridging gaps between surface and canopy fuels.

P9. Trees with low leaf-litter production over summer

Ensure that trees which are selected do not she leaves under drought conditions.

C4.5.3. Landscape Maintenance

Regular maintenance actions are essential if the landscape is to provide positive benefits during the fire season:

- Clear ground fuel from underneath plants, on and around the house, and
- Prune plants with low-hanging branches, providing separation of at least 2 metres above the ground, and
- Replace plants that die or become diseased, and
- Maintain pathways to ensure access is available to all areas of the APZ, and
- Keep plants well hydrated through watering and mulch. Watering less frequently but for longer encourages the plants to develop deep roots reducing moisture loss during dry periods, and
- Replace or cover organic mulch such as woodchips, straw or dead plant matter with non-flammable mulches, and
- Remove other flammable objects from your defendable space, and
- Remove any fine, dead material that has accumulated in plants, and
- Remove weeds from the APZ as these often contribute to high fuel loads.

Part C4 - Table 1(a) - (e) provides guidance on some suitable species to be used within Asset Protection Zones.

Note: Links to PlantNet, Sydney Royal Botanic Garden, or Australian Native Plants Society are provided for the indigenous species listed, where images and descriptions are available.

Part C4 - Table 1(a): Native Species (Most are locally indigenous)

PROTEACEAE family: One of the most suitable groups of plants for use within the asset protection zone

Botanic Name	Common Name
Banksia marginata	Silver Banksia
Banksia serrata	Old-man Banksia, Wiriyagan (Cadigal)
Banksia spinulosa	Hairpin Banksia
Conospermum taxifolium	Variable Smoke-bush
Conospermum tenuifolium	Sprawling Smoke-bush, Slender Wire Lily
Grevillea buxifolia	Grey Spider Flower
Grevillea laurifolia	Laurel-leaf Grevillea
Grevillea linearifolia	Linear-leaf Grevillea
Grevillea mucronulata	Green Spider Flower
Grevillea sericea	Pink Spider Flower
Grevillea shiressii	
Grevillea speciosa	Red Spider Flower
Hakea dactyloides	Finger Hakea, Broad-leaved Hakea
Hakea gibbosa	Needlebush
Hakea sericea	Needlebush
Lambertia formosa	Mountain Devil
Lomatia ilicifolia	Holly Lomatia, Native Holly
Lomatia myricoides	River Lomatia, Long-leaf Lomatia
Lomatia silaifolia	Crinkle Bush, Fern-leaved Lomatia, Parsley Bush
Persoonia lanceolata	Lance Leaf Geebung
Persoonia laurina	Laurel Geebung
Persoonia levis	Broad-leaved Geebung
Stenocarpus salignus	Scrub Beefwood,
Telopea speciosissima	Waratah
Xylomelum pyriforme	Woody Pear

Part C4 - Table 1(b): Other suitable open forest species which provide good ground cover or low shrubbery

Botanic Name	Common Name
Acacia oxycedrus	Spike Wattle
Dampiera purpurea	Purple Dampiera
Dianella longifolia	Blue Flax-Lily, Blueberry Lily
Dianella prunina	Purple leaf Dianella
Epacris pulchella	Wallum Heath
Goodenia ovata	Hop Goodenia
Hibbertia aspera	Rough Guinea Flower
Hibbertia dentata	Trailing Guinea Flower
Hibbertia saligna	Twining Guinea Flower
Indigofera australis	Australian indigo
Pimelea linifolia	Slender Rice Flower
Pittosporum revolutum	Rough fruit Pittosporum
Pratia purpurascens	Whiteroot
Dichondra repens	Kidney Weed
Viola hederacea	Native Violet

Part C4 - Table 1(c): Other 'mesic' species (trees and shrubs) which perform well in the local area

Botanic Name	Common Name
Backhousia myrtifolia	Grey myrtle, Ironwood
Callicoma serratifolia	Black Wattle, Silver-leaf Butterwood
Ceratopetalum gummiferum	New South Wales Christmas Bush
Doryanthes excelsa	Giant Lily Gymea Lily
Doryphora sassafras	Sassafras
Elaeocarpus reticulatus	Blueberry Ash, Fairy Petticoats
Eucryphia moorei	Eastern Leatherwood, Plumwood
Ficus coronata	Sandpaper Fig
Ficus rubiginosa	Port Jackson Fig, Rusty Fig

Hymenosporum flavum	Native Frangipani
Myrsine howittiana (syn. *Rapanea howittiana*)	Brush Muttonwood
Myrsine variabilis (syn. *Rapanea variabilis*)	Variable Muttonwood
Tristania neriifolia	Water Gum
Tristaniopsis collina	Mountain Water Gum, Hill Kanuka
Tristaniopsis laurina	Water gum, Kanooka, Kanuka

Part C4 - Table 1(d): Other landscape species (exotic and cultivated native trees and taller shrubs) which may be suitable for bushfire prone locations:

Botanic Name	Common Name
Acmena smithii "Minor" cultivars	Lilli Pilli
Camellia japonica	Japanese Camellia
Camellia sasanqua	Sasanqua Camellia
Carpinus japonica	Japanese Hornbeam
Lagerstroemia indica cultivars	Crepe Myrtles
Liquidambar styraciflua	Sweet Gum
Liriodendron tulipifera	Tulip Tree
Magnolia species deciduous or evergreen	
Malus floribunda	Showy Crabapple
Photina species and cultivars	
Quercus rubra	Red Oak
Rhododendron cultivars	Rhododendron and Azalea
Tibouchina species and cultivars	Lasiandra
Viburnum davidii	Davids Viburnum

Part C4 - Table 1(e): Exotic and cultivated native low shrubs and ground covers

Botanic Name	Common Name
Abelia x grandiflora	Glossy Abelia
Ajuga reptans	Blue Bugle
Buxus sepervirens	(Dutch Box)
Ceanothus "Blue Pacific"	
Correa "Dusky Bells" and similar cultivars	
Daphne odora	(Common Daphne)
Escallonia (Pink Pixie)	
Gardenia species and cultivars	
Hebe oval leaf species and varieties	
Hydrangea varieties	
Myoporum parvifolium	Creeping Boobialla
Pieris japonica	(Lily-of-the-valley-bush)
Rosa cultivars	

PART C5
TREE AND VEGETATION PRESERVATION



Introduction

Collectively, all individual trees within both private and public lands, whether indigenous or introduced, form the urban forest of the Blue Mountains LGA. Urban forests provide a number of benefits, including influence on micro-climate, soil and water management and carbon storage, as well as cultural and aesthetic value.

Single trees within the urban forest and patches of indigenous vegetation may represent corridors through developed areas which link the wider forested landscape. These links contribute to biodiversity conservation and genetic exchange capabilities in a landscape. The conservation and appropriate management of these trees is therefore important.

This section is made in accordance with State Environmental Planning Policy (Biodiversity and Conservation) 2021 (the Biodiversity and Conservation SEPP) and prescribes the trees and vegetation to which the Biodiversity and Conservation SEPP and/or Clause 5.10 of Blue Mountains LEP applies and the applicable approval process.

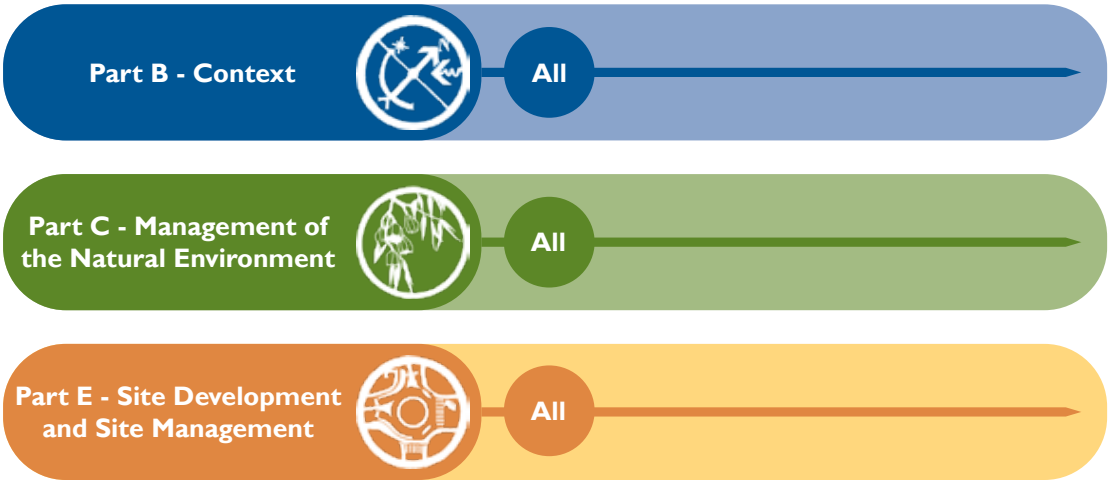
A permit is required to remove prescribed trees and vegetation.

Trees and other vegetation proposed for removal as a component of development are assessed via the development application process, and require more detailed assessment than is generally required for the purpose of securing a Tree Management Permit.

Note 1: This Part should be read in conjunction with the 10/50 Vegetation Clearing Code of Practice prepared by the Commissioner of the NSW Rural Fire Service, available at www.rfs.nsw.gov.au.

Note 2: In the assessment of applications for tree removal or pruning permits, the Council takes guidance from judgements delivered by the Land and Environment Court in matters heard under the Trees (Disputes Between Neighbours) Act 2006.

Read in conjunction wtih:



Submission requirements:

Applications for a permit to prune or remove prescribed trees or vegetation are to include the following supporting information:

- *Written consent from the tree or vegetation owner giving permission for the application to be lodged;*
- *A plan of the site sufficient to identify the trees or vegetation to be pruned or removed,*
- *Reasons for the proposed tree or vegetation pruning or removal;*
- *Information in support of the reasons for pruning or removal,*
- *Proposed replacement planting (where required)*

Relationship to other documents:

Where relevant, this part of the DCP should be read in conjunction with the following documents:

- *Blue Mountains City Council Street Tree Master Plan*
- *Australian Standard AS4373 – Pruning Of Amenity Trees*
- *Australian Standard AS4970 – Protection of Trees on Development Sites*

C5.1. Tree and vegetation preservation objectives

O1. To foster and develop the visual, cultural and landscape amenity of the Blue Mountains through the protection and enhancement of trees and other vegetation.

O2. To maintain and enhance the ecological value of Blue Mountains native and exotic canopy species and ensure their long-term survival.

O3. To improve Urban Forest Management to maximise the benefit to the community based on the principles of Ecologically Sustainable Development (ESD).

Note: Supporting documentation can be accessed via the Local Government Association website.

O4. To minimise the risk of injury to people or damage to property from trees through a formal assessment process.

O5. To ensure sufficient compensatory replanting is specified where appropriate and undertaken as a component of the tree permit approval process.

O6. To integrate the tree permit approval process, as far as practical, with the 10/50 Vegetation Clearing Code of Practice prepared by the Commissioner of the NSW Rural Fire Service

C5.2. Prescribed trees

Explanation

In accordance with the Biodiversity and Conservation SEPP, this part of the DCP prescribes trees which require development consent or a permit granted by the Council to undertake any of the activities identified by the SEPP.

Controls

C1. For the purposes of this DCP a prescribed tree is:

(a) a long lived woody perennial plant with one or relatively few main stems, being of any species whether indigenous, exotic or introduced and that is 4m or more in height and/or has a crown spread of more than 4m, or

(b) any mallee formed *Eucalyptus* species, irrespective of size, with such a tree having multiple leader form seldom exceeding five metres in height, originating from a lignotuber at or near ground level.

Note: Where land is subject to mapping contained within State Environmental Planning Policy (Biodiversity and Conservation) 2021, additional preservation controls may apply.

C5.3. Trees which are not prescribed (exempt)

Explanation

In accordance with the Biodiversity and Conservation SEPP, this part of the DCP describes trees which are not prescribed by this development control plan, and therefore do not require development consent or a permit granted by the Council to undertake any of the activities identified by the SEPP.

Controls

C1. Trees which are not prescribed (exempt) include:

(a) Any tree species listed in 'Weeds of the Blue Mountains' in Part C2.3 of this DCP, except:

i. if it exceeds the nominated height limitation in Part C2.3, or

ii. if it is located on any land to which Part D Heritage Management of this DCP applies.

(b) Any non-indigenous species located on a Council owned and maintained bushland reserve, irrespective of height, where a Council directed bushland restoration program is operational.

(c) Any tree which poses an imminent danger to human life, buildings or other property as a result of a bush fire burning in the vicinity of the land on which the tree is situated.

(d) Any tree that complies with the 10/50 Vegetation Clearing Code of Practice prepared by the Commissioner of the NSW Rural Fire Service.

(e) Trees that are dead or dying and are not required as the habitat of native animals or pose a risk to property or life (if not part of mapped biodiversity areas).

(f) Vegetation removal as part of traditional Aboriginal cultural activities.

Note: Where a tree is pruned or removed under the exemption provisions, evidence of the condition or characteristics of the tree is to be retained by the owner for a minimum of six months and presented to the Council upon request. That evidence may include:

- detailed photos of the tree; or
- where the photographs do not provide adequate or clear description of the circumstances, a declaration from a third party which corroborates the facts regarding the removal.

Note: All exempt vegetation must meet the documentation requirements specified in State Environmental Planning Policy (Biodiversity and Conservation) 2021

C5.4. Prescribed vegetation

Explanation

In accordance with the Biodiversity and Conservation SEPP, this part of the DCP prescribes vegetation which requires development consent or a permit granted by the Council to undertake any of the activities identified by the SEPP.

Controls

C1. For the purposes of this DCP vegetation is defined as:

(a) Vegetation that is either a remainder of the natural vegetation of the land or, if altered, is still representative of the structure and/or floristics of the natural vegetation, and

(b) Young age class and regenerating indigenous trees not of a size sufficient to be classed as a 'tree', plus shrub and ground layer vegetation.

C2. Prescribed Vegetation for the purposes of this DCP is all vegetation, except vegetation that is exempt under this DCP.

Note: Where land is subject to mapping contained within State Environmental Planning Policy (Biodiversity and Conservation) 2021, additional preservation controls may apply.

C5.5. Vegetation which is not prescribed (exempt)

Explanation

In accordance with the Biodiversity and Conservation SEPP, this part of the DCP describes vegetation which is not prescribed by this development control plan, and therefore does not require development consent or a permit granted by the Council to undertake any of the activities identified by the SEPP.

Controls

C1. Vegetation which is not prescribed includes:

(a) Any vegetation species listed in 'Weeds of the Blue Mountains' in Part C2.3, and

(b) Any introduced vegetation (but not trees), for the maintenance of an established exotic garden, and

(c) Native vegetation previously planted for agriculture, agroforestry, forestry, horticulture or woodlot purposes, and

(d) Regrowth native vegetation (but not trees) in areas previously cleared with consent, and

(e) Vegetation removed for the purpose of the ongoing maintenance of fuel load limits within approved asset protection zones, but only in accordance with the terms and conditions of any issued development consent, and

(f) Native vegetation removed or cleared in bushland for the purposes of creating gardens where:

i. the work does not involve the removal of trees, and

ii. the area to be cleared is within 35 metres of a dwelling for which consent or a building approval has been granted on the same allotment as the proposed clearing, and

iii. the area to be cleared is not within Environmentally Sensitive Land, and

iv. the work is undertaken in such a manner that it does not result in injury to trees which may result in their premature decline and/or death, and

v. the removal of native vegetation is not contrary to any condition of consent or any plan approved by the Council, and

vi. there is to be no planting of any species listed as a 'Weed of the Blue Mountains' in Part C2.3 of this DCP.

(g) Any vegetation that complies with the 10/50 Vegetation Clearing Code of Practice prepared by the Commissioner of the NSW Rural Fire Service, and

(h) Vegetation removal as part of traditional Aboriginal cultural activities.

Note: Where vegetation is pruned or removed under the exemption provisions, evidence of the condition or characteristics of the vegetation is to be retained by the owner for a minimum of six months and presented to the Council upon request. That evidence may include:

- detailed photos of the vegetation; or
- where the photographs do not provide adequate or clear description of the circumstances, a declaration from a third party which corroborates the facts regarding the removal.

Note: All exempt vegetation must meet the documentation requirements specified in State Environmental Planning Policy (Biodiversity and Conservation) 2021.

C5.6. Tree and vegetation preservation on heritage items and heritage conservation areas

Explanation

The Biodiversity and Conservation SEPP allows for 'activities of a minor nature or for the maintenance of the heritage item' to occur to trees and vegetation within a heritage item or a heritage conservation area, or to an Aboriginal object or within a place of Aboriginal significance. The following objectives and controls provide guidance on what works can be considered 'of a minor nature'. Note this refers to the minor nature of heritage impacts, rather than minor nature in terms of arboricultural impacts.

Where this clause does not or cannot apply, development consent is required, and reference should be made instead to Clause 5.10 Heritage Conservation of LEP 2015, and Part D Heritage of this DCP.

Objectives

O1. To preserve heritage significant landscapes, gardens, trees and vegetation, where these elements have significance in their own right, and/or contribute to the significance of heritage items, heritage conservation areas, Aboriginal objects or places of Aboriginal significance.

O2. To ensure a clear distinction is made between works of minor heritage significance and those that would require a development application.

Controls

C1. Works of a minor nature in terms of heritage impacts will generally need to be considered on a case-by-case basis under the Biodiversity and Conservation SEPP. Examples of works that may be considered of a minor nature include:

(a) Exotic trees and/or vegetation that have self-seeded (and thus do not form part of any garden or planting scheme);

(b) Trees and/or vegetation that are causing significant damage to heritage buildings, structures and/or elements (and thus some immediate adverse heritage impacts may be mitigated by greater heritage benefits in the long term);

(c) Trees and/or vegetation that are inappropriately located in terms of size or form (and thus longer-term heritage considerations must be taken into account).

Note: Replacement plantings may be required to maintain heritage values. Selection of replacement plantings may require advice from a heritage consultant.

C5.7. Requirements for tree removal

Explanation

Where a permit or development consent has been provided for the removal of a tree, any potential impacts on adjacent trees and the associated landscape are to be appropriately considered. This is both to ensure the safety and integrity of the trees which remain, as well as to mitigate any impacts on the streetscape as a result of the tree removal.

Controls

C1. Approved tree removal is to be undertaken in a manner which prevents significant damage to the root zone, trunk or major branch scaffold of any adjacent trees which are being retained.

C2. Trees (and tree stumps) are not to be pushed, pulled or mechanically extracted during tree removal adjacent to other trees, as doing so may cause an unacceptable level of root damage to trees that remain. Under these circumstances stumps may be ground to 100mm below ground level or retained in situ.

C3. Compensatory replanting following tree removal may be a condition of the permit, unless remaining trees and other vegetation satisfy any landscape outcomes relevant to the site. Replacement trees, capable of achieving a minimum height specified by the permit, will need to be provided under the following circumstances:

(a) where approval is granted for the removal of trees within the front set back to a dwelling, or

(b) where approval is granted for the removal of significant streetscape trees, or

(c) where approval is granted for the removal of trees which represent a component of a fauna corridor, or

(d) where approval is granted for the removal of trees which represent the last of the canopy components on a site, or

(e) where approval is granted for the removal of trees which have been damaged by landscape, service installation or other similar works.

C4. Replacement trees are to be planted and maintained in a healthy condition until they reach maturity. Plants that die, or become unviable due to poor condition, health, and/or architectural form are to be replaced with the same species and maintained as above.

C5. Replanting is to be completed within 12 months from the date of tree removal.

C6. Where vegetation removal impacts areas which are mapped as having high biodiversity value under State Environmental Planning Policy (Biodiversity and Conservation) 2021, applicants must provide a Biodiversity Development Assessment Report (BDAR) in accordance with the requirements of Part C1.2 of this DCP.

C5.8. Pruning

Explanation

This part of the DCP describes the circumstances where development consent or a permit granted by the Council is not required to prune prescribed trees and vegetation.

Controls

C1. Development consent or a permit granted by the Council is not required for the following pruning works to prescribed trees and vegetation:

(a) the removal of structurally failed storm-damaged branches, and

(b) 'deadwooding', being the removal of dead wood from the tree, provided the branch being removed is not a branch with hollows on a likely habitat tree, and

(c) the seasonal pruning of fruit trees for the purposes of maintaining fruit production. This exemption does not apply to the following native species *Acmena* spp. or *Syzgium* spp. (all Lilli Pilli), nor to *Elaeocarpus* spp. (all Blueberry Ash) *Ficus coronata* (Sandpaper Fig) *Ficus rubiginosa* (Rusty or Port Jackson Fig) or *Macadamia* spp. (Macadamia nut), and

(d) selective pruning, being only pruning to remove branches back to the nearest branch collar or junction to clear a roof, where trees directly overhang the roof of a dwelling, garage or commercial building, and only if the owner of the land where the centre of the tree originated or where the majority of the tree is growing, is in agreement, and

Note: Where the tree owner does not consent to the pruning works, the disagreement remains a civil matter between neighbours. Civil matters may be resolved through mediation services independent of Council or through *Trees (Disputes Between Neighbours) Act 2006* proceedings initiated by the aggrieved neighbour.

(e) target pruning of branches overhanging public pathways from trees originating on private property, to a maximum height of 2.5m above natural ground level.

C2. Any pruning that complies with the 10/50 Vegetation Clearing Code of Practice prepared by the Commissioner of the NSW Rural Fire Service, on land to which that code applies.

C3. Pruning exemptions are valid only if conducted in compliance with Australian Standard AS 4373.

C4. All pruning and shaping of trees undertaken in accordance with the provisions of this exemption, is to ensure that the structural and aesthetic integrity of the tree is maintained.

PART C6
WATER MANAGEMENT



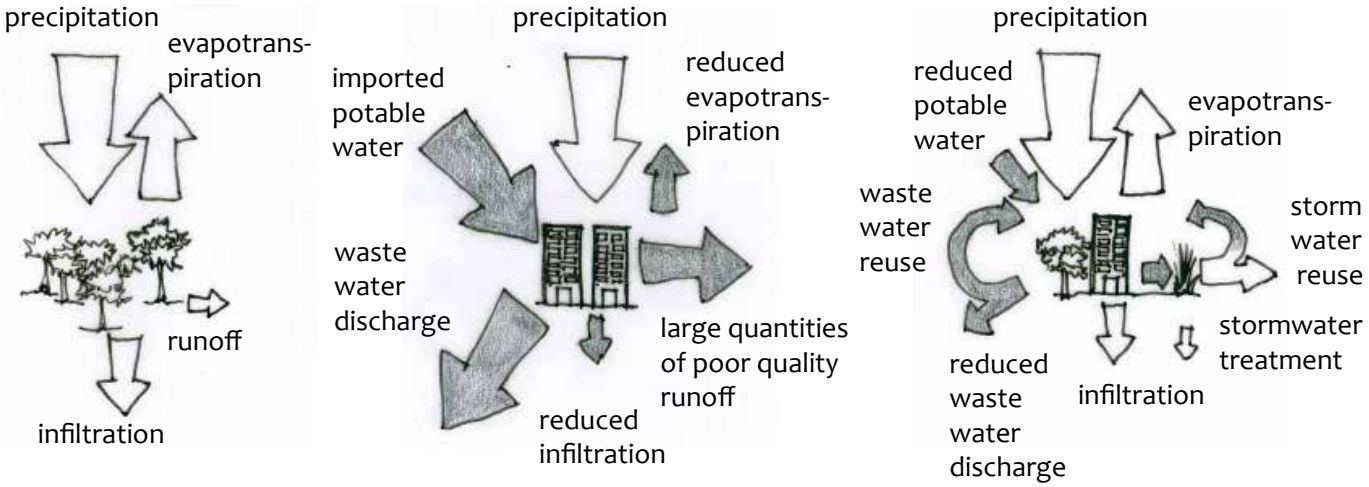
Introduction

Stormwater runoff from the impervious surfaces created by development has the potential to cause adverse impacts to the receiving natural environment and the natural water cycle, as a result of increased runoff volumes and peaks, reduced infiltration and reduced water quality. The appropriate treatment of stormwater is therefore important, and particularly so in the Blue Mountains due to its location adjacent to World Heritage listed national park, and partly within the Sydney drinking water catchment.

Water sensitive urban design (WSUD) seeks to restore the natural water balance within the urban environment by promoting the retention and reuse, evapotranspiration, and infiltration of rainfall, and conveyance of stormwater runoff within systems that enable natural filtering and cleaning processes (*Part C6 - Figure 1*).

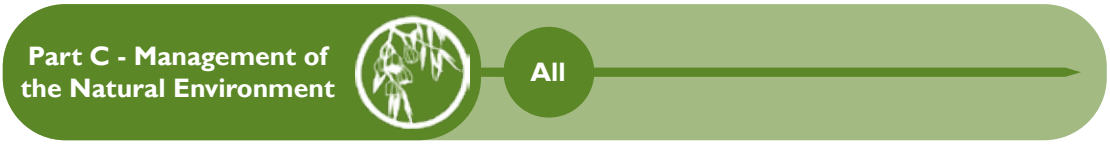
With the practice of water sensitive urban design, stormwater management measures can be undertaken in a manner that protects natural hydrological processes, conserves biodiversity and is integrated within landscape outcomes to achieve improved urban amenity.

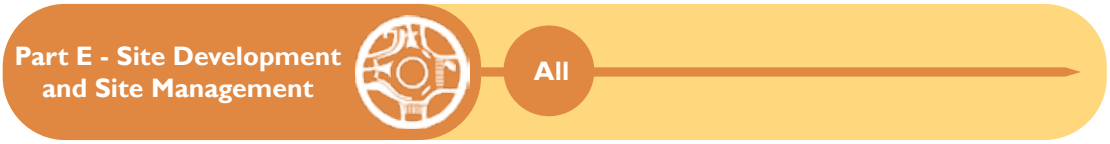
The consideration of all opportunities and constraints for the cycling of water through the urban environment at the early planning stage will provide for an integrated approach to best practice water cycle management.



Part C6 - Figure 1: *The urban water cycle showing changes to the natural water cycle with traditional urban development and with WSUD. Source: Water by Design (2009) Concept Design Guidelines for Water Sensitive Urban Design Version 1, South East Queensland Healthy Waterways Partnership, Brisbane, March 2009)*

Read in conjunction with:





Submission Requirements

One or more of the following types of reports may need to accompany a development application:

Small-scale developments

- *Concept Stormwater Management Plan*

Supporting documentation including:

- *Proposed and existing impervious area calculation*
- *Small scale stormwater quality model (s3qm) certificate (if applicable)*

Large-scale developments

- *Concept Stormwater Management Plan*
- *Water Sensitive Urban Design (WSUD) strategy*

Supporting documentation including:

- *Proposed and existing impervious area calculation*
- *Pre and post development water quality analysis*
- *Geo-technical investigation*

All development within the Sydney drinking water catchment

- *WaterNSW information requirements to demonstrate a neutral or beneficial effect on water quality*

C6.1. Water Sensitive Urban Design and Stormwater Management

C6.1.1. Key principles and design outcome objectives for all development

Explanation

The location, topography and environmental sensitivity of the Blue Mountains are such that the measures for water conservation and reuse, and the treatment of stormwater are particularly important. A lack of appropriate measures can result in adverse impacts within and beyond the development site.

This part of the DCP details Water Sensitive Urban Design (WSUD) guidelines for all development, to minimise any adverse impacts of stormwater on the natural environment through the implementation of best practice principles and techniques. Different types of development will require different solutions. The following objectives and controls detail the requirements for all development, appropriate for the scale and potential impact of that development proposal.

Objectives

The following objectives are applicable to all forms of development, where any water conservation and reuse or stormwater treatment is required or proposed to be modified.

O1. To ensure that water management measures for development incorporate key principles of WSUD, being to:

(a) protect existing hydrological and ecological processes of natural features and systems including watercourses, wetlands, lagoons and aquatic, riparian and groundwater dependant ecosystems; and

(b) maintain the natural hydrologic behaviour of catchments, and

(c) protect the water quality of surface and ground waters, and

(d) minimise demand on reticulated water supply system, and

(e) minimise wastewater impacts to the natural environment, and

(f) integrate water into the landscape to enhance ecological, visual, social, economic and cultural values.

O2. To integrate the management of all parts of the urban water cycle (rainwater, groundwater, potable water, stormwater, wastewater, grey water) in a sustainable way.

O3. To promote the reuse of rainwater and promote the harvesting and reuse of stormwater where appropriate.

O4. To provide a stormwater management system that implements a best practice WSUD strategy to treat the range of expected nutrients and pollutants for the proposed development.

O5. To maximise opportunities for absorption, infiltration and evapotranspiration to reduce runoff and promote groundwater recharge.

O6. Implement stormwater management devices, including outlets that are appropriate for the site and environmental conditions, in suitable locations and with adequate measures to protect the natural environment.

O7. To provide improved visual, recreational and urban amenity and integrate the landscaping design with water cycle management measures.

O8. To provide feasible and cost effective operation, access and maintenance for all devices comprising the stormwater management system.

O9. To ensure that development in the Sydney drinking water catchment demonstrates a neutral or beneficial effect on water quality.

C6.1.2. Application of requirements for Water Sensitive Urban Design

Explanation

The scale and type of development contribute to the potential for stormwater impacts. This therefore leads to the requirement for different levels of assessment and treatment, to ensure that the site, adjoining land and the broader receiving environment are adequately protected.

Development within the Sydney drinking water catchment is also regulated by Water NSW (formerly the Sydney Catchment Authority) and must have a neutral or beneficial effect on water quality in accordance with the requirements of the State Environmental Planning Policy (Sydney Drinking Water Catchment) 2011.

All developments within the drinking water catchment must provide the information requirements outlined within the Water NSW publication "Developments in the Sydney drinking water catchment – water quality information requirements" and refer to the Neutral or Beneficial Effect on Water Quality Assessment Guideline, both found at the following link:

www.waternsw.com.au/water-services/catchment-protection/building-and-development

C1. The controls in this part of the DCP apply to all types of development, including additions and alterations, demolition and rebuild, and new development on vacant land.

C2. All applications are to submit a *Concept Stormwater Management Plan* to demonstrate compliance with the relevant WSUD and stormwater management provisions.

C3. Small scale development includes:

(a) all single dwelling and dual occupancy developments, or

(b) where the total impervious area of the development is less than 1000m^2.

C4. Small scale developments are to provide appropriate stormwater quality treatment and quantity control devices to provide water conservation and re-use, detention, retention, filtering, infiltration and disposal in accordance with the small scale development provisions at Part C6.1.3 of this DCP.

C5. Large scale development is development that exceeds any of the provisions in control C3 above.

C6. Large scale developments are to provide appropriate stormwater quality treatment and quantity control devices to provide water conservation and re-use, detention, retention, filtering, infiltration and disposal in accordance with the large scale development provisions at Part C6.1.4 of this DCP.

C6.1.3. Water Sensitive Urban Design for Small Scale Development

Controls

Water Conservation and Re-use

C1. Maximise on-site retention and re-use of stormwater runoff through measures such as rainwater tanks, dual plumbing and accredited grey water use.

Note: Any proposed grey water treatment is to utilise a system that has been accredited by NSW Health and approved by Council.

C2. Rainwater tanks shall:

(a) have a collection capacity of not less than 4,000 litres per 100m^2 of proposed additional impervious area, for the reuse of water drained from those surfaces, for development within zones C3 Environmental Management and C4 Environmental Living, and

(b) have a collection capacity of not less than 1000 litres per 100m^2 of proposed additional impervious area, for the re-use of water drained from those surfaces for development within other zones, and

Note: If the development is subject to other rainwater tank capacity requirements to comply with the BASIX Building Sustainability Index or NSW Rural Fire Service conditions for bushfire protection, then the largest of all applicable rainwater tank capacity requirements is to be provided.

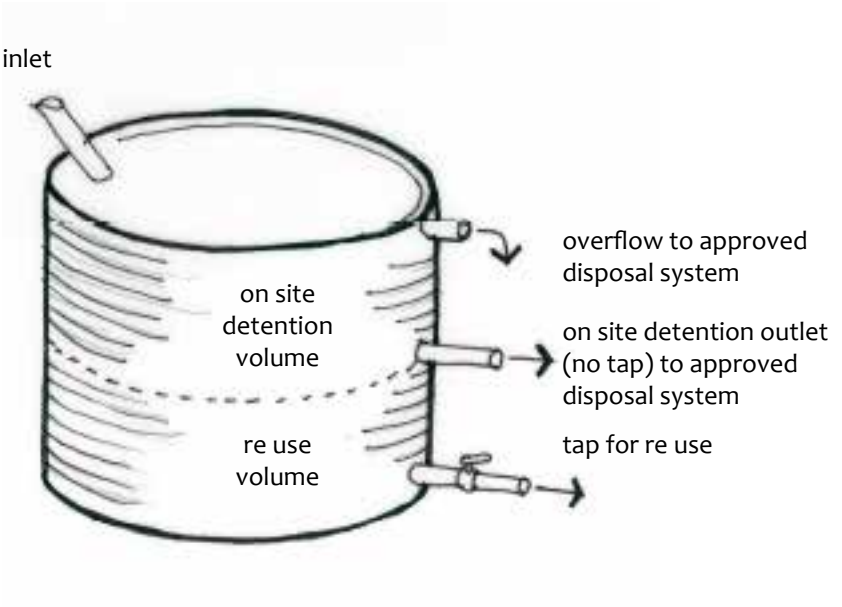
(c) include a first flush device to maximise water quality. The first flush device is to divert the first 1mm of roof water and include an outlet for drainage of the first flush device between rain events. The first flush water is to be directed to a stable vegetated area, rain garden or other absorption or infiltration measure, and

(d) be located and designed to maximise the percentage of roof area directed to the tank(s), and

(e) ensure overflows are drained to an adequately sized absorption or infiltration device, rain garden or bio-retention system or equivalent device, and

(f) incorporate additional storage volume for (temporary) on-site detention of stormwater where required.

C3. Where rainwater tanks are used for retention and reuse, as well as on-site detention system, the tank shall include an outlet (without a tap) to ensure that the on-site detention volume is available for the next storm. Refer to *Part C6 - Figure 2*.



Part C6 - Figure 2: Combined re-use and detention tank.

C4. Industrial and commercial developments that are not subject to the requirements of the BASIX Building Sustainability Index are to:

(a) demonstrate that consideration has been given to the interaction of all elements of the urban water cycle (potable water, stormwater and wastewater), and

(b) identify how each water source is appropriate for its end use, and

(c) ensure any water use fittings demonstrate compliance with the minimum standards defined by Water Efficiency Labelling and Standards (WELS) Scheme, and

(d) provide alternative sources (e.g. rainwater, stormwater, recycled water, grey water) to meet demand for non-potable uses where practicable.

Stormwater Design

C5. All stormwater management measures are to be located within the site boundaries and are not to result in impacts on adjoining properties.

C6. Minimise hydrologic impacts by:

(a) ensuring that the design, location and function of stormwater management devices protects, restores or replicates the natural water cycle for the site, and

(b) maintaining pre-development surface and groundwater regimes, run-off volumes and natural flow velocities, rates, peaks, and durations, and

(c) maintaining the sites natural topography, landforms, catchment areas, drainage patterns and flow paths by minimising cut and fill and avoiding redirection of surface flows, and

(d) retaining pervious areas and preserving the natural interactions between surface and groundwater, including soil permeability, groundwater seepage and recharge areas, and

(e) minimising the size of impervious areas and the amount of runoff generated from the site, and

(f) maximising opportunities to provide on-site absorption and infiltration where site conditions are suitable, and

(g) locating stormwater treatment devices to maximise the area of impervious areas that are treated.

C7. Minimise environmental impacts by:

(a) ensuring that all stormwater management measures and water sensitive urban design devices do not extend below the groundwater table or into bedrock and are appropriate for the given soil permeability and depth, and

(b) locating devices wholly outside of environmentally sensitive areas and existing vegetated areas, unless no alternative is available and it is demonstrated that minimal adverse impact will result, and

(c) designing stormwater treatment measures to cater for high flows above the devices' design size and locating them outside of the 1:100 year flow path to prevent the release of previously trapped sediment or pollutants by high flows or flood waters, and avoid damage to vegetation components, and

(d) keeping infiltration devices and rain gardens or bio-retention systems separate from on-site effluent disposal/management areas, and

(e) ensuring that developments with potential to generate pollutants such as fuels, heavy metals, oils, solvents, detergents, fertilisers, pesticides and the like, demonstrate appropriate containment and operational methods (e.g. bunding, first flush systems, dedicated vehicle washing bay), and provide appropriate water quality treatment and management measures to mitigate impacts of stormwater runoff. Such developments are to provide contingency management measures in the event of an accidental spill, and

Note: Environment pollution licenses may apply under the Protection of the Environment Operations Act, 1997.

(f) providing disposal, scour protection and dissipation mechanisms compatible with protecting the natural environment, by:

i. preferentially disposing of stormwater runoff within the site boundaries by re-use, absorption, and infiltration or evapotranspiration measures to the maximum extent that the expected site usage and site conditions allow, and

ii. minimising the area of impervious areas directly connected by piped or sealed drainage infrastructure to the receiving natural environment, and

iii. providing level spreading overflows to stable vegetated areas, and

iv. minimising the number of concentrated stormwater outlets, and

v. providing scour protection using environmentally sensitive, soft-engineered methods such as a rough and permeable rock riprap surface combined with native revegetation, rather than gabions, concrete surfaces or concrete grouting, and

vi. providing grassed filter strips or vegetated buffer areas to protect the receiving natural environment from stormwater runoff, and

vii. disposing of stormwater design overflows from absorption, infiltration, rain garden or bio-retention devices, which may not be otherwise disposed of within the site, to the street gutter where the site falls to the street, and

viii. disposing of stormwater design overflows from absorption, infiltration, rain garden or bio-retention devices via a level spreading overflow to a stable vegetated area, and any other measures necessary to protect the receiving natural environment where the site falls away from the street and towards bushland or natural areas, and

ix. disposing of stormwater design overflows from absorption, infiltration, rain garden or bio-retention devices to an interallotment drainage system where the site falls away from the street and towards areas that are not bushland or natural areas, and

Note: See Part C6.5 regarding connections to interallotment drainage systems and alternate methods of stormwater disposal.

x. not piping or otherwise discharging stormwater directly to a watercourse unless no alternative is available.

Stormwater Infiltration

C8. To provide appropriate stormwater infiltration:

(a) stormwater runoff from proposed impervious areas less than 50m^2 in area may be directed to adjacent stable vegetated areas via single cross fall design or other appropriate method to distribute flows evenly and allow on-site absorption, infiltration and dissipation, and

Note: Stormwater runoff from impervious areas less than 50m^2 as referred to above, may be a sub-catchment of a larger impervious area.

(b) stormwater runoff from proposed impervious areas greater than or equal to 50m^2 in area and all rain tank overflows are to be absorbed and infiltrated, to the maximum extent possible, via an appropriately designed and adequately sized device(s) such as infiltration trench, absorption trench or pit, proprietary infiltration tunnel, an un-lined rain garden or bio-retention system or equivalent device, and

Note: infiltration may be achieved within an appropriately designed water quality treatment device such as an un-lined rain garden or bio-retention system (if required) rather than providing two separate devices.

(c) all absorption and infiltration devices are to be located at least 4m from any property boundary, retaining wall or building, and

(d) for single dwellings and dual occupancies, absorption trenches shall measure at least 600mm wide x 600mm deep x 2m long for each down pipe, and

(e) all absorption and infiltration devices are to provide suitable collection and disposal of design overflows to a stormwater drainage system (street gutter, inter-allotment drainage system or drainage easement), unless these overflows can be otherwise safely disposed of within the site.

Stormwater Quality

C9. To ensure the quality of surface or groundwater is not reduced in the short or long term, all single dwelling or dual occupancy developments that exceed the maximum site coverage or that have less than the minimum pervious area provisions in Part B2.4 of the DCP, are to provide one of the following measures:

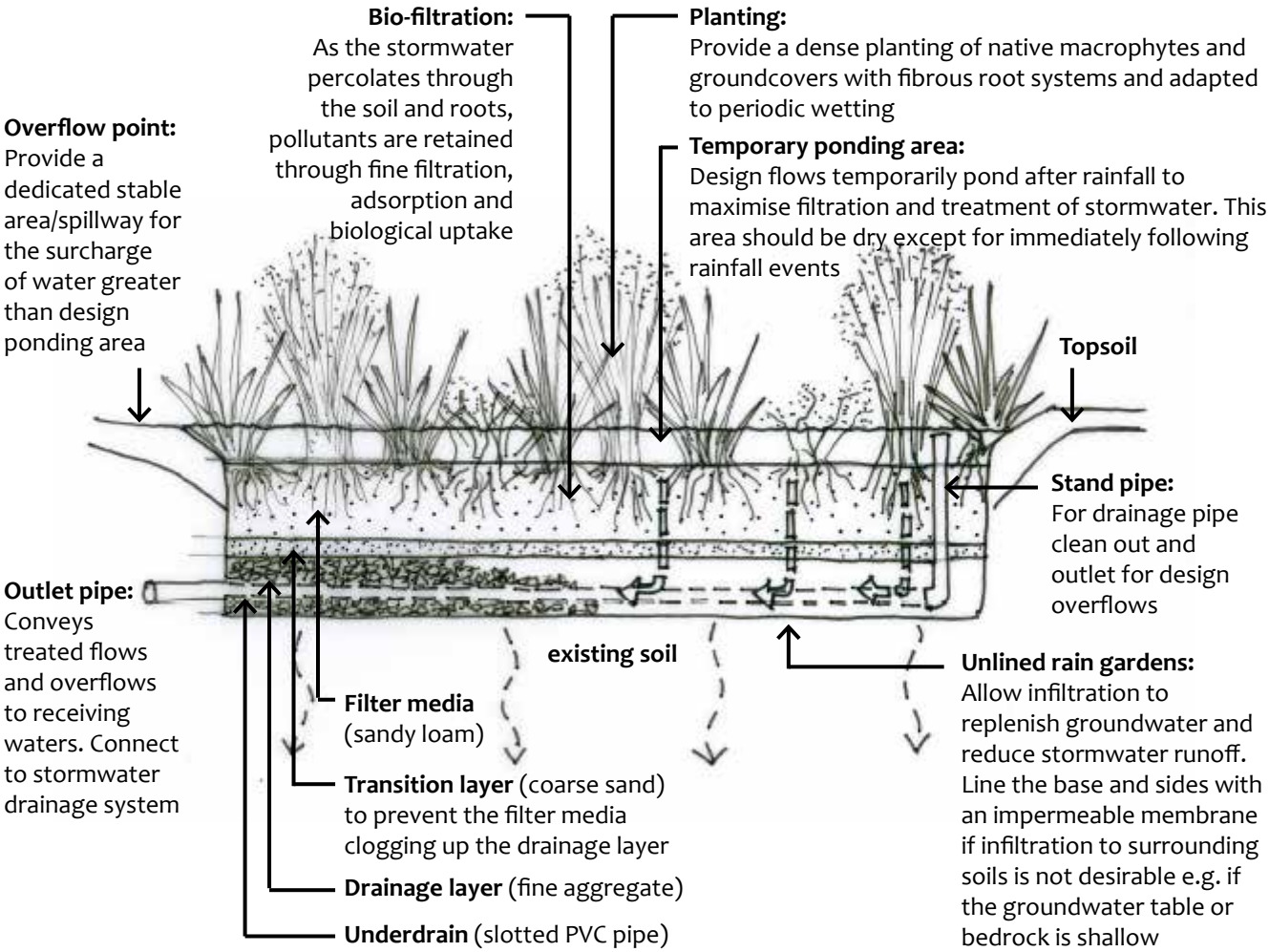
(a) a rain garden or bio-retention system with the filter surface area sized to be at least 2% of the total proposed increase in impervious area, or

Note: A rain garden is comprised of a soil based filter (specified sandy loam material), with a sand transition layer and gravel trench underneath that contains a slotted drain pipe (ag pipe) that is connected to the drainage system. A typical system (looking through the side of the filter) is shown below.

(b) a stormwater treatment train certified by the WaterNSW Small Scale Stormwater Quality Model (s3qm) that demonstrates the achievement of a neutral or beneficial effect on water quality (NorBE), or

Note: A certificate from an assessment completed using s3qm must be submitted with these proposals. The s3qm model is available at www.s3qm.com.au and information about s3qm is available at the WaterNSW website www.waternsw.com.au.

(c) an alternative solution with equivalent stormwater quality treatment performance outcomes designed and certified by a suitably qualified person and that demonstrates the achievement of NorBE.



Note: all development within the Sydney drinking water catchment is required by the State Environmental Planning Policy (Biodiversity and Conservation) 2021 to achieve a NorBE (www.waternsw.com.au/water-services/catchment-protection/building-and-development).

Part C6 - Figure 3: Typical rain garden system.

C10. To ensure the quality of surface or groundwater is not reduced in the short or long term, all small scale developments (other than single dwellings and dual occupancies) where there is a proposed increase in impervious area of 250m^2 or more, are to provide either one of the measures referred to in C9 (b) or C9 (c).

Landscape, Visual and Social Amenity

C11. The development and its associated water sensitive urban design measures are to enhance the landscape, visual, social and recreational amenity by:

(a) incorporating stormwater treatment and control measures into the urban and landscape design where appropriate, and

(b) incorporating local indigenous plant species into vegetated stormwater management devices to enhance the natural attributes of the Blue Mountains local government area, and

(c) ensuring the location and design of stormwater treatment and control devices do not compromise the achievement of landscape and streetscape outcomes and the protection of natural features, and

(d) maximising the retention and restoration of vegetated areas and deep-rooted trees to maintain the water table conditions, stabilise the soil, decrease runoff volumes and velocities, capture sediment and filter nutrients.

Operation and Maintenance

C12. Provide for the ongoing operation and maintenance of WSUD devices by locating and designing devices with appropriate access for all maintenance activities necessary to ensure their effective operation to meet the required performance and treatment targets over the life of the development.

Note: The designer of proposed stormwater treatment measures should prepare a maintenance schedule for each measure.

C6.1.4. Water Sensitive Urban Design for Large Scale Development

Explanation

Large scale development includes any development other than single dwellings and dual occupancies, where the total impervious area of the development is equal to, or greater than, 1000m^2.

Controls

Water Sensitive Urban Design Strategy

C1. A WSUD strategy is to be submitted with the development application for all large scale development to provide for water conservation and re-use, on-site detention, retention, water quality treatment, infiltration and safe disposal to the receiving natural environment.

The WSUD strategy is to demonstrate that the proposed development will achieve the water sensitive urban design objectives.

Note 1: Applicants and developers are to employ the services of appropriately qualified and experienced practitioners for the development of appropriate WSUD plans and strategies.

Note 2: Prior to lodgement of a development application, the applicant is encouraged to submit a pre-lodgement application with Council to seek advice regarding stormwater and WSUD requirements, particularly for large or complex development proposals.

Guidelines

C2. The WSUD strategy and stormwater management system is to be appropriately designed and sized in a manner consistent with the latest guidelines and current best management practices for WSUD.

Water Conservation and Re-use

C3. Demonstrate that consideration has been given to the interaction of all elements of the urban water cycle (potable water, stormwater and wastewater).

C4. Identify and implement all opportunities for on-site retention, water conservation, harvesting and re-use through measures such as rainwater tanks and dual plumbing and accredited grey water use.

Note: Any proposed grey water treatment is to utilise a system that has been accredited by NSW Health and approved by Council.

C5. Provide alternative sources to meet demand for non-potable uses where practicable, such as recycled water, roof water and harvested stormwater for vehicle washing, irrigation, water features.

C6. The water source used for a particular end use should reflect the quality required for that end use (i.e. be fit for purpose).

C7. Ensure any water use fittings demonstrate compliance with the minimum standards defined by Water Efficiency Labelling and Standards (WELS) Scheme.

C8. Rainwater tanks shall:

(a) for development on land zoned C3 Environmental Management and C4 Environmental Living have a collection capacity of not less than 4,000 litres per 100m^2 square metres of proposed impervious area, for the reuse of water drained from those surfaces, and

(b) for development on land in other zones have a collection capacity of not less than 1,000 litres per 100m^2 of proposed impervious area, for the re-use of water drained from surfaces, and

(c) provide pre-screening of organic matter (leaf litter) prior to the collection of water in the rainwater tank, and

(d) include a first flush device to maximise water quality. The first flush device is to divert the first 1mm of roof water and contain an outlet for drainage of the first flush device between rain events. The first flush water is to be directed to a stable vegetated area, rain garden or other absorption or infiltration measure, and

(e) be located and designed to maximise the percentage of roof area directed to the tank(s), and

(f) ensure overflows are drained to an adequately sized absorption or infiltration device, rain garden or bio-retention system or equivalent device.

Stormwater Treatment Performance Targets

C9. The development must demonstrate the achievement of the following stormwater treatment performance targets.

(a) Water quality

The WSUD strategy and stormwater management system is to provide appropriate water quality treatment to ensure that the quality of surface or ground water leaving the site is not to be reduced in the short or long term. The development must achieve a neutral or beneficial effect (NorBE) on water quality.

The achievement of this performance objective is to be demonstrated by using the Model for Urban Stormwater Improvement Conceptualisation (MUSIC) developed by the Cooperative Research Centre for Catchment Hydrology (now eWater) or an equivalent industry standard water quality model.

For development within the Sydney drinking water catchment, the WaterNSW (2023) 'Using MUSIC in the Sydney Drinking Water Catchment' guideline applies.

(b) Water quantity

The WSUD strategy and stormwater management system is to take all reasonable management actions appropriate for the site conditions to minimise adverse hydrologic impacts by:

i. ensuring that the design, location and function of stormwater management devices protects, restores or replicates the natural water cycle for the site, and

ii. maintaining pre-development surface and groundwater regimes, run-off volumes and natural flow velocities, rates, peaks, and durations, and

iii. maintaining the sites natural topography, landforms, catchment areas, drainage patterns and flow paths by minimising cut and fill and avoiding redirection of surface flows, and

iv. retaining pervious areas and preserving the natural interactions between surface and groundwater, including soil permeability, groundwater seepage and recharge areas, and

v. maintaining the physical nature of pre-development flows into receiving environments (i.e. where the receiving environment receives dispersed flows, concentration of flows is to be avoided), and

vi. reducing runoff volumes and preserving natural surface and groundwater hydrologic regimes by incorporating the following measures into the WSUD strategy:

- identifying and implementing all opportunities for water conservation, harvesting and re-use, and
- minimising the size of impervious areas and the amount of runoff generated from the site, and
- locating stormwater quantity control devices to maximise the area of impervious areas that are treated, and
- maximising on-site absorption, infiltration and evapotranspiration by:
  - » maximising the provision of permeable surfaces, and
  - » minimising the size of impervious areas that are directly connected by piped or sealed drainage infrastructure to the receiving environment by directing runoff to infiltration and evapotranspiration mechanisms or to other grassed, vegetated or landscaped areas designed to accept these flows.

Environmental and Site Constraints and Opportunities

C10. The stormwater management system is to be appropriate for the given site conditions and address the following potential environmental and site constraints:

(a) the soil depth, character and permeability/hydraulic conductivities, and

(b) the presence of rock outcropping and depth to bedrock, and

(c) any impermeable clay layers, and

(d) the depth to groundwater, perched groundwater tables, groundwater seepage and recharge areas, and

(e) any watercourses, riparian land or groundwater dependant ecosystems, and

(f) any steep slopes, and

(g) any environmentally sensitive areas.

Note: The following *Part C6 - Table 1* indicates the site constraints that should be considered for a range of WSUD measures

Part C6 - Table 1: Site constraints for various WSUD measures.
Source: Adapted from Water by Design (2009) Concept Design Guidelines for Water Sensitive Urban Design Version 1, South East Queensland Healthy Waterways Partnership, Brisbane, March 2009

WSUD Measure	Steep site	Shallow bedrock	Low permeability soil e.g. clay	High permeability soil e.g. sand	High water table	High sediment input
Gross pollutant capture devices	Y	D	Y	Y	D	Y
Sediment basins	D	D	Y	D	D	Y
Grass or vegetated swales	X	D	Y	D	X	D
Sand filter	Y	D	Y	Y	X	D
Bioretention systems	D	D	Y	D	X	D
Constructed wetlands	X	D	Y	D	D	D
Rainwater tanks	Y	Y	Y	Y	Y	Y
Porous pavements	X	X	X	Y	X	X
Infiltration measures	X	X	X	Y	X	X

X = Constraint may preclude use

D = Constraint may be overcome through appropriate design

Y = Generally not a constraint

C11. A geotechnical investigation and report is required to be undertaken by a qualified geotechnical professional with demonstrated experience to investigate the current site conditions and suitability of the site to accommodate WSUD devices to provide stormwater water quality control, infiltration and treatment systems.

C12. The results of the investigation are to inform the WSUD strategy and confirm the nature of WSUD devices are suitable and effective for the site.

Stormwater Design Principles

C13. All stormwater management measures are to be located within the site boundaries and are to not cause nuisance to adjoining properties.

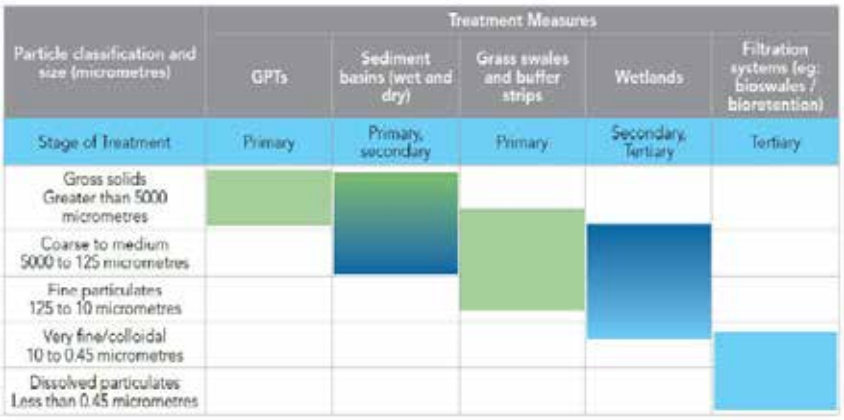
C14. Locate devices wholly outside of environmentally sensitive areas, unless no alternative is available and it is demonstrated that minimal adverse impact will result.

C15. The WSUD strategy is include a treatment train that is to comprise all necessary stormwater treatment devices, to provide the retention, filtering, infiltration and safe disposal necessary to meet the required stormwater treatment performance outcomes.

C16. The selection and order of stormwater management devices in the treatment train is to provide the necessary primary, secondary and tertiary treatment, in the appropriate sequence, to effectively remove the expected range of pollutants (including but not necessarily limited to sediment, gross pollutants, Total Phosphorous, Total Nitrogen, Total Suspended Solids, heavy metals, oils & greases, hydrocarbons) to achieve the required treatment.

C17. The coarse fraction of pollutants (i.e. litter and coarse sediment) must generally be removed first to ensure subsequent finer and dissolved pollutant treatments are effective.

C18. The stormwater treatment device(s) within the treatment train must be suitable for the characteristics of the target pollutants being generated from the development.



Part C6 - Figure 4: Pollutant ranges for best management practice treatment measure

Source: Blacktown City Council (adapted from Ecological Engineering (2003) Landcom Water Sensitive Urban Design Strategy - Design Philosophy and Case Study Report, Landcom NSW).

C19. The treatment train is to prioritise the use of filtration and infiltration source controls where ever possible over a single end-of-line measure.

C20. Sediment traps, vegetated filter strips or equivalent measures are to be provided upstream of porous paving to reduce sediment inputs and minimise likelihood of clogging.

C21. To minimise site disturbance, consideration is to be given to the achievement of multiple stormwater management objectives in one device (e.g. water quality treatment, retention and detention).

C22. To prevent the release of previously trapped sediment or pollutants by high flows or flood waters, and avoid damage to vegetation components, stormwater treatment measures are to be designed to cater for high flows above the devices' design size and located outside of the 1:100 year flow path.

C23. Any bio-retention filter areas must be the minimum size required to achieve the required stormwater treatment performance criteria and to minimise site disturbance. The minimum size required is to be demonstrated by a supporting stormwater quality analysis (e.g. via MUSIC model).

Note 1: The planting schedule is to ensure vegetative components within the bio-retention system(s) include local macrophytes and groundcovers that:

a. are adapted to local climatic conditions, and

b. are adapted to the expected high and variable nutrient and moisture conditions, and

c. are suitable for the bio-retention basin's expected management and maintenance requirements, and

d. include plants with extensive fibrous root systems, spreading rhizomatous or suckering habitat in preference to plants with a clumped habit, and

e. are to be established at a minimum density of at least 8 plants per square metre across the base and any side batters of the bio-retention systems, and

f. ensure complexity and year round coverage by including at least 4 different macrophyte or groundcover species within the basin area.

C24. Roof gardens and wall gardens are to be considered and incorporated as a measure to maximise water re-use opportunities and enhance landscape outcomes where feasible to do so, particularly for developments with minimal pervious areas.

C25. Impermeable liners are to be utilised to prevent infiltration from any infiltration system or bio-retention device within areas of shallow groundwater or where potential groundwater contamination is identified.

C26. Developments with potential to generate pollutants such as fuels, heavy metals, oils, solvents, detergents, fertilisers, pesticides and the like, must demonstrate appropriate containment and operational methods (e.g. bunding, first flush systems, dedicated vehicle washing bay), and provide appropriate water quality treatment and management measures to mitigate impacts of stormwater runoff. Such developments are to provide contingency management measures in the event of an accidental spill to protect the natural environment.

Note: Environment pollution licenses may apply under the *Protection of the Environment Operations Act, 1997.*

C27. All absorption and infiltration, devices are to be located at least 4m from any property boundary, retaining wall or building.

C28. All absorption and infiltration devices are to provide suitable collection and disposal of design overflows to a stormwater drainage system (street gutter, inter-allotment drainage system or drainage easement), unless these overflows can be otherwise safely disposed of within the site.

C29. Stormwater disposal outlets are to provide scour protection and dissipation mechanisms compatible with protecting the natural environment, by:

(a) preferentially disposing of stormwater runoff within the site boundaries by re-use, absorption, and infiltration or evapotranspiration measures to the maximum extent that the expected site usage and site conditions allow, and

(b) minimising the area of impervious areas directly connected by piped or sealed drainage infrastructure to the receiving natural environment, and

(c) providing level spreading overflows to stable vegetated areas, and

(d) minimising the number of concentrated stormwater outlets, and

(e) providing scour protection using environmentally sensitive soft-engineered methods such as a rough and permeable rock riprap surface combined with native revegetation, rather than gabions, concrete surfaces or concrete grouting, and

(f) providing grassed filter strips or vegetated buffer areas to protect the receiving natural environment from stormwater runoff, and

(g) disposing of stormwater design overflows from absorption, infiltration, rain garden or bio-retention devices, which may not otherwise be disposed of within the site to the street gutter, where the site falls to the street, and

(h) disposing of stormwater design overflows from absorption, infiltration, rain garden or bio-retention devices via a level spreading overflow to a stable vegetated area and any other measures necessary to protect the receiving natural environment where the site falls away from the street and towards bushland or natural areas, and

(i) disposing of stormwater design overflows from absorption, infiltration, rain garden or bio-retention devices to an interallotment drainage system where the site falls away from the street and towards areas that are not bushland or natural areas, and

Note: See Part C6.5 regarding connections to interallotment drainage systems and alternate methods of stormwater disposal.

(j) not piping or otherwise discharging stormwater directly to a watercourse unless no alternative is available.

Landscape, visual and social amenity

C30. The development and its associated water sensitive urban design measures are to enhance the landscape, visual, social and recreational amenity by:

(a) incorporating stormwater treatment and control measures into the urban and landscape design where appropriate, and

(b) incorporating local indigenous plant species into stormwater and landscaping design to enhance the natural attributes of the Blue Mountains LGA, and

(c) ensuring the location and design of stormwater treatment and control devices do not compromise the achievement of landscape outcomes and protection of environmentally sensitive areas, and

(d) maximising the retention and restoration of vegetated areas and deep-rooted trees to maintain the water table conditions, stabilise the soil, decrease runoff volumes and velocities, capture sediment and filter nutrients, and

(e) considering, and where appropriate providing adaptability to, climate change in the design and planning of vegetation in the landscape.

Operation and maintenance – requirements, feasibility and costs

C31. Provide for the ongoing operation and maintenance of WSUD devices by locating and designing devices with appropriate access for all maintenance activities necessary to ensure their effective operation to meet the required performance and treatment targets over the life of the development.

Assets proposed to be transferred to the Council

C32. Stormwater assets proposed to be transferred to Council are to relate to the treatment of runoff from public land only.

C33. Any proposal to transfer stormwater treatment assets to Council must be demonstrated to be an effective sound solution and in the public interest.

C34. Any stormwater assets transferred to Council must be designed and located in a manner that ensures effective ongoing operation and maintenance.

C6.2. On-site stormwater detention

Explanation

On-site Stormwater Detention (OSD) temporarily stores excess stormwater on a site. It acts to restrict the rate that the stormwater leaves the site with the aim of better managing the rate of stormwater entering the drainage system, to reduce the risk of downstream scouring and flooding effects.

Objectives

O1. To control the rate of release of private stormwater into Council's drainage system so as to not exceed the hydraulic and environmental capacity of the receiving system.

Controls

C1. OSD is required for all developments, other than single dwellings, that increase the impervious area of a site by more than 100m^2.

C2. OSD will be required for single dwelling developments where there is a restriction on the title of the land requiring OSD, or where increases in stormwater discharge rates are likely to result in adverse environmental impacts or where there are known downstream hydraulic capacity constraints.

C3. OSD systems shall be designed and constructed to achieve post development discharge rates that are no greater than the pre-development discharge rates for all storms up to and including the 100 year event.

C4. All development applications that require OSD are to be supported by a *Concept Stormwater Management Plan* (SMP) that sizes and locates the proposed OSD system. The SMP is to include calculations justifying the sizing of the OSD system and specifying the 100 year event top water level in the system.

C5. Where a site discharges to separate sub-catchments pre-development, discharge into those sub-catchments and discharge rates are to be maintained post development.

C6. OSD storage is to be located at a level that is above the 1:5 ARI flood level.

C7. OSD proposed in car parking areas may be a combination of above and below ground storage. All storage in car parks is to be below ground for all storms up to the 20 year event, and the above ground storage is to be no greater than 200mm in the 100 year event.

C8. Below ground OSD tanks are to be a minimum internal depth of 800mm to allow access for maintenance.

C9. OSD systems are to be provided with an emergency overflow/ surcharge path to a public road, or to a watercourse within the site. The overflow/surcharge path is to have the capacity for the post-development 100 year event.

C6.3. Groundwater

Explanation

Sustainable development requires sound groundwater management that protects groundwater regimes, water quality and groundwater dependent ecosystems.

Groundwater is an integral part of the water cycle and is an important resource and natural asset. While water availability in a river can change very quickly because of rainfall or dry conditions, groundwater availability is more stable.

Groundwater can be found almost everywhere but the depth of the water table may vary. Groundwater supplies are replenished, or recharged, by rain. During heavy rain the water table may rise and during extended dry weather may fall.

Water in aquifers may be brought to the surface naturally via a spring or discharge into a local creek or river via the water table.

Perched aquifers are a type of aquifer of generally limited extent that occurs where an impermeable layer prevents the downward infiltration of groundwater. The formation of perched aquifers may occur within soils or rocks, but are most commonly presented as groundwater retained upon clay layers within unconfined alluvial aquifers. Perched aquifers are common in the upper Blue Mountains and support groundwater dependent ecosystems such as Blue Mountains Swamps.

Springs occur where groundwater intersects with the land surface. Such springs are generally developed from shallow, perched groundwater systems that are of limited lateral extent and can be depleted rapidly in the absence of continuing recharge.

This section applies to all development sites where ground water is evident and excavations may interact with the groundwater table, and also applies to all development proposals other than single dwellings, with an excavation greater than 1m.

Objectives

O1. To protect and maintain groundwater regimes and natural hydrological processes.

O2. To protect and maintain groundwater quality.

O3. To protect and maintain groundwater dependant ecosystems and their supporting groundwater systems.

O4. To ensure appropriate site investigations are undertaken to identify the potential for a development to affect groundwater.

Controls

C1. Development is to be designed, constructed and operated to ensure no adverse impacts to the nature, extent and behaviour of natural groundwater systems. Where an adverse impact cannot be avoided, it is to be minimised, mitigated and offset.

C2. The design and construction methods of all development shall avoid the need for aquifer interference and sub-surface drainage.

C3. If interference with the water table is unavoidable, all groundwater management systems are to be located and designed so that they are easily maintained.

C4. The development is to demonstrate that there will be no adverse impact on surrounding or adjacent properties, infrastructure, natural groundwater systems or groundwater dependant ecosystems as a result of changes in the nature or behaviour of groundwater on site or in the surrounding area caused by:

(a) the design of the constructed form, and

(b) the method of construction chosen, and

(c) the groundwater management system implemented, and

(d) the cumulative impact the development may have on groundwater.

Note: Groundwater dependant ecosystems are common in the Blue Mountains LGA.

C5. All development proposals where ground water is evident and excavations may interact with the groundwater table, and all development proposals other than single dwellings, with an excavation greater than 1m, are to undertake a Groundwater Assessment (or Geotechnical Investigation) to establish whether the development may impact upon groundwater.

C6. If proposed development is determined to impact upon groundwater, then the development application is to be supported by a Groundwater Management Plan.

C7. Where an impediment to the natural flow paths is created as a result of the nature of the construction methods utilised or the bulk of the below-ground structure, a temporary or permanent groundwater management system may be utilised. These systems may only be utilised where it can be demonstrated that the natural groundwater flow regime is restored both up-gradient and down-gradient of the site, without any adverse effects on surrounding property, infrastructure, natural groundwater systems or nearby groundwater dependant ecosystems.

C8. All development applications involving excavations that require temporary de-watering are to include a statement prepared by a suitably qualified engineer experienced in the construction of structures below a water table. The statement shall:

(a) outline the proposed method of construction and dewatering, and

(b) outline the proposed method to reintroduce the extracted water back into the water table, and

(c) confirm that the development can be feasibly constructed without causing unreasonable impacts on the groundwater system or neighbouring properties.

Note: Dewatering activities may require concurrence from the NSW Office of Water/State Government Approval Body. The NSW State Government must be consulted if dewatering is proposed prior to lodgement of the application. If State Government approval is required, then the development application must be lodged as Integrated Development - *Water Management Act 2000*.

C9. Where the development is to involve construction of permanent structures below the water table, details of the method of construction, other than pile or footing installation are to be provided. The details provided must be prepared by a suitably qualified engineer experienced in the construction of structures below a water table and be sufficient to demonstrate compliance with the following:

(a) all components of the structure including subsoil drainage must be located entirely within the property boundary, and

(b) disposal of collected sub soil water must be achieved through a gravity drainage system, and

(c) the groundwater management system must not have an adverse impact on the natural hydrological regime of the locality.

C6.4. Flooding

Explanation

Flooding in NSW is managed in accordance with the NSW Government's Flood Prone Lands Policy, which aims to reduce the impact of flooding and flood liability on individual owners and occupiers of flood prone property, and to reduce private and public losses resulting from floods, utilising ecologically positive methods wherever possible.

Blue Mountains City Council is in the process of undertaking Floodplain Risk Management Studies and Plans for its catchments to quantify flooding risks and potential measures in accordance with the NSW Government's Floodplain Development Manual. This part of the DCP is to be read in conjunction with the City of Blue Mountains Flood Liable Land Policy.

This part of the DCP applies to:

- all development on land below the 1% Annual Exceedance Probability (AEP) flood plus the required freeboard, and
- development of critical facilities on land below the Probable Maximum Flood (PMF) plus the required freeboard, and
- development on all land affected by or adjacent to a stormwater drainage system, and
- development on land in a catchment that has not been the subject of a Floodplain Risk Management Study, but due to its location and topography, it is considered reasonable to expect that the land may be subject to an overland flow path and/or flood impacts.

Applicants are encouraged to liaise with Council early in the design process to identify any applicable flooding implications.

Objectives

O1. To control development at risk of flooding in accordance with the NSW Government's Floodplain Development Manual.

O2. To ensure that the economic and social costs which may arise from damage to property due to flooding is minimised and can be reasonably managed by the property owner and general community.

O3. To reduce the risk to human life and damage to property caused by flooding by controlling development on land impacted by potential floods.

O4. To ensure that development is appropriately sited and designed according to the site's sensitivity to flood risk.

C6.4.1. Flood Studies and Plans

Objectives

O1. To ensure that development addresses any relevant flood studies, and is consistent with the requirements of any floodplain risk management studies or plans.

Controls

C1. Development applications are to identify any flood related information including flood levels, locations of floodways or overland flow paths impacting the site. Where the site is affected, the application is to be supported by a site specific flood study or other calculations to demonstrate there is no adverse impact on the development from flooding or overland flow paths in a 1% AEP event.

C2. The development proposal is to comply with any catchment-specific controls in an adopted Floodplain Risk Management Plan in addition to the controls in this section.

Note: As the Council prepared flood studies are broad scale assessments generally utilising aerial photography contours, a site specific flood study may be required even if the site is located in a catchment where a Council flood study has already been prepared.

C6.4.2. Flood effects

Objectives

O1. To ensure that development, either individually or cumulatively, minimises adverse impacts on flooding, conveyance of floodwaters and floodplain storage volume.

O2. To ensure that floodways and overland flow paths are not obstructed by development.

Controls

C1. The development shall not increase flood effects elsewhere, having regard to loss of flood storage, changes in flood levels and velocities and the cumulative impact of multiple potential developments, for floods up to and including the 1% AEP flood.

C2. Floodways and overland flow paths must not be obstructed or diverted onto adjoining properties.

C3. Areas identified as flood storage areas must not be filled unless compensatory excavation is provided to ensure that there will be no net loss of floodplain storage volume below the 1% AEP flood.

C6.4.3. Floor levels

Objective

O1. To ensure that floor levels are set at an appropriate height to reduce the frequency of inundation of structures and floors to an acceptable probability.

Note: Floor levels refer to the minimum required building floor levels. For development such as basements, the floor level refers to the lowest level at each access point.

Controls

C1. Building floor levels shall comply with the *Part C6 - Table 2*.

C2. For the purposes of this part of the DCP, non-habitable floors include areas such as laundries or sheds, but exclude garages. All other floor spaces are habitable areas.

Part C6 - Table 2: Floor levels for buildings

Scenario	Floor Level
Habitable Floors - all development (excluding critical facilities)	
Inundated by flooding	1% AEP + 0.5m freeboard
Inundated by overland flow path	1% AEP + 0.5m freeboard
Habitable floors - Critical facilities	
Inundated by flooding	PMF + 0.5m freeboard
Inundated by overland flow path	1% AEP + 0.5m freeboard
Non-habitable floors – residential outbuildings (excluding garages)	
Gross floor area less than or equal to 10 square metres.	1% AEP but not less than 0.15m above surrounding ground level
Gross floor area greater than 10 square meters.	The applicable habitable floor level
Non-habitable floors – Industrial and commercial	
Located on flooding or over-land flow path	1% AEP but not less than 0.15m above surrounding ground level

Material storage locations – all development	
Materials sensitive to flood damage, or which may cause pollution or be potentially hazardous during flooding	1% AEP + 0.5m freeboard

Note: Floor levels for car parking areas are covered in *Part C6 - Table 3.*

C6.4.4. Building components

Objective

O1. To ensure the structure and construction of development is compatible with flooding up to the applicable floor level.

Controls

C1. All development shall have flood compatible building components below the floor levels identified in *Part C6 - Table 2*.

C2. All structures within the flood plain shall be constructed to withstand the forces of floodwater, debris and buoyancy up to and including the floor levels identified in *Part C6 - Table 2*.

C3. All development that is in 'cut' is to address the impacts of overland flow path from upstream catchments, and ensure that there is adequate provision for flows paths around the structure, stepping up at least 225mm from the finished ground level to the floor level of the structure.

C6.4.5. Driveway access and car parking

Objectives

O1. To ensure car parking and site access is constructed to an acceptable flood standard.

O2. To require appropriate protection measures for warning and safe evacuation from basement car parking.

O3. To minimise the likelihood of cars or other objects becoming floating debris during a flood.

Controls

C1. Car parking floor levels shall comply with *Part C6 - Table 3.*

C2. Vehicular access is to be located where the road level is greater than or equal to the required floor level for the car park. Where road access above the required floor level is not available, the vehicular access is to be located at the highest feasible location.

C3. The level of the driveway between the road and car park shall be no lower than 0.3m below the 1% AEP flood, or such that the depth of inundation during the 1% AEP flood is not greater than the depth of flooding at either the car park or the road where the site is accessed.

C4. Underground car parking accommodating more than three vehicles shall have warning signage to ensure safe evacuation.

C5. Barriers are to be provided to prevent floating vehicles leaving the site during a 1% AEP flood if the depth of flooding exceeds 0.3m.

C6. Vehicle access to critical facilities that have an emergency function must be achieved for floods up to the PMF.

Part C6 - Table 3: Floor Levels for Car Parking

Scenario	Floor Level
Above ground level open car parking, car ports and garages	
Open car parking spaces and car ports	5% AEP flood
Residential garages with up to two spaces	1% AEP but not less than 0.15m above surrounding ground level
Residential garages with more than two spaces	Applicable residential habitable floor level requirement (Table 2)
Enclosed industrial/ Commercial parking spaces	Applicable industrial/commercial floor level requirement (Table 2)
Underground car park (where floor level is more than 0.8m below surrounding ground level)	
All driveways	1% AEP plus 0.3m freeboard at its highest point
All emergency exits	All underground garages and car parks to have emergency exits protected from inundation up to the 1% AEP flood plus 0.5m freeboard with a minimum of 0.2m freeboard from vehicle entry point.
All other openings inundated by flooding or local overland flow path	All openings to be sealed up to 1% AEP + 0.5m freeboard with a minimum of 0.3m above the surrounding ground level

C6.5. Stormwater infrastructure

Explanation

This section applies to all development in proximity to public stormwater infrastructure or inter-allotment drainage, and all development requiring new connections to the Council's drainage system.

Objectives

O1. To ensure stormwater infrastructure is designed and constructed to an acceptable standard.

O2. To prevent adverse impacts of development on the performance, serviceability and integrity of interallotment drainage lines and publicly owned stormwater systems.

O3. To ensure that private stormwater systems discharge to the public stormwater system in an acceptable manner.

Controls

C1. Natural flow paths and discharge points from the site are to be retained and directed to its natural catchment.

C2. Structures are not to be constructed above public stormwater infrastructure or inter-allotment drainage.

C3. Redevelopment of existing structures above public stormwater infrastructure or inter allotment drainage shall occur only where:

(a) relocation of the stormwater conduit or structure is not feasible, and

(b) the conduit is reconstructed to meet relevant standards and to ensure structural soundness and serviceability for the life of the structure and the life of the conduit, and increased in capacity to accommodate the 100 year event upstream runoff, with a 50% blockage factor, if there is no safe alternate surcharge path.

C4. Stormwater is to be gravity drained to Council's drainage system or a natural watercourse. This may require the provision of an inter-allotment drainage system

C5. Where the private stormwater drainage system is unable to connect to the road gutter within the frontage of the site due to differences in level, the piped road drainage may be extended to the site to facilitate direct connection to the new road drainage pit.

C6. Where an inter-allotment drainage easement is required to facilitate a development, the applicant is to negotiate the creation of the easement over downstream properties for drainage purposes prior to the lodgement of the development application. A letter of consent from the owners of the affected downstream properties is to be submitted with the development application.

Note: If consent is granted to the development, the registration of the easement/s will be required as a deferred commencement consent matter, prior to the issue of an operational consent.

C7. Alternate solutions to interallotment drainage will only be considered where the applicant can demonstrate that they have undertaken a thorough and complete negotiation process with the owners of downhill properties, and have written evidence of that negotiation and the downhill property owners' refusal to grant the easement. The negotiation process is to include a reasonable offer of compensation based upon advice from a licensed land valuer.

C8. Alternate solutions to interallotment drainage, in descending order of preference are:

(a) a combination of onsite detention, retention and absorption, or

(b) a charged drainage line to the road gutter, or

(c) an onsite sump and pump to the road gutter.

C9. Charged drainage lines are to incorporate a mechanism to permit cleaning of material from the lowest point of the drainage line.

C10. Sumps for onsite sump and pump system are to have the capacity to hold runoff from the 4 hour, 1 in 100 year ARI storm event (to allow for loss of power during storm events).